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The role of neoantigens and tumor mutational burden in cancer immunotherapy: advances, mechanisms, and perspectives

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Abstract

Cancer immunotherapy has revolutionized oncology by leveraging the immune system to combat tumors. Among various biomarkers, neoantigens and tumor mutational burden (TMB) have emerged as critical factors in tailoring personalized treatments. Neoantigens are tumor-specific peptides displayed on cancer cell surfaces, derived from somatic mutations. Recognized as "non-self" by the immune system, they trigger T-cell responses and enable therapies like personalized vaccines and adoptive T-cell transfer. Critically, neoantigen potential correlates with TMB, which quantifies the total somatic mutations within a tumor genome. A higher TMB generally correlates with a greater likelihood of generating immunogenic neoantigens, making it a predictive biomarker for the efficacy of immune checkpoint inhibitors (ICI). Progress in high-throughput sequencing, bioinformatics, and immuno-peptidomics has significantly enhanced the accuracy of neoantigen prediction, including assessments of major histocompatibility complex (MHC) binding affinity and T-cell receptor recognition. Clinically, neoantigen-based therapies have shown efficacy in early trials, with strategies such as mRNA vaccines demonstrating synergy with ICI by boosting T-cell activation and overcoming immune suppression. Combining neoantigen-based therapies with chemotherapy and radiotherapy harnesses synergistic mechanisms to enhance efficacy, overcome resistance, and emerge as a pivotal oncology research focus. The integration of TMB into clinical practice has received regulatory approval as a biomarker for stratifying patients for ICI therapies. Furthermore, advanced methodologies like liquid biopsy and single-cell technologies have streamlined TMB measurement, improving its predictive value for personalized immunotherapy. Collectively, neoantigens and TMB have optimized the evolution of precision immuno-oncology by providing frameworks that maximize therapeutic efficacy, overcome resistance mechanisms, and advance durable cancer remission.

Keywords Tumor neoantigens, Tumor mutational burden, Immunotherapy, Liquid biopsy, Clinical trials

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Introduction

Cancer immunotherapy represents a transformative approach in cancer treatment, leveraging the host immune system to target and eliminate malignant cells [1]. The conceptual foundation of this approach dates back to the late nineteenth century, when William B. Coley, often regarded as the father of immunotherapy, utilized bacterial infections to induce an immune response against tumors [2, 3]. Over the past century, immunotherapy has evolved to encompass a variety of strategies, including immune checkpoint inhibitors, oncolytic viruses (OV), cytokines, and cancer vaccines [4]. These modalities operate through distinct mechanisms, targeting the complex interactions between immune cells and neoplastic cells within the tumor microenvironment (TME) [5, 6]. The effectiveness of immunotherapy hinges on the ability of T lymphocytes to identify tumor-specific antigens. This principle has guided the development of immune checkpoint blockade (ICB) therapies, including programmed cell death-1 (PD-1)/programmed cell death ligand-1 (PD-L1) inhibitor and cytotoxic T-lymphocyte-associated protein 4 (CTLA-4) inhibitor, both of which have achieved remarkable success in treating diverse cancers [7, 8]. However, the response rates to immunotherapy remain limited, with only a subset of patients deriving benefit due to tumor-intrinsic and extrinsic resistance mechanisms [7]. Consequently, the exploration of robust biomarkers in cancer immunotherapy is of paramount importance. Biomarkers can predict patient responsiveness to specific immunotherapeutic agents, thereby significantly enhancing the precision and efficacy of treatment [9]. They are also critical in identifying individuals at risk for severe adverse effects, thus improving the safety profile of these therapies [10]. Moreover, novel biomarkers can elucidate the mechanisms of resistance and toxicity, facilitating the development of innovative therapeutic strategies [9, 11]. The identification and validation of superior biomarkers are essential for broadening the therapeutic impact of immunotherapy to a wider patient population.

The emergence of neoantigens and tumor mutational burden (TMB) as biomarkers in cancer immunotherapy signifies a substantial advancement in the field. The key development events about neoantigen, TMB and cancer immunotherapy are shown in Fig. 1. Neoantigens, arising from somatic mutations unique to individual tumors, have become crucial targets for personalized immunotherapies [8, 12]. Their identification is based on the premise that these tumor-specific antigens can be recognized by the host immune system, facilitating the development of tailored treatments to enhance anti-tumor immunity [13]. This progress has led to innovative

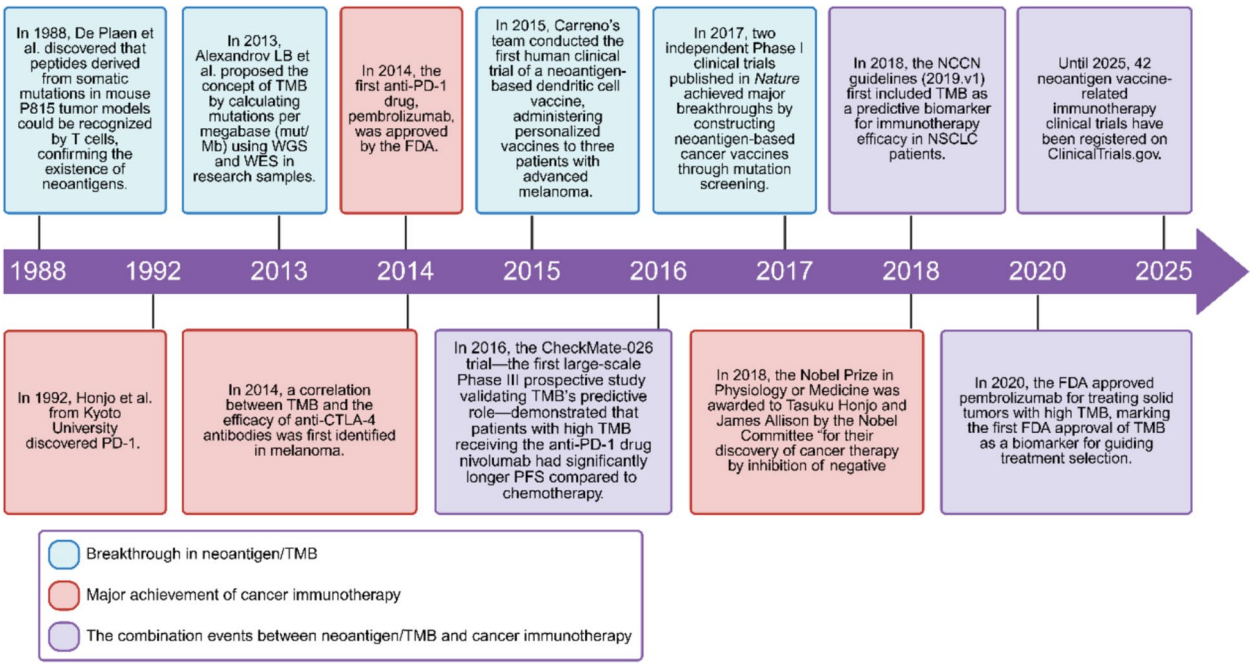

Fig. 1 Milestone events of neoantigen, TMB and cancer immunotherapy. The breakthrough in neoantigen or TMB, the major achievement of cancer immunotherapy and their combination events were reviewed retrospectively. CTLA-4: cytotoxic T-lymphocyte-associated protein 4, FDA: Food and Drug Administration, NCCN: National Comprehensive Cancer Network, NSCLC: non-small cell lung cancer, PD-1: programmed cell death-1, PFS: progression-free survival, TMB: tumor mutational burden, WES: whole-exome sequencing, WGS: whole-genome sequencing. Created with BioRender.com

therapeutic strategies, including personalized vaccines and adoptive T-cell therapies, which leverage neoantigens to provoke a targeted immune response [14, 15]. On the other hand, TMB quantifies the total number of somatic non-synonymous mutations within a tumor's genome, serving as an indicator of the potential neoantigen landscape [16]. The concept of TMB gained attention with the recognition that a higher mutational load might correlate with an increased likelihood of response to immune checkpoint inhibitors (ICI) [17]. This correlation has been validated by both retrospective and prospective studies, establishing TMB as a predictive biomarker for ICI response across various cancer types [18–20]. The incorporation of TMB into clinical practice represents a significant milestone in the application of biomarkers in immunotherapy. However, the journey of neoantigens and TMB has encountered several challenges. The heterogeneity of tumor mutations, the complexity of the TME, and the inherent variability in immunogenicity have posed significant obstacles to accurately predicting treatment response [16, 21]. Despite these obstacles, the role of neoantigens and TMB in immunotherapy is increasingly recognized, with ongoing research focused on refining their predictive capabilities and expanding their clinical applications [22]. As the field advances, the integration of neoantigen and TMB data with other biomarkers, such as PD-L1 expression and microsatellite instability [23, 24], is expected to further enhance the precision of immunotherapy, ultimately aiming to fulfill the full potential of this transformative treatment approach.

In the present review, we elucidate the definitions and interrelationship between neoantigens and TMB. The advancements in neoantigen-based treatments and the application of TMB in predicting immunotherapy responses are also explored. Additionally, we discuss the detailed mechanisms by which neoantigens and TMB contribute to cancer immunotherapy. This review also summarizes future challenges and potential strategies involving neoantigens and TMB, aiming to enhance neoantigen prediction and treatment efficacy in cancer immunotherapy.

Neoantigen and TMB: definition, interrelationship and assessment

Neoantigen and TMB are two pivotal concepts in cancer immunotherapy. Understanding their definitions and interrelationship is fundamental for grasping the mechanisms of immune recognition and therapeutic response (Fig. 2). Neoantigens are derived from non-synonymous somatic mutations that are unique to individual tumors and absent in the normal germline genome [25, 26]. These neoantigens arise from alterations in the DNA sequence, leading to the production of novel or altered proteins that can be recognized as foreign by the immune system, particularly by T cells [27, 28]. The generation of neoantigens is a direct result of the genetic instability inherent in cancer cells, which accumulate a high number of mutations over time [29]. The immune system has the potential to recognize these neoantigens as non-self, triggering an anti-tumor immune response [30]. Conversely, TMB quantifies the total number of somatic non-synonymous mutations within a tumor's genome [16]. It serves as a proxy for the potential neoantigen landscape of a tumor, as a higher mutational load suggests a greater likelihood of presenting a broader array of neoantigens to the immune system [31]. The relationship between neoantigens and TMB is intricate and complementary. While TMB provides a broad estimation of the neoantigen potential within a tumor, specific neoantigens offer a more targeted approach to understanding an individual tumor's immunogenicity [32, 33]. The interplay between these factors is critical in developing personalized cancer vaccines and selecting patients most likely to benefit from ICI. TMB-high tumors are often associated with a better response to ICI [32]; however, the presence of specific neoantigens can further refine patient stratification, potentially leading to more precise and effective immunotherapeutic strategies [34]. Thus, neoantigens and TMB are closely linked in cancer immunotherapy. Neoantigens represent personalized antigenic targets that can be exploited for targeted immunotherapy, while TMB serves as a predictive biomarker gauging the overall immunogenic potential of a tumor [35, 36]. The combination of these parameters holds promise for advancing precision medicine in oncology, allowing for the development of more tailored and effective treatment approaches.

Neoantigen identification and prediction methods have become increasingly sophisticated, facilitating the precise targeting of tumor-specific antigens in cancer immunotherapy [25]. The latest advances in the technologies used for defining or characterizing neoantigens are listed in Table 1 [22, 37–49].The process typically involves high-throughput sequencing to detect somatic mutations within the tumor genome, which are then correlated with the potential to generate neoantigens [50, 51]. Next-generation sequencing (NGS) serves as a cornerstone technology for neoantigen discovery by enabling comprehensive profiling of the tumor exome and identification of non-synonymous mutations that could lead to neoantigen production [52–54]. Besides, bioinformatics tools are essential for predicting mutations that can generate immunogenic peptides. Algorithms such as NetMHCpan and NetMHCIIpan predict the binding affinity of potential neoantigens to major histocompatibility complex

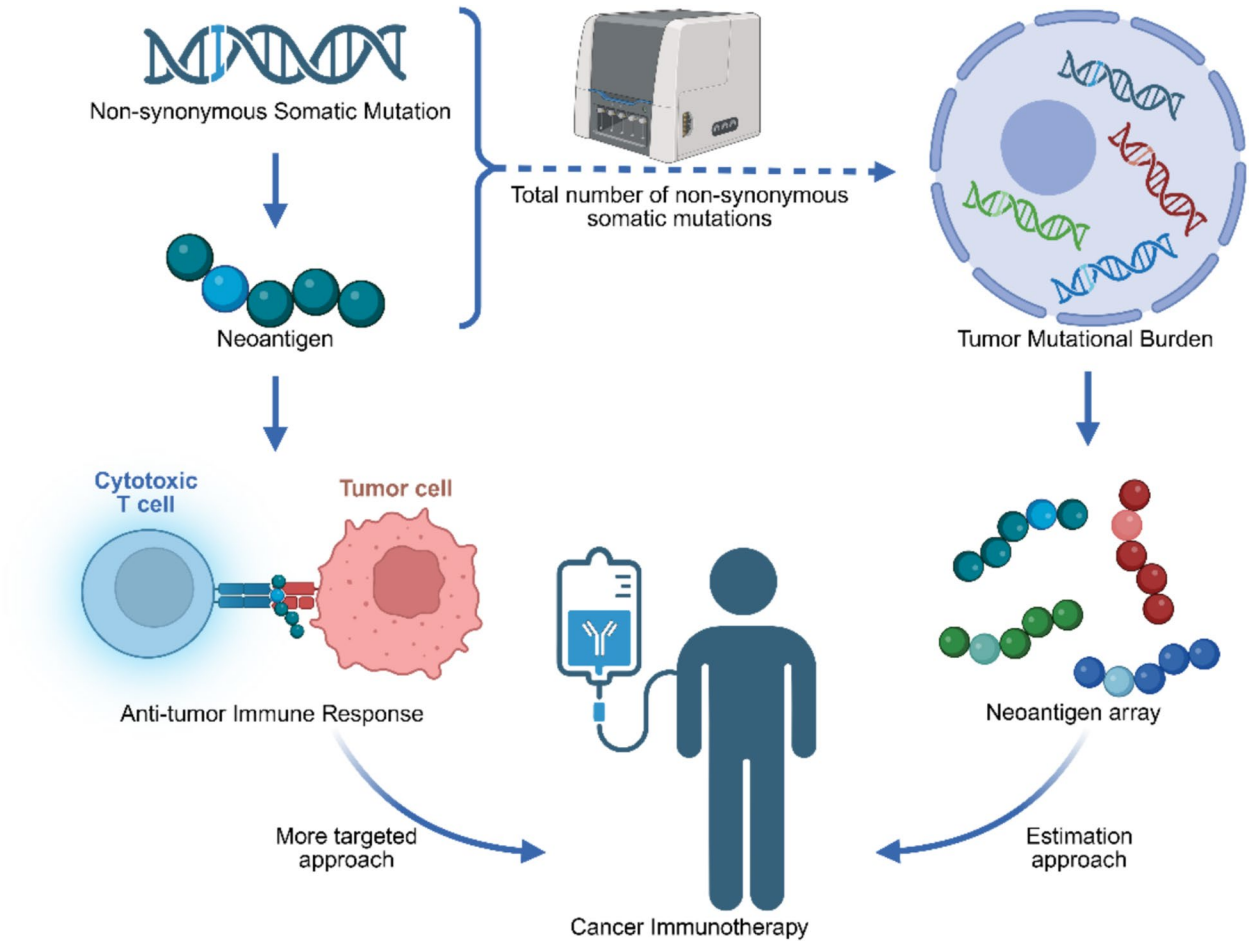

Fig. 2 The interrelationship between neoantigen, TMB and cancer immunotherapy. Neoantigens are derived from non-synonymous somatic mutations, leading to the production of novel or altered proteins that can be recognized as foreign by the immune system. When recognizing these neoantigens, the immune system triggers an anti-tumor immune response which represent personalized antigenic targets that can be exploited for targeted immunotherapy. On another hand, TMB quantifies the total number of somatic non-synonymous mutations within a tumor's genome. Higher mutational load suggests a greater likelihood of presenting a broader array of neoantigens, thereby providing a broad estimation of the neoantigen potential within a tumor. As a result, specific neoantigens offer a more targeted approach, while TMB serves as a predictive biomarker gauging the overall immunogenic potential of a tumor. Created with BioRender.com

(MHC) molecules [40, 55], a critical step in presenting antigens to T cells. These tools integrate peptide-MHC binding motifs and experimental binding data to predict the likelihood of a peptide binding to a specific MHC allele. Mass spectrometry (MS)-based approaches complement sequencing methods by directly analyzing peptides presented on the tumor cell surface, allowing for the identification of actual neoantigens [56]. This immunopeptidomic approach offers a more accurate assessment of the tumor's immunopeptidome, which comprises the peptides loaded onto MHC molecules and presented to the immune system [57]. Integrating NGS with MS data has also significantly improved neoantigen prediction accuracy [45, 58]. By comparing the tumor's mutational landscape with the peptides identified by MS, researchers can pinpoint which mutations are likely to be presented as neoantigens [59]. This integrated approach helps narrow down potential neoantigens to those both expressed and presented on the tumor cell surface, making them more likely to be recognized by the immune system [60]. In summary, the identification and prediction of neoantigens involve combining NGS to detect tumor-specific mutations, bioinformatics tools to predict peptide-MHC binding, and MS to confirm neoantigen presence in the tumor's immunopeptidome. These methods are essential for developing personalized immunotherapies that can effectively target the unique neoantigens expressed by an individual's tumor.

The measurement of TMB is typically conducted using NGS technologies, with whole-exome sequencing (WES)

Table 1 The latest advances in the technologies used for defining or characterizing neoantigens

Classification	Name	Input	Output	Neoantigen Source	Characteristics and utility	Ref
Genomic and Transcriptomic Approaches	NGS	WES and RNA-seq FASTQ	Features & ranks	SNV and Indel	Identifies patient-specific somatic mutations absent in normal tissues. Enables predicting neoantigen candidates for personalized vaccine design	[37]
	WES	Genomic DNA and FASTQ	Features & ranks	SNV and Indel	Identifies patient-specific somatic mutations, particularly missense SNV and Indel, in expressed genes. Enables prediction of unique, potentially immunogenic neoantigen profiles for personalized breast cancer immunotherapy and biomarker discovery in specific populations	[38]
	pVAC-Seq	Annotated non-synonymous somatic mutations, Patient-specific HLA alleles, and RNA-Seq expression data	Features & ranks	SNV	Integrates tumor mutation data with HLA binding affinity and RNA expression for streamlined analysis. Prioritizes high-confidence neoantigen candidates for personalized cancer vaccines and immunotherapy screening	[39]
	NetMHC	Peptide sequences, MHC allele(s), and Peptide length	Features & ranks	SNV and Indel	Employs concurrent motif deconvolution and integration of MS eluted ligand data for state-of-the-art predictive performance. Enables identification of potential T-cell epitopes and MHC ligands for applications like vaccine design and immunotherapy	[40]
	TSNAD	WGS FASTQ, WES FASTQ, and Targeted panel sequencing FASTQ	Features	SNV and Indel	Provides an integrated pipeline with a GUI for users without programming experience and predicts neoantigens from both membrane protein extracellular mutations and MHC-I presented peptides. Identifies potential tumor-specific neoantigens for use as targets in antibody-based or T cell-based cancer immunotherapy	[41]

Table 1 (continued)

Classification	Name	Input	Output	Neoantigen Source	Characteristics and utility	Ref
	CloudNeo	VCF file (for mutations) and BAM file (for HLA typing)	Features	SNV	Cloud-based pipeline providing scalable computation and eliminating the need for local infrastructure. Enables identification of patient-specific tumor neoantigens from large-scale sequencing datasets accessible on the cloud	[42]
	TIminer	RNA-seq reads and Somatic DNA mutations	Features	SNV and Indel	Integrates somatic mutation data with HLA types. Enables comprehensive characterization of the tumor-immune interface for cancer immunology research	[43]
	MixMHCpred2.2 and PRIME2.0	Naturally presented HLA-I ligands, Immunogenic neo-epitopes, and non-immunogenic peptides	Features & ranks	SNV	MixMHCpred2.2 integrates peptide length distributions and binding motifs; PRIME2.0 uses neural networks to correlate TCR-facing residues with HLA-I affinity. Identifies potent $CD8^+$ T-cell epitopes for vaccine design and immunotherapy	[44]
MS-based Immunopeptidomics	EDGE model	HLA peptide mass spectrometry datasets, HLA types, Transcriptome RNA-seq data, and genomic data	Features & ranks	SNV and Indel	Uses deep learning on tumor HLA peptidomics data to model HLA antigen presentation, integrating peptide sequence, flanking regions, gene expression, and per-gene presentation propensity. Significantly improves positive predictive value for identifying immunogenic neoantigens and enables their detection using routine clinical samples and small peptide sets for immunotherapy development	[45]
	SHERPA	HLA binding pocket, Peptide ligand sequence, Peptide length, Flanking regions, Source protein transcript abundance, Gene propensity score, and Hot spot score	Features & ranks	SNV and Indel	Integrates large-scale monoallelic/multiallelic immunopeptidomics, binding affinity data, and empirically derived antigen-processing features. Enhances precision for clinical applications like personalized cancer vaccines and immunotherapy biomarkers, outperforming existing tools by up to 1.44-fold	[46]

Table 1 (continued)

Classification	Name	Input	Output	Neoantigen Source	Characteristics and utility	Ref
Bioinformatics and Computational Models	RPEMHC	Amino acid sequences of MHC molecules and peptides	Features	SNV and Indel	Uses residue-residue pair encoding to capture critical interaction patterns between MHC and peptides. Enhances cancer vaccine development by accurately predicting immunogenic neoantigens for personalized immunotherapy	[47]
	UniPMT	Peptide sequences (P), MHC class I pseudosequences (M), and TCR β-chain CDR3 sequences (T)	Features	Experimentally validated neoantigens	Integrates P-M-T, P-M, and P–T binding information within a unified multitask learning framework. Identifies neoantigen-specific TCR for applications like T cell therapy and correlates with T cell clonal expansion	[48]
	TEIM-Res	Sequences of TCR CDR3β and epitopes	Features	Linear peptide sequences (epitopes) presented by MHC-I	Uses few-shot learning to overcome structural data scarcity by pre-training on sequence-level binding data. Enables residue-level interaction analysis for optimizing TCR-based immunotherapy and antigen discovery	[49]
	Proteomic and immunopeptidomic profiles	MS spectra of HLA-I and HLA-II peptides and personalized proteome reference	Features	SNV, Viral antigens, and TSA	Directly identifies naturally presented HLA-bound peptides from tumors. Validates in silico predictions and pinpoints non-canonical neoantigens for therapeutic targeting	[22]

BAM Binary Alignment/Map format, *CDR3* complementarity determining region 3, *FASTQ* Fast Quality format, *GUI* graphical user interface, *HLA* human leukocyte antigen, *Indel* insertions/deletions, *MHC* major histocompatibility complex, *MS* mass spectrometry, *NGS* next-generation sequencing, *pVAC-Seq* personalized Variant Antigens by Cancer Sequencing, *RNA-seq* RNA sequencing, *SHERPA* Systematic HLA Epitope Ranking Pan Algorithm, *SNV* single-nucleotide variant, *TCR* T cell receptor, *TEIM-Res* TCR–Epitope Interaction Modelling at Residue Level, *TIminer* Tumor Immunology miner, *TSA* tumor-specific antigen, *TSNAD* tumor-specific neoantigen detector, *VCF* variant call format, *WES* whole exome sequencing

considered the gold standard [16, 61]. WES provides a comprehensive landscape of the coding mutations within the tumor genome [62]. However, due to its high cost and lengthy turnaround time, targeted panel sequencing is often used as a more practical alternative in clinical settings [61]. These panels sequence specific genes or regions of interest, offering a more cost-effective and expedited approach compared to WES [63, 64]. Clinically, TMB has emerged as a predictive biomarker for response to ICI. Therapies such as pembrolizumab have received FDA approval for use in patients with TMB-high tumors, regardless of tumor type, signifying a pivotal step towards personalized medicine and the utilization of biomarkers to direct treatment decisions [20, 65]. Nevertheless, the implementation of TMB measurement in routine clinical practice faces challenges, including the standardization of methods, the determination of optimal cut-off, and the need for rigorous analytical and clinical validation [16]. Current research aims to refine TMB measurement and combine it with other biomarkers to enhance the precision of immunotherapy. The combination is expected to further improve patient selection for immunotherapies, potentially leading to more effective and personalized treatment strategies.

Advances of neoantigen in cancer immunotherapy

Neoantigens, arising from tumor-specific somatic mutations, represent promising targets in cancer immunotherapy due to their potential for personalized tumor targeting [66]. Advances in genomic and computational technologies have deepened insights into how neoantigens enhance antitumor immunity. Mechanistically, neoantigens enhance T cell activation and cytotoxicity while potentiating ICI efficacy [67]. The concept of tumor neoantigen burden (TNB) has further emerged as a biomarker, correlating strongly with improved immunotherapy outcomes [68]. Clinically, neoantigen-based therapies, such as personalized vaccines and adoptive T cell therapies, have demonstrated efficacy in early-phase trials [30, 69, 70]. These advances highlight the transformative potential of neoantigens in precision oncology, leveraging the immune system's specificity to eradicate malignant cells. In this section, we elucidate the mechanisms by which neoantigens amplify T cell responses, synergize with immune checkpoint blockade, and guide the development of combination therapies to circumvent treatment resistance.

Mechanisms underlying neoantigen-based immunotherapy

The efficacy of cancer immunotherapy hinges on the immune system's ability to discriminate tumor cells from healthy tissues, primarily mediated by tumor-specific antigens recognition [71]. Unlike shared tumor-associated antigens, neoantigens are exclusively expressed by malignant cells, minimizing off-target toxicity and serving as ideal targets for T cell-based antitumor immunity [72]. The mechanisms underlying neoantigen-based immunotherapy are explained as follows (Fig. 3). Recognition of neoantigen-MHC complexes by T cells triggers immune cascades that drive therapeutic responses [25, 73]. $CD8^+$ cytotoxic T lymphocytes (CTL) are pivotal in this process, inducing tumor cell apoptosis via perforin and granzymes release upon binding neoantigen peptides presented on MHC class I molecules [74]. $CD4^+$ helper T cells augment this response by secreting cytokines which enhance CTL activation and prolong effector functions within the TME [75, 76]. Neoantigen immunogenicity is determined by their capacity to bind MHC molecules and engage T cell receptors (TCR) [77, 78]. High-affinity neoantigen-MHC binding is essential for robust T cell activation, with clinical studies demonstrating that tumors harboring neoantigens with strong MHC affinity exhibit improved responses to ICI [79, 80]. Additionally, neoantigens structurally analogous to pathogen-derived epitopes may leverage pre-existing T cell memory, facilitating cross-reactive immune responses that bolster tumor elimination [78, 81].

Neoantigen-based therapies critically depend on functional T cell responses, yet are frequently compromised by T cell exhaustion—a dysfunctional state driven by persistent antigen exposure and immunosuppressive microenvironmental signals [82, 83]. This exhaustion manifests as progressive functional impairment, initiated by sustained overexpression of co-inhibitory receptors (e.g., PD-1, CTLA-4), which deliver intracellular inhibitory signals to paralyze T cell activation [84]. These defects are reinforced through profound epigenetic reprogramming, where stable chromatin modifications silence key transcription factors, locking in exhaustion-associated transcriptional programs [85]. Concurrently, metabolic dysregulation exacerbates dysfunction via mitochondrial impairment, compromised nutrient utilization, and accumulation of inhibitory metabolites, collectively impairing bioenergetic capacity [85]. Ultimately, these mechanisms induce hierarchical effector failure: beginning with loss of cytokine secretion, progressing to diminished proliferative potential, and eventually leading to irreversible decrease of cytotoxic granule exocytosis and target cell elimination. Notably in acute myeloid leukemia (AML), the bone marrow microenvironment actively drives T cell exhaustion through abundant inhibitory cytokines, immunosuppressive cells and metabolites [82, 86]. Solid tumors similarly foster exhaustion via hypoxia, nutrient depletion, and acidic conditions [87].

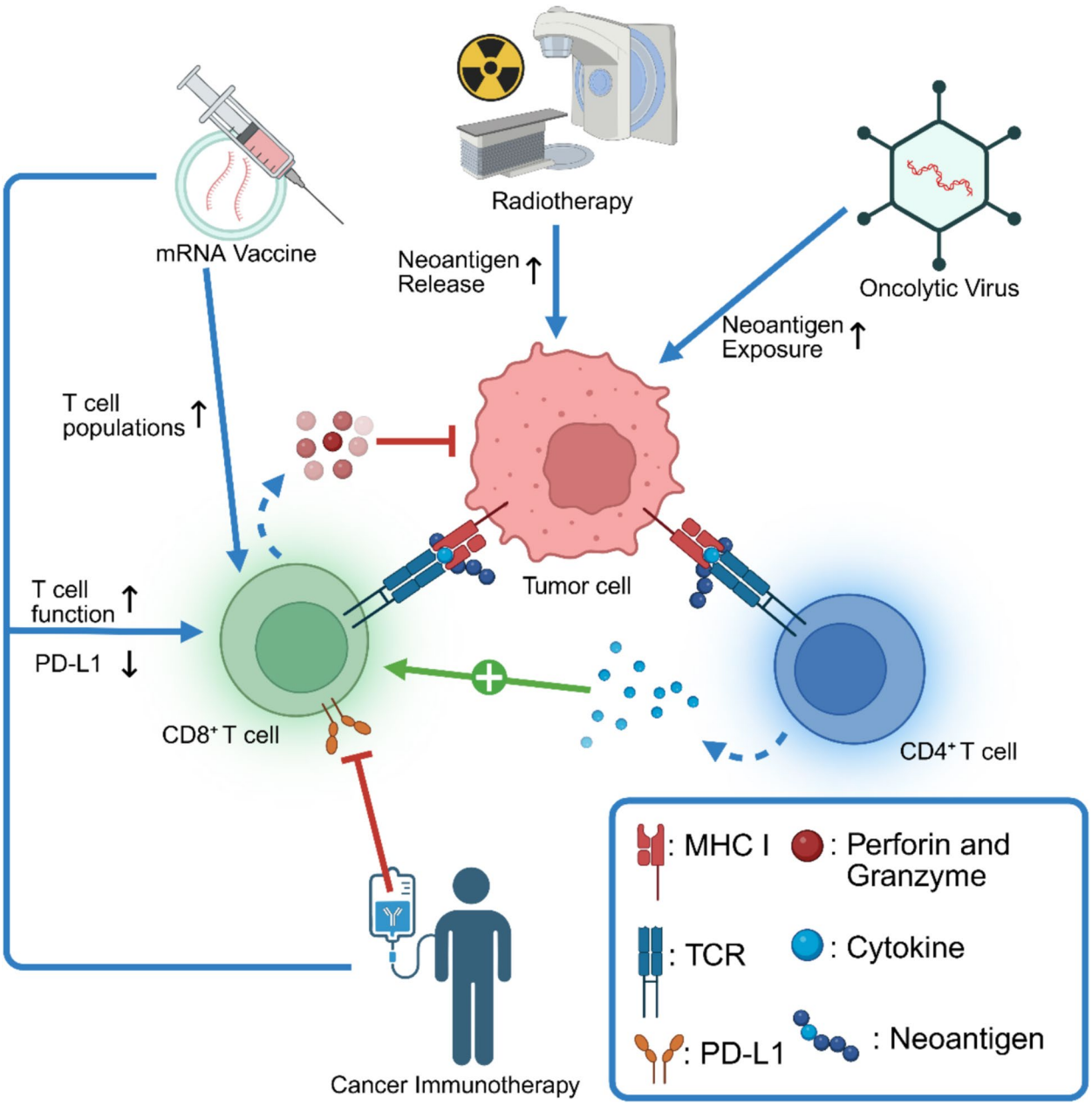

Fig. 3 Mechanisms underlying neoantigen-based immunotherapy. Recognition of neoantigen-MHC complexes by T cells triggers immune cascades that drive therapeutic responses. CTL are pivotal in this process, inducing tumor cell apoptosis via perforin and granzymes release. $CD4^+$ helper T cells augment this response by secreting cytokines which enhance CTL activation and prolong effector functions within the TME. Combination therapies further exploit neoantigens to amplify T cell activity. Radiotherapy releases neoantigens by inducing immunogenic cell death. OV similarly enhance neoantigen exposure by lysing tumor cells and activating innate immune pathways. Neoantigen-based mRNA vaccines directly expand tumor-reactive T cell populations. When combined with ICI, these vaccines counteract immunosuppressive mechanisms such as PD-L1 upregulation and sustain T cell effector functions. MHC I: major histocompatibility complex I, PD-L1: programmed cell death ligand-1, TCR: T cell receptor. Created with BioRender.com

Immune checkpoint blockade amplifies neoantigen-based T cell responses by blocking inhibitory signals [8]. Anti-PD-1/PD-L1 antibodies reinvigorate exhausted T cells within the TME, thereby restoring their cytotoxic function and proliferative capacity [88]. CTLA-4 inhibitors similarly enhance T cell priming in lymphoid organs by disrupting co-inhibitory signals, broadening the diversity and magnitude of neoantigen-specific T

cell clones [88]. Clinical evidence indicates that tumors with high TMB, a predictor of an expanded neoantigen repertoire, respond more favorably to ICI, highlighting the synergistic relationship between neoantigen abundance and checkpoint modulation [32, 65, 89]. Beyond ICI, complementary approaches to reverse exhaustion include cytokine-based interventions, such as interlukin (IL)−7, IL-12, or IL-15 administration, which promote T cell survival, proliferation, and functional recovery, with IL-15 showing particular promise in AML for its effects on NK and T cells [90–92]. Epigenetic modifiers, like histone deacetylase (HDAC) inhibitors or DNA methyl transferase (DNMT) inhibitors, aim to reset the exhaustion-associated epigenetic landscape and restore T cell functionality in AML [93–95]. Additionally, targeting alternative inhibitory pathways, such as transforming growth factor-β (TGF-β) signaling or adenosine receptors, represents an active area of investigation applicable to both solid tumors and hematologic malignancies [96, 97]. Combination therapies further exploit neoantigens to amplify T cell activity. Radiotherapy induces immunogenic cell death, releasing neoantigens that act as in situ vaccines to prime naïve T cells and recruit effector T cells to distant tumor sites (abscopal effect) [98, 99]. OV similarly enhance neoantigen exposure by lysing tumor cells and activating innate immune pathways, thereby promoting a pro-inflammatory TME that facilitates T cell infiltration [100, 101]. Neoantigen-based vaccines, which deliver personalized mutant peptides or RNA-encoded epitopes, directly expand tumor-reactive T cell populations [101]. When combined with ICI, these vaccines counteract immunosuppressive mechanisms such as PD-L1 upregulation and sustain T cell effector functions [102, 103]. In summary, neoantigens serve as molecular beacons that direct T cell-mediated tumor destruction. Their integration of neoantigen-targeted strategies with checkpoint blockade and adjunct therapies represents a paradigm shift in oncology, leveraging the precision of adaptive immunity to achieve durable antitumor responses.

Tumors also evade immune recognition through two primary mechanisms: downregulation of human leukocyte antigen (HLA) molecules and disruption of antigen processing [104, 105]. In HLA downregulation, tumor cells employ genetic deletions, transcriptional repression, or epigenetic silencing to reduce surface HLA expression, which directly impairs $CD8^+$ T cell recognition of tumor neoantigens [106, 107]. Concurrently, tumors disrupt antigen processing by compromising multiple steps: impaired immunoproteasome function limits antigenic peptide generation; defects in peptide transporters hinder peptide translocation into the endoplasmic reticulum; and molecular chaperone deficiencies destabilize HLA-peptide complexes [106, 108, 109]. Together, these defects create an "immunological invisibility" state by preventing functional antigen-HLA complexes from reaching the cell surface. Therapeutically, restoring antigen presentation leverages complementary strategies targeting distinct layers of tumor-induced suppression. Interferon-γ (IFN-γ) priming acts as a master transcriptional activator: by engaging Janus kinase-signal transducer and activator of transcription (JAK-STAT) signaling, it directly induces the expression of HLA class I molecules, antigen-processing machinery components, and peptide-loading chaperones [110–113]. These mechanisms re-establish the functional capacity for antigen processing and surface presentation. Epigenetic modulators operate at a foundational level, reversing tumor-driven epigenetic silencing of HLA and antigen-processing genes [95]. By demethylating promoters or enhancing histone acetylation, they restore basal transcriptional accessibility, enabling cells to respond to immunomodulatory signals like IFN-γ. Critically, these approaches act in concert—epigenetic reprogramming primes gene responsiveness, while IFN-γ drives robust expression—creating a synergistic restoration of antigen presentation that prevents tumor immune evasion.

Increasingly, TNB has emerged as a biomarker demonstrating potential to optimize precision oncology and therapeutic efficacy in cancer [68]. TNB quantifies immunogenic neoantigens arising from tumor-specific somatic mutations, including nonsynonymous single-nucleotide variants (SNV), insertions/deletions (indel), and gene fusions [27, 114]. These neoantigens are processed and presented by MHC molecules to trigger T cell-mediated antitumor immunity. Although TNB correlates with TMB, this correlation varies significantly across cancer types and even within individual tumors. TMB serves as a crude proxy for the potential neoantigen load, reflecting the raw number of mutations, while TNB aims to capture the actual immunogenic burden by predicting which mutations are likely to generate antigens presented by the patient's specific HLA alleles and capable of eliciting an immune response. Consequently, only a subset of mutations generates functional neoantigens, as their immunogenicity hinges on MHC binding affinity, TCR recognition, and the immunosuppressive TME [77, 78, 115]. Consequently, TNB is superior to TMB in predicting immunotherapy efficacy. For instance, in non-small cell lung cancer (NSCLC) and melanoma, patients exhibiting high TNB demonstrate enhanced responses to PD-1/CTLA-4 inhibitors and prolonged progression-free survival (PFS) [67, 116]. Tumors with DNA mismatch repair deficiency (dMMR) or high microsatellite instability (MSI-H) accumulate elevated TNB due to frameshift-derived neoantigens, which foster pro-inflammatory

microenvironments accompanied by robust T cell infiltration [117].

TNB has demonstrated several clinical applications to date. For instance, TNB serves as a predictive biomarker for immunotherapy response by identifying patients most likely to benefit from ICI [79]. In metastatic melanoma, high TNB is associated with enhanced cytolytic activity and durable clinical responses [79]. Additionally, TNB-guided neoantigen selection facilitates the development of personalized RNA or peptide-based vaccines [118, 119]. When combined with ICI, these vaccines augment T cell clonality and overcome ICI resistance [118, 119]. TNB can also be integrated synergistically with conventional therapies. Radiotherapy and chemotherapy induce immunogenic cell death, releasing neoantigens that augment the efficacy of ICI [120, 121]. For example, neoadjuvant chemoradiotherapy in rectal cancer increases neoantigen diversity and stimulates antitumor immunity [122]. Furthermore, TNB can be longitudinally monitored through liquid biopsy. Liquid biopsy is a minimally invasive diagnostic approach that analyzes circulating biomarkers in bodily fluids (such as plasma, urine, or saliva) to detect cancer and other diseases [123]. Circulating tumor DNA (ctDNA) analysis enables tracking of TNB dynamics, permitting real-time assessment of treatment efficacy and early detection of immune escape mechanisms such as neoantigen loss or HLA defects [124]. Despite its potential, challenges remain in standardizing TNB quantification, addressing tumor heterogeneity, and identifying high-quality neoantigens. Future efforts must integrate multi-omics data to refine TNB-based precision immunotherapy strategies.

Advances of neoantigen-based immunotherapy in clinical trials

The exceptional specificity of neoantigens has established them as critical targets in the rapidly advancing field of cancer immunotherapy [25]. Building upon the foundational understanding of neoantigen biology and the synergistic interaction with TMB, recent clinical research has transitioned from theoretical exploration to practical application [16]. The clinical trials of neoantigen-based immunotherapy in recent 3 years are shown in Table 2. These advancements aim to leverage the immune system's capacity to recognize neoantigens as foreign, thereby triggering robust, tumor-specific responses. Clinical trials have increasingly concentrated on four primary therapeutic strategies: personalized neoantigen vaccines, adoptive cell therapies, monoclonal antibodies, and OV [125]. Each of these approaches harnesses distinct mechanisms to enhance neoantigen immunogenicity, overcome immunosuppressive barriers, and improve therapeutic precision. For example, personalized vaccines deliver patient-specific mutant peptides or RNA-encoded epitopes to prime and expand neoantigen-reactive T cell populations [126, 127]. Adoptive cell therapies, such as engineered TCR or chimeric antigen receptor (CAR)-T cells, directly infuse neoantigen-targeted lymphocytes to mediate tumor eradication [128]. Monoclonal antibodies, particularly when combined with ICI, enhance T cell activation by blocking co-inhibitory signals in the tumor microenvironment [129, 130]. OV further complement these strategies by inducing immunogenic cell death and releasing neoantigens to stimulate systemic immunity [131]. This section delves into the clinical progress of these neoantigen-based strategies, underscoring their transformative potential in precision oncology.

The clinical translation of personalized neoantigen vaccines has advanced rapidly due to breakthroughs in RNA technology and computational neoantigen prediction [132]. Building on the foundational understanding of neoantigen-T cell interactions, recent trials have demonstrated significant synergy between these vaccines and ICI [133]. A landmark phase 2b trial (KEYNOTE-942) showed that combining the personalized mRNA vaccine mRNA-4157 (V940) with pembrolizumab in resected stage III/IV melanoma reduced recurrence risk by 44% compared to pembrolizumab monotherapy, with 18-month recurrence-free survival rates of 78.6% versus 62.2% [134]. This trial underscored the synergy between neoantigen vaccines and ICI, where vaccines expand tumor-reactive T cell clones while ICI reverses T cell exhaustion [134, 135]. Notably, the vaccine induced $CD8^+$ T cell responses against multiple neoantigens, with clonal expansion correlating with prolonged survival [134, 135]. In pancreatic ductal adenocarcinoma (PDAC), a traditionally immunologically "cold" tumor, Rojas et al. reported that adjuvant mRNA neoantigen vaccines combined with PD-L1 blockade and chemotherapy elicited durable $CD8^+$ T cell responses in 50% of patients, with vaccine-expanded T cell clones detectable for up to 1.9 years post-treatment [35]. Remarkably, patients with vaccine-induced T cells showed delayed recurrence, challenging the assumption that low-TMB tumors are resistant to vaccine [35]. This study also highlighted the importance of neoantigen quality—defined by MHC binding affinity and dissimilarity to wild-type peptides—in determining immunogenicity, with only 24% of predicted neoantigens eliciting detectable T cell responses [35]. Advances in RNA vaccine technology have further improved efficacy. For instance, codon optimization, nucleoside modifications, and lipid nanoparticle (LNP) delivery systems enhance mRNA stability, translational efficiency, and lymph node-targeted antigen presentation [136–138]. Using unmodified RNA in cancer vaccines, as opposed to modified RNA in pathogen vaccines, retains

Table 2 Representative clinical trials of neoantigen-based cancer immunotherapy in recent 3 years (selected from ClinicalTrials.gov). Trials of lacking messages, unrelated to cancer or immunotherapy, status being "not yet recruiting, unknown, and withdrawn", and estimated enrollment < 20 were excluded

Neoantigen name	ClinicalTrials.gov ID	Study Title	Phase	Status	First Location	Country	Actual/Estimated Study Start Date	Estimated Enrollment	Conditions	Interventions	Intervention Model	Primary outcomes	Secondary outcomes
Neoantigen mRNA Vaccine	NCT07067385	Personalized mRNA Cancer Vaccine for Gastrointestinal Solid Tumor Treatment	Phase1	Recruiting	Ruijin Hospital Affiliated to Shanghai Jiaotong University School of Medicine	China	2025/7/1	40	Gastrointestinal Tumors	Biological: Neoantigen Personalized Cancer Vaccine	Single Group Assignment	AE, SAE, Immunogenicity of a personalized cancer vaccine (Up to 2.5 years)	Cohort 1: ORR, DOR, DCR, PFS, OS (Up to 2.5 years) Cohort 2: 18-month RFS rate, 18-month OS rate (Up to 2.5 years)
	NCT06995105	Personalized Tumor Neoantigen mRNA Therapy for Advanced Hepatocellular Carcinoma	Phase1 Phase2	Recruiting	The First Affiliated Hospital, Zhejiang University School of Medicine	China	2025/4/9	30	Advanced Hepatocellular Carcinoma	Procedure: TACE Drug: Sintilimab combined with Bevacizumab Biological: individualized anti-tumor new antigen iNeo-Vac-R01 injection	Single Group Assignment	AE (21 ± 3 days after the last dose)	ORR, DCR, Conversion rate, PFS, OS (Up to 2 years); PFS Rate (1-Y-PFS%, 2-Y-PFS%,3-Y-PFS%), OS Rate (1-Y-OS%,2-Y-OS%,3-Y-OS%) (Up to 3 years)
	NCT06980155	XP-005 Personalized Vaccine Alone or in Combination with Toripalimab for the Prevention of Relapse After Remission in Acute Myeloid Leukemia	Early Phase1	Recruiting	Shanghai Jiao Tong University School of Medicine	China	2025/5/30	60	AML	Biological: Personalized neoantigen tumor vaccine Biological: PD-1 inhibitor	Single Group Assignment	DLT & MTD, BED (Day 1 to Day 21); Percentage of Participants with AE (Up to 52 weeks)	MRD negativity rate, Degree of MRD decrease, Duration of MRD response, RFS, OS, Levels of tumor neoantigen-specific $CD4^+$ and $CD8^+$ T lymphocytes, Changes in lymphocyte subset levels, Serum cytokine test, Peripheral blood TCR sequences and abundance, Single-cell sequencing results (Up to 52 weeks)
	NCT06956716	Personalized Tumor Neoantigen mRNA Therapy for Advanced Intrahepatic Cholangiocarcinoma	Phase1 Phase2	Recruiting	The First Affiliated Hospital, Zhejiang University School of Medicine	China	2025/4/9	30	Advanced Intrahepatic Cholangiocarcinoma	Biological: individualized anti-tumor new antigen iNeo-Vac-R01 injection Drug: Gemcitabine + cisplatin (GP) Drug: Sintilimab injection	Single Group Assignment	Safety and tolerable dose (21 ± 3 days after the last dose)	ORR, DCR, PFS, OS (Up to 2 years); OS Rate (1-Y-OS%,2-Y-OS%,3-Y-OS%) (Up to 3 years)

Table 2 (continued)

Neoantigen name	ClinicalTrials.gov ID	Study Title	Phase	Status	First Location	Country	Actual/Estimated Study Start Date	Estimated Enrollment	Conditions	Interventions	Intervention Model	Primary outcomes	Secondary outcomes
	NCT06623422	A Study of Pembrolizumab (MK-3475) With or Without Intismeran Autogene (V940) in Participants with Non-small Cell Lung Cancer (V940-009/INTerpath-009)	Phase3	Recruiting	Banner MD Anderson Cancer Center	United States	2024/10/21	680	NSCLC	Biological: Pembrolizumab Drug: Cisplatin Drug: Carboplatin Drug: Pemetrexed Drug: Gemcitabine Drug: Paclitaxel Biological: Intismeran autogene Other: Placebo	Parallel Assignment	DFS (Up to ~ 97 months)	OS, DMFS, DFS2, LCSS, Number of participants with ≥ 1 AE, Number of participants discontinuing from study therapy due to AE (Up to ~ 129 months); Change from Baseline in EORTC QLQ-C30 Global Health Status/Quality of Life, Physical Functioning and Role Functioning (Baseline and up to ~ 129 months)
	NCT06577532	Study of KRAS Neoantigen mRNA Vaccine (ABO2102) in Patients with KRAS -Mutated Solid Tumors	Early Phase1	Recruiting	Ruijin Hospital Affiliated to Shanghai Jiaotong University School of Medicine	China	2024/9/24	56	Pancreatic Neoplasms Other Solid Tumors	Drug: ABO2102 Drug: Toripalimab	Sequential Assignment	Part I: DLT (21 days after the first dose of study treatment); TEAEs, TESAEs (from the first dose of study treatment to 30 days after the last dose of monotherapy or to 90 days after the last dose of the combination therapy) Part II: ORR (from the first dose of study treatment to up to 2 years)	\
	NCT06497010	An Exploratory Study of Individualized Neoantigen mRNA Cancer Vaccine InnoPCV in Advanced Solid Tumor Treatment	Early Phase1	Recruiting	The Affiliated Hospital of Guizhou Medical University	China	2024/8/29	40	Advanced Solid Tumor	Drug: Biological: PD-1 Drug: Biological: InnoPCV	Sequential Assignment	RP2D (Baseline through 90 days after last InnoPCV dose)	Number of Participants with AEs (Baseline through 90 days after last tislelizumab or sintilimab dose); Neoantigen-specific T cell Response rate (Baseline through 26 weeks after last InnoPCV dose); ORR, DoR, PFS (up to approximately 2 years); OS (From baseline to approximately 2 years)

Table 2 (continued)

Neoantigen name	ClinicalTrials.gov ID	Study Title	Phase	Status	First Location	Country	Actual/Estimated Study Start Date	Estimated Enrollment	Conditions	Interventions	Intervention Model	Primary outcomes	Secondary outcomes
	NCT06496373	Clinical Study of mRNA Vaccine Combined With PD-1 Inhibitor as Adjuvant Therapy for Postoperative Pancreatic Cancer	Early Phase1	Recruiting	Ruijin Hospital Affiliated to Shanghai Jiaotong University School of Medicine	China	2024/10/16	20	Pancreatic Cancer	Biological: Fixed neoantigen tumor vaccine Biological: personalized neoantigen tumor vaccine Drug: PD-1 inhibitor	Single Group Assignment	Drug related toxicity (18 months)	RFS, OS, Reaction of antigen-specific T cells in peripheral blood (Up to 18 months)
	NCT06295809	A Study of (Neo) Adjuvant Intismeran Autogene (V940) and Pembrolizumab in Cutaneous Squamous Cell Carcinoma (V940-007)	Phase2 Phase3	Active, not recruiting	USC/Norris Comprehensive Cancer Center	United States	2024/4/18	1012	Squamous Cell Carcinoma Skin Neoplasms	Biological: Pembrolizumab Biological: Intismeran autogene Procedure: Surgery	Parallel Assignment	EFS (Up to ~ 59 months)	ORR, FFS rate, pCR rate, mPR rate (Up to ~ 38 months); DFS, OS, Percentage of participants who experience an AE (Up to ~ 59 months); Percentage of participants who discontinue study intervention due to AEs (Up to ~ 19 months); Change from baseline in Global health status, Physical functioning score and Role functioning score of QLQ-C30 (Baseline and up to ~ 38 months)
	NCT05933577	A Clinical Study of Intismeran Autogene (V940) Plus Pembrolizumab in People with High-Risk Melanoma (V940-001)	Phase3	Active, not recruiting	Highlands Oncology Group	United States	2023/7/19	1089	Melanoma	Biological: Intismeran autogene Biological: Pembrolizumab Other: Placebo	Parallel Assignment	RFS (Up to approximately 74 months)	DMFS, OS (Up to approximately 85 months); Number of Participants Who Experience an AE (Up to approximately 68 weeks); Number of Participants Who Discontinue Study Treatment Due to an AE (Up to approximately 56 weeks); Change from Baseline in EORTC QLQ-C30 Global Health Status/Quality of Life, Physical Functioning and Role Functioning Combined Score (Baseline and up to approximately 85 months)
	NCT05916261	Personalized Tumor Vaccines and Pabolizumab in Patients with Advanced Pancreatic Cancer	Early Phase1	Recruiting	Ruijin-Hainan Hospital Affiliated to Shanghai Jiaotong University School of Medicine	China	2023/4/26	54	Advanced Pancreatic Cancer	Biological: Personalized neoantigen tumor vaccine	Single Group Assignment	MTD (If MTD is not reached, BED) or DLT, Reaction of antigen-specific T cells in peripheral blood (Up to 27 weeks); ORR, DCR (Up to 105 weeks)	Number of participants with TRAE, PFS (Up to 105 weeks); OS (Up to 3 years)

Table 2 (continued)

Neoantigen name	ClinicalTrials.gov ID	Study Title	Phase	Status	First Location	Country	Actual/Estimated Study Start Date	Estimated Enrollment	Conditions	Interventions	Intervention Model	Primary outcomes	Secondary outcomes
	NCT05916248	Combination Therapy of Personalized mRNA-0217 Vaccines and Pembrolizumab in Patients with Advanced Solid Tumors	Early Phase1	Recruiting	Ruijin Hospital Affiliated to Shanghai Jiaotong University School of Medicine	China	2023/12/20	34	Advanced Solid Tumor	Biological: Personalized neoantigen tumor vaccine Biological: PD-1 inhibitor	Single Group Assignment	MTD or DLT, BED (Day1 to Day21); Percentage of Participants with AEs (Up to 54 weeks)	PFS, OS, Reaction of antigen-specific T cells in peripheral blood, ORR, DCR (Up to 54 weeks)
	NCT05579275	Evaluate the Safety and Tolerability of JCXH-212 Monotherapy and Combined with Toripalimab in the Treatment of Malignant Solid Tumors	Early Phase1	Recruiting	Peking University Cancer Hospital & Institute	China	2023/2/6	24	Malignant Solid Tumors	Biological: JCXH-212 Injection	Single Group Assignment	DLT (Up to 21 days after first dose every subject); Analysis of primary efficacy indicators (At the end of the time 3–6 months after last subject last dose)	Peripheral blood neoantigen-specific T cell responses before and after administration (The day of administration of the last subject)
Neoantigen-based Peptide Vaccine	NCT05269381	Personalized Neoantigen Peptide-Based Vaccine in Combination with Pembrolizumab for Treatment of Advanced Solid Tumors	Phase1 Phase2	Recruiting	Mayo Clinic in Florida	United States	2022/3/31	36	Solid Tumor	Drug: Cyclophosphamide Biological: Neoantigen Peptide Vaccine Biological: Pembrolizumab Biological: Sargramostim	Single Group Assignment	Incidence of adverse events (Up to 2 years from first vaccine administration)	The number and percentage of participants who completed the sequencing with satisfactory data quality registration and identified at least 10 actionable peptides, meet the eligibility criteria for registration, and able to initiate vaccine production Immunogenicity responders (Up to 16 weeks); Immunogenicity responders (Within 24 weeks)
	NCT05141721	A Study of a Patient-Specific Neoantigen Vaccine in Combination with Immune Checkpoint Blockade for Patients with Metastatic Colorectal Cancer	Phase2 Phase3	Active, not recruiting	Banner MD Anderson Cancer Center	United States	2022/2/12	700	Colorectal Neoplasms	Drug: GRT-C901 Drug: GRT-R902 Drug: Atezolizumab Drug: Ipilimumab Drug: Fluoropyrimidine plus leucovorin Drug: Bevacizumab	Parallel Assignment	Phase 2: Molecular response defined as ≥ 30% decrease from baseline in ctDNA (Baseline and up to 27 months) Phase 3: PFS per iRECIST as assessed by blinded IRC (Up to 60 months)	Phase 2 and 3: Incidence of TEAEs, immune-related AEs, TRAEs, SAEs, AEs leading to death, AEs leading to dose delays, and AEs leading to discontinuation of study treatment, PFS and iRECIST, OS, ORR, DOR, CBR, Deepening of Response (Phase 2 up to 27 months, Phase 3 up to 60 months); The feasibility of manufacturing a patient-specific vaccine (Up to 28 days before Day 1 of study drug administration) Phase 3: PFS per RECIST v1.1 as assessed by blinded IRC (Up to 60 months)

Table 2 (continued)

Neoantigen name	ClinicalTrials.gov ID	Study Title	Phase	Status	First Location	Country	Actual/Estimated Study Start Date	Estimated Enrollment	Conditions	Interventions	Intervention Model	Primary outcomes	Secondary outcomes
	NCT05111353	Neoantigen Vaccines in Pancreatic Cancer in the Window Prior to Surgery	Phase1	Active, not recruiting	Washington University School of Medicine	United States	2022/12/15	33	Pancreatic Cancer	Biological: Optimized neoantigen synthetic long peptide vaccine Biological: Poly-ICLC	Parallel Assignment	Safety of neoantigen SLP vaccine as measured by number of subjects experiencing each type of AE (Through 4 weeks after completion of last vaccination)	Immunogenicity of neoantigen peptide vaccine as measured by the number and phenotype of neoantigen-specific T cells (only Arm 1 and Arm 2) (Through approximately 2 years and 78 days)
	NCT05098210	Personalized Neo-Antigen Peptide Vaccine for the Treatment of Stage IIIC-IV Melanoma, Hormone Receptor Positive HER2 Negative Metastatic Refractory Breast Cancer or Stage III-IV Non-Small Cell Lung Cancer	Phase1	Recruiting	Fred Hutch/University of Washington Cancer Consortium	United States	2022/6/9	25	Stage IIIC-IV Melanoma Hormone Receptor Positive HER2 Negative Metastatic Refractory Breast Cancer Stage III-IV NSCLC	Biological: Neoantigen Peptide Vaccine Biological: Nivolumab Drug: Poly ICLC Procedure: Echocardiography Procedure: Multigated Acquisition Scan Procedure: Biopsy Procedure: Biospecimen Collection Procedure: Computed Tomography Procedure: Positron Emission Tomography	Single Group Assignment	Incidence of AEs (1 year post first vaccination)	Number of formulated and administered personalized neoantigen vaccines, Number of formulated personalized neoantigen vaccines with at least five vaccine peptides (Week 48); Number of formulated personalized neoantigen vaccines in less than 16 weeks since screening visit biopsy (16 weeks); Evaluation of target lesion, best overall response, PFS (1 year post first vaccination)
	NCT05013216	Mutant KRAS -Targeted Long Peptide Vaccine for Patients at High Risk of Developing Pancreatic Cancer	Phase1	Recruiting	Sidney Kimmel Comprehensive Cancer Center	United States	2022/4/11	37	Pancreatic Cancer	Drug: Cohort A: Patients at high risk of developing pancreatic cancer Drug: Cohort B: Patients must have evidence of a pancreatic cystic neoplasm	Sequential Assignment	Number of participants experiencing study drug-related toxicities (1.5 years); Maximal percentage of change of IFN-γ producing mutant-KRAS-specific $CD8^+$ and $CD4^+$ T cells (17 weeks)	Fold change in interferon-producing mutant-KRAS-specific $CD8^+$ and $CD4^+$ T cells at 5, 13 and 17 weeks (Baseline, 5, 13 and 17 weeks)

Table 2 (continued)

Neoantigen name	ClinicalTrials.gov ID	Study Title	Phase	Status	First Location	Country	Actual/Estimated Study Start Date	Estimated Enrollment	Conditions	Interventions	Intervention Model	Primary outcomes	Secondary outcomes
	NCT04943848	rHSC-DIPGVax Plus Checkpoint Blockade for the Treatment of Newly Diagnosed DIPG and DMG	Phase1	Recruiting	Children's Health Orange County	United States	2022/1/10	36	Diffuse Glioma	Biological: rHSC-DIPGVax Drug: Balstilimab Drug: Zalifrelimab	Sequential Assignment	Safety and Tolerability: Dose limiting toxicities of rHSC-DIPGVax (DLT period of 28 days for rHSC-DIPGVax monotherapy); Safety and Tolerability: Dose limiting toxicities of rHSC-DIPGVax plus BALSTILIMAB (DLT period of 28 days for Part A); Safety and Tolerability: Dose limiting toxicities of rHSC-DIPGVax plus BALSTILIMAB and ZALIFRELIMAB (DLT period of 42 days for Part B)	Total number of DLT's for ZALIFRELIMAB at RP2D in combination with rHSC-DIPGVax and BALSTILIMAB (On-going during 1 year of therapy plus 3 month follow up); To evaluate the efficacy of the combination of rHSC-DIPGVax, BALSTILIMAB, and ZALIFRELIMB in pediatric subjects with DIPG and DMG as measured by overall survival at 12 months and time-to-progression as measured from time of diagnostic imaging (On-going during 1 year of therapy plus 3 month follow up); Overall survival at 1 year (On-going during 1 year of therapy); Time to progression (On-going during 1 year of therapy plus up to 5 years off treatment)
	NCT04930783	Neo-Vax + CDX-301 and Nivolumab or Pembrolizumab in Melanoma	Phase1	Recruiting	Dana Farber Cancer Institute	United States	2022/1/3	30	Melanoma	Drug: CDX-301 Biological: NEOVAX Drug: Nivolumab Drug: Pembrolizumab	Single Group Assignment	Rate DLT (Toxicities experienced within 49 days/7 weeks of Neoantigen Vaccine treatment initiation); Recommended MTD (Up to 12 weeks for each dosing cohort)	Neoantigen-specific cellular immune responses, estimate rates of disease progression/recurrence depending on whether patient had all melanoma resected or has measurable disease per RECIST 1.1 (Enrollment to end of treatment up to 24 weeks)
	NCT03361852	Personalized Neoantigen Cancer Vaccine + Pembrolizumab After Rituximab for Follicular Lymphoma	Phase1	Active, not recruiting	Dana Farber Cancer Institute	United States	2022/3/14	20	Follicular Lymphoma	Drug: Rituximab Biological: Neo Vax Drug: Pembrolizumab	Single Group Assignment	Feasibility of Neovax following 4 weekly doses of rituximab (2 years)	Proportion of participants who achieve an IFN-γ T cell response to one or more of the peptide pools, proportion of participants who convert from PR to CR, SD to PR/CR, Best Objective Response, to describe the safety and tolerability of NeoVax following 4 weekly doses of rituximab in patient with previously untreated follicular lymphoma (2 years)

Table 2 (continued)

Neoantigen name	ClinicalTrials.gov ID	Study Title	Phase	Status	First Location	Country	Actual/Estimated Study Start Date	Estimated Enrollment	Conditions	Interventions	Intervention Model	Primary outcomes	Secondary outcomes
Neoantigen-loaded DC vaccine	NCT06675201	Neoantigen Vaccines in Esophageal Squamous Cell Carcinoma	Phase2	Recruiting	West China Hospital of Sichuan University	China	2024/10/1	165	ESCC	Biological: neoantigen-loaded dendritic cell vaccine (NeoDC-Vac)	Parallel Assignment	2-year OS rate (24 months)	PFS, TRAEs (12 months)
	NCT06522919	Sequential Immuno-chemotherapy Treatment with Pembrolizumab Plus Dendritic Cell (DC) Vaccine Followed by Trifluridine/Tipiracil Plus Bevacizumab in Refractory Mismatch-repair-proficient (pMMR) or Microsatellite-stable (MSS) Metastatic Colorectal Cancer	Phase2	Recruiting	IRCCS Istituto Romagnolo per lo Studio dei Tumori "Dino Amadori" -IRST S.r.l	Italy	2025/1/7	36	CRC	Biological: Autologous Dendritic Cell (DC) Vaccine	Single Group Assignment	ORR (2 years)	PFS, Safety evaluation, OS, In vivo immunomonitoring through DTH test, Characterization of patient's HLA class I, HLA class II and HLA-restricted immune response, Definition of the prognostic and predictive role of peripheral immune cell subsets and soluble factors, In situ characterization of pMMR/MSS mCRC through Next Generation Sequencing analysis, In situ protein validation at the single cell level, Identification of tumor infiltrating immune cell subsets (2 years)
	NCT06329908	DC Combined with ICIs in the Treatment of Advanced Lung Cancer Resistant to ICIs	Early Phase1	Recruiting	West China Hospital of Sichuan University	China	2023/9/27	20	Lung Cancer	Drug: LG002	Single Group Assignment	The safety of Neo-DCVac combined with ICIs (2 years)	The efficacy of Neo-DCVac combined with ICIs (2 years)
	NCT05886439	LK101 Combined With PD-1 or PD-L1 Monoclonal Antibody in the Treatment of Lung Cancer	Phase1	Recruiting	Cancer hospital Chinese Academy of Medical Sciences	China	2023/5/11	40	Advanced Lung Carcinoma	Drug: LK101 injection (personlized neoantigen pulsed DC vaccine) Drug: Pembrolizumab Drug: Durvalumab	Parallel Assignment	DLT, AEs, SAEs, irAEs (Continuously throughout the study until 90 days after Termination of the treatment)	ORR (accessed up to 24 months from baseline); DoR, DCR, TTR, TTP, PFS, OS, Immune response evaluation (24 months)
Neoantigen DNA Vaccine	NCT06631079	Personalised Neoantigen-targeting Cancer Vaccine NECVAX-NEO1 in Anti-PD-1/PD-L1 Therapy in Patients With Solid Tumors	Phase1 Phase2	Recruiting	Comprehensive Cancer Center	Germany	2024/10/7	20	Solid Tumor	Biological: NECVAX-NEO1	Single Group Assignment	AEs, SAEs, Change from Baseline in Laboratory Parameters and Electrocardiograms (Up to week 24 plus 24 months)	Antitumor activity (Up to 24 weeks), ORR, PFS, TTP, OS (Up to week 24 plus 24 months)

Table 2 (continued)

Neoantigen name	ClinicalTrials.gov ID	Study Title	Phase	Status	First Location	Country	Actual/Estimated Study Start Date	Estimated Enrollment	Conditions	Interventions	Intervention Model	Primary outcomes	Secondary outcomes
Unknown Neoantigen Vaccine	NCT06344156	Neoantigen Vaccine Plus Anti-PD1 and Chemotherapy as an Adjuvant Therapy for Pancreatic Cancer	Phase1	Recruiting	West China Hospital of Sichuan University	China	2024/4/1	43	Pancreatic Cancer	Drug: Neoantigen Vaccine Plus Anti-PD1 and Chemotherapy	Single Group Assignment	18-month RFS (through study completion, an average of 2 year); Incidence of Treatment-Related Adverse Events [Safety and Tolerability] (3 months after the last administration of neoantigen vaccine)	18-month OS (through study completion, an average of 3 year)
Neoantigen specific T cell	NCT05576077	A Study of TBio-4101 (TIL) and Pembrolizumab in Patients With Advanced Solid Tumors	Phase1	Terminated	University of California Irvine	United States	2023/1/17	30	Breast Cancer CRC Uveal Melanoma Cutaneous Melanoma NSCLC HNSCC	Biological: TBio-4101 Drug: Pembrolizumab	Parallel Assignment	Safety and tolerability (25 months)	ORR, DCR, DoR (25 months)
	NCT05194735	Phase I/II Study of Autologous T Cells to Express T-Cell Receptors (TCRs) in Subjects with Solid Tumors	Phase1 Phase2	Active, not recruiting	Banner MD Anderson Cancer Center	United States	2022/4/4	180	Solid Tumor	Biological: Neoantigen specific TCR-T cell drug product Biological: Aldesleukin (IL-2)	Sequential Assignment	Phase I: DLT, MTD or RP2D of T-Cell Receptor T cells (Approximately one month) Phase II: ORR, Incidence of AEs (Up to 2 years)	Phase I: Feasibility of neoantigen-specific TCR-T cells manufacturing, Translational hypotheses related to TCR-T cell persistence without IL-2 (Arm A) or with IL-2 (Arm B) (Approximately one month) Phase II: To confirm Phase I results of translational hypotheses related to TCR-T cell persistence without IL-2 (Arm A) or with IL-2 (Arm B) (Approximately one month)

AE adverse event, *AML* acute myeloid leukemia, *BED* biologically effective dose, *CBR* clinical benefit rate, *CR* complete response, *CRC* colorectal cancer, *ctDNA* circulating tumor DNA, *DC* dendritic cell, *DCR* disease control rate, *DFS* disease-free survival, *DFS2* disease-free survival 2, *DIPG* diffuse intrinsic pontine glioma, *DLT* dose-limiting toxicity, *DMFS* distant metastasis-free survival, *DMG* diffuse midline glioma, *DoR* duration of response, *DTH* delayed type hypersensitivity, *EFS* event free survival, *ESCC* esophageal squamous cell carcinoma, *EORTC* the European Organization for Research and Treatment of Cancer, *FFS* freedom from surgery, *HER2* human epidermal growth factor receptor 2, *HLA* human leukocyte antigen, *HNSCC* head and neck squamous cell carcinoma, *ICI* immune checkpoint inhibitor, *IFN-γ* interferon-γ, *IL-2* interlukin-2, *irAE* immune-related adverse event, *IRC* independent review committee, *iRECIST* immune-based Response Evaluation Criteria in Solid Tumors, *KRAS* Kirsten rat sarcoma viral oncogene homolog, *LCSS* lung cancer specific survival, *mCRC* metastasis colorectal cancer, *mPR* major pathological response, *MRD* minimal residual disease, *MSS* microsatellite-stable, *MTD* maximum tolerated dose, *NSCLC* non-small cell lung cancer, *ORR* overall response rate, *OS* overall survival, *pCR* pathological complete response, *PD-1* programmed cell death-1, *PD-L1* programmed cell death ligand-1, *PFS* progression free survival, *pMMR* mismatch-repair-proficient, *PR* partial response, *QLQ-C30* Quality of Life Questionnaire-Core 30, *RECIST* the Response Evaluation Criteria in Solid Tumors, *RFS* recurrence-free survival, *RP2D* recommended phase 2 dose, *SAE* severe adverse event, *SD* stable disease, *SLP* synthetic long peptide, *TACE* transhepatic arterial chemotherapy and embolization, *TCR* T cell receptor, *TEAE* treatment-emergent adverse event, *TESAE* treatment-emergent serious adverse event, *TIL* tumor infiltrating lymphocyte, *TRAE* treatment-related adverse event, *TTP* time to progression, *TTR* time to remission

intrinsic adjuvanticity by activating Toll-like receptors (TLR) and retinoic acid-inducible gene I (RIG-I) pathways, which promote robust type-I interferon responses critical for dendritic cell maturation and cross-priming [138]. Additionally, incorporating MHC-II neoantigens into vaccines has shown promise in preclinical models by broadening $CD4^+$ T cell help and sustaining $CD8^+$ T cell memory [139]. Parallel advances in DNA vaccine platforms, exemplified by liposome-encapsulated multiepitope neoantigen constructs, have demonstrated potent tumor regression and reduced lung metastasis in preclinical melanoma models by enhancing intratumoral $CD8^+$ T cell infiltration and cytotoxicity [140]. In personalized T cell therapies, neoantigen-expanded autologous T cells induced polyclonal TCR repertoires and tumor regression in metastatic ovarian cancer, with durable TCR clonotypes persisting in circulation for over 15 months [141]. For DC-based vaccines, neoantigen-pulsed dendritic cells combined with immune adjuvants (e.g., TLR agonists) or checkpoint inhibitors elicited antigen-specific T cell responses in pancreatic cancer clinical trials, showing safety and potential synergy with chemotherapy [142]. In summary, personalized neoantigen vaccines represent a paradigm shift in precision oncology, with early clinical successes in both "hot" and "cold" tumors [138]. Future trials focused on minimally diseased hosts, combination therapies, and iterative vaccination strategies hold promise for overcoming resistance and achieving durable antitumor immunity.

Adoptive cell therapy, particularly CAR-T cell therapy, has emerged as a groundbreaking strategy for targeting neoantigens in solid tumors [143]. Unlike hematologic malignancies, solid tumors present distinct challenges, including antigen heterogeneity, immunosuppressive microenvironments, and the risk of off-target toxicities [144, 145]. Recent advancements in CAR-T engineering and neoantigen selection have begun to address these hurdles, demonstrating promising clinical outcomes [143, 146]. A key focus has been the targeting of clonal neoantigens derived from driver mutations, such as epidermal growth factor receptor variant III (EGFRvIII) in glioblastoma (GBM) [147]. This constitutively active variant of the epidermal growth factor receptor (EGFR), characterized by an extracellular domain deletion, functions as a tumor-specific antigen [148]. Early-phase trials of EGFRvIII-directed CAR-T cells demonstrated antigen reduction in post-treatment resections [149]. However, limited clinical efficacy highlighted issues such as adaptive immune resistance and regulatory T cell infiltration [143]. To enhance specificity, combinatorial antigen-sensing systems, such as synthetic Notch (SynNotch) receptors, have been engineered [150]. These logic-gated CAR require dual antigen recognition to activate cytotoxicity, reducing off-target effects while improving tumor discrimination [150]. Preclinical models demonstrated that SynNotch circuits enhance CAR-T cell persistence and mitigate tonic signaling, offering a blueprint for clinical translation [151]. Another breakthrough involves targeting public neoantigens, such as tumor-associated mucin 1 (Tan-MUC1), which exhibits aberrant glycosylation in cancers [152]. Early trials of Tan-MUC1-directed CAR-T cells reported stable disease in patients with solid tumors without severe toxicity, underscoring its potential as a pan-cancer target [143]. Additionally, disialoganglioside (GD2), an oncofetal antigen re-expressed in neuroblastoma and diffuse midline glioma, has demonstrated remarkable efficacy [153]. In a phase I trial, third-generation GD2 CAR-T cells achieved a 63% overall response rate in neuroblastoma, with complete responses in 33% of patients [154]. As synthetic biology and multi-omics converge, CAR-T therapies targeting neoantigens are poised to redefine precision immunotherapy for solid tumors.

Monoclonal antibodies, particularly ICI, have shown significant potential in neoantigen-based immunotherapy by enhancing T cell responses against tumor-specific mutations [155]. Recent clinical trials have advanced the understanding and application of neoantigen-based immunotherapy in combination with ICI. For instance, a phase 1 trial of a shared neoantigen vaccine combined with immune checkpoint blockade in patients with advanced metastatic solid tumors demonstrated promising antitumor activity, with some patients achieving objective responses and a manageable safety profile [156]. In advanced hepatocellular carcinoma, a phase 1/2 trial of personalized neoantigen vaccine combined with pembrolizumab induced robust T cell responses and showed encouraging antitumor activity, with some patients experiencing partial responses [157]. Additionally, a phase 1b trial investigated the use of neoadjuvant nivolumab or nivolumab plus lymphocyte activation gene-3 (LAG-3) inhibitor relatlimab in resectable esophageal/gastroesophageal junction cancer [158]. This trial revealed promising pathological responses, with 2-year recurrence-free survival (RFS) and overall survival (OS) rates observed in 72.5% of patients receiving nivolumab monotherapy and 82.6% in the combination arm, as well as a high R0 resection rate of 100% [158]. Lastly, autogene cevumeran is a uridine messenger RNA lipoplex-based individualized neoantigen-specific immunotherapy designed from tumor-specific somatic mutation data [159]. A phase 1 trial of autogene cevumeran, with or without atezolizumab, in advanced solid tumors demonstrated the feasibility and immunogenicity of this approach, with some patients experiencing durable clinical benefits, including a patient with NSCLC who had a durable response (>1 year on treatment) [159].

Collectively, these studies highlight the potential of combining neoantigen-based therapies with ICI to enhance antitumor immunity and improve patient outcomes across various cancer types.

Recent clinical trials have also demonstrated significant progress in OV-based therapies, particularly through strategic engineering and combination approaches [131]. T-VEC (talimogene laherparepvec), a herpes simplex virus (HSV)−1-derived OV expressing granulocyte–macrophage colony-stimulating factor (GM-CSF), remains a cornerstone, delivering durable responses in advanced melanoma [160]. In 2021, Japan approved DELYTACT (G47Δ-modified HSV-1) for malignant glioma, marking the first OV approval for brain tumors [161]. A phase 1/2 trials with rQNestin34.5v.2, another oncolytic herpes simplex virus (oHSV), revealed prolonged survival in glioma patients, with a median OS of 12.2 months compared to a historical median of 5.6 months. Combination therapies have further amplified efficacy [131]. A Phase 1b trial combining T-VEC with anti-PD-1 antibody pembrolizumab achieved overall and complete response rates of 62% and 33%, respectively, in metastatic melanoma, highlighting synergistic immune activation [162]. Similarly, JX-594 (a vaccinia virus expressing GM-CSF) demonstrated survival benefits in colorectal cancer and hepatocellular carcinoma via intravenous delivery [163, 164]. Adenoviral vectors, such as DNX-2401, showed tumor reduction in pediatric diffuse intrinsic pontine glioma when paired with radiotherapy, with 9 out of 12 patients exhibiting immune activation [165]. Recently, emerging strategies have focused on arming OV with immunomodulators. For instance, oHSV-IL-12 enhanced TME infiltration of effector T cells and natural killer (NK) cells in preclinical models, while IL-15-armed OV improved CAR-NK cell persistence in glioblastoma [166, 167]. These advancements underscore the potential of OV to transform "cold" tumors into immunogenic hotspots, paving the way for next-generation combinatorial regimens with ICI, CAR-T/NK cells, and neoantigen-based therapies [168].

The applications of TMB in cancer immunotherapy

Mechanisms of TMB in Predicting Immunotherapy Response

The predictive value of TMB in immunotherapy depends on its mechanistic association with neoantigen generation and subsequent immune activation [28]. As a surrogate marker for tumor immunogenicity, TMB reflects the probability of generating neoantigens capable of eliciting T cell-mediated antitumor responses [21]. This relationship can be delineated into four interrelated processes: neoantigen generation, antigen presentation, immune recognition, and host immune competency (Fig. 4) [28, 30].

TMB quantifies somatic non-synonymous mutations within a tumor genome [16]. Elevated TMB increases the probability of immunogenic neoantigen generation, as coding-region mutations may produce altered peptides perceived as"non-self"by the immune system [30]. These neoantigens arise from diverse mutational processes, such as SNV, indel, and gene fusions [30]. For instance, tumors with dMMR or polymerase epsilon/delta (POLE/POLD1) mutations exhibit hypermutated genomes enriched in frameshift-derived neoantigens——structurally distinct from self-peptides and inherently immunogenic [169, 170]. Mechanistically, neoantigen abundance correlates with TMB, expanding the antigenic repertoire available for immune recognition [21]. However, only mutations exhibiting high binding affinity to MHC molecules and sufficient dissimilarity from self-antigen confer functional immunogenicity [77, 171].

Neoantigen presentation via MHC class I and II molecules is critical for T cell activation [172]. Elevated TMB increases the probability of neoantigen-MHC binding with sufficient affinity for surface presentation [173, 174]. Defects in antigen presentation machinery, such as loss of β2-microglobulin (B2M) or MHC downregulation, can disrupt this process even in TMB-high tumors [28, 175]. Additionally, tumor heterogeneity, marked by spatial and temporal variations in neoantigen expression, may limit presentation consistency [176, 177]. Clonal evolution under therapeutic pressure further modulates immunogenicity through dynamic neoantigen presentation [178]. In TMB-high tumors, neoantigen diversity also increases the probability of TCR recognition [179]. Neoantigen-specific T cells infiltrate the TME and initiate cytotoxic responses. Pre-existing neoantigen-specific T cell clones in peripheral blood or tumor-infiltrating lymphocytes (TIL) correlate with improved ICI efficacy [180–182]. Conversely, tumors with low neoantigen clonality often evade immune detection due to suboptimal T cell activation [183]. Structural similarity between neoantigens and pathogen-derived epitopes may also engage cross-reactive memory T cells, amplifying antitumor immunity [184]. For instance, viral peptide-like neoantigens can recruit pre-existing memory T cell populations, accelerating immune responses [185, 186].

Host immune competency, governed by systemic and local factors, determines the response efficacy of immunotherapy [187]. Elevated TMB alone is insufficient in immunosuppressive TME characterized by PD-L1 expression, regulatory T cell (Treg) infiltration, myeloid-derived suppressor cells (MDSC) or inhibitory cytokine [188–191]. ICI like anti-PD-1/PD-L1 agents restore exhausted T cell cytotoxicity by blocking inhibitory

signals [192]. However, TMB-high tumors with low $CD8^+$ T cell infiltration ("immune-excluded"phenotypes) frequently resist ICI, highlighting the necessity of a permissive TME [193]. Germline polymorphisms in immune-related genes, including HLA alleles, further influence outcomes [194]. Broad peptide-binding HLA class I supertypes enhance neoantigen presentation, whereas HLA loss of heterozygosity (LOH) compromises immunity [194, 195].

Immunotherapy efficacy hinges on the interplay between TMB and these four pillars. TMB-high tumors with intact antigen presentation, robust T cell infiltration, and favorable immune contexts are more responsive to ICI. Discordances arise when TMB overlooks neoantigen quality or immune evasion mechanisms. For instance, hypermutated tumors with dMMR may resist therapy due to impaired antigen presentation or dominant immunosuppressive [196, 197]. Thus, TMB requires integration with complementary biomarkers—such as PD-L1 expression, immune gene signatures, and HLA status—to enhance predictive accuracy and predicts immunotherapy response by approximating neoantigen-driven immunogenicity, contingent on host immune competence. Future research should prioritize multi-omics integration to elucidate tumor-immune dynamics and optimize patient stratification.

Detection of TMB in cancer immunotherapy

Recent advancements in TMB measurement have prioritized enhancing accuracy, scalability, and clinical applicability, especially in immunotherapy contexts. Although WES remains the gold standard for comprehensive TMB assessment, emerging technologies overcome its limitations, such as high costs and lengthy processing times, while improving compatibility with low-input or low-purity clinical samples [16, 61]. The primary detection methods of TMB are shown in Fig. 5.

Liquid biopsy has emerged as a non-invasive alternative to tissue-based TMB analysis, particularly for inaccessible tumors or those with spatially heterogeneous mutational profiles [198]. Blood TMB (bTMB) measures somatic mutations in ctDNA, capturing the cumulative mutational load across primary and metastatic sites [199]. Recent studies have validated assays such as GuardantOMNI and PredicineATLAS for bTMB estimation, showing correlation with tissue TMB in different cancers [200, 201]. For example, synthetic reference standards generated by spiking tumor cell line DNA into donor-matched lymphoblastoid DNA at low-tumor-fraction thresholds (0.5%–2%) have enabled bTMB assay calibration, mitigating challenges like ctDNA fragmentation and low variant allele frequencies (VAF) [202]. Advanced bioinformatics pipelines now employ noise-reduction algorithms to filter artifacts from clonal hematopoiesis or sequencing errors, improving specificity for true tumor-derived mutations [202]. Next-generation targeted panels have also advanced to enhance sensitivity and broaden genomic coverage. Hybrid capture-based panels paired with unique molecular identifiers (UMI) improve detection of low-VAF mutations (< 1%) in low-tumor-content samples [202, 203]. Furthermore, multiplex PCR-based approaches, including the Oncomine Tumor Mutation Load Assay, facilitate rapid and cost-effective TMB estimation by targeting mutation hotspots and immune-relevant genomic regions [204]. These panels are increasingly integrated with machine learning algorithms to predict neoantigen load and immunogenicity, associating TMB quantification with functional immune response metrics. Single-cell sequencing, a newly developed high-throughput technology, allows investigation of genomics, transcriptomics, and epigenetics on a single-cell level [205]. Single-cell DNA sequencing (scDNA-seq) and single-cell RNA sequencing (scRNA-seq) are transforming TMB analysis by dissecting intratumoral heterogeneity and distinguishing clonal neoantigens [206, 207]. Spatial transcriptomics, such as Visium Spatial Gene Expression, provides spatial context for TMB within the TME, correlating mutational hotspots with immune-excluded regions or PD-L1 expression [208]. These methods elucidate interactions between TMB and local immune activity, refining its role as a dynamic biomarker.

Standardization efforts have resulted in synthetic reference materials, incorporating predefined mutations at specified VAF for cross-platform validation and reduced interlaboratory variability. Furthermore, the Friends of Cancer Research TMB Harmonization Consortium has developed guidelines for panel design, bioinformatics pipelines, and clinical reporting, standardizing criteria [209]. In summary, contemporary TMB measurement methods prioritize precision, scalability, and integration with complementary biomarkers. Liquid biopsy, ultra-sensitive targeted panels, and single-cell technologies are broadening TMB's clinical utility, while standardization initiatives mitigate reproducibility challenges. Future directions will likely emphasize dynamic TMB assessment through ctDNA analysis and multi-omics integration to optimize immunotherapy stratification.

Clinical applications of TMB in predicting immunotherapy response

The clinical utility of TMB as a predictive biomarker for ICI response has been validated across diverse cancer types, with growing evidence supporting its role in patient stratification. Clinical trials of cancer immunotherapy closely related to TMB are shown in Table 3

[210–212]. The FDA's 2020 tumor-agnostic approval of pembrolizumab for advanced solid tumors with TMB ≥ 10 mutations per megabase (mut/Mb), based on the phase II KEYNOTE-158 trial, constituted a pivotal milestone [20]. This approval underscores TMB's potential as a pan-cancer biomarker, though its application remains context-dependent and requires histology-specific integration.

In NSCLC, TMB has demonstrated robust predictive value. Retrospective analyses of the CheckMate-026 and CheckMate-227 trials revealed that patients with TMB-high tumors (≥ 10 mut/Mb) treated with nivolumab or ipilimumab demonstrated significantly improved PFS and objective response rates (ORR) compared to chemotherapy [213, 214]. Similarly, in melanoma, TMB correlates with durable responses to anti-PD-1/CTLA-4 therapies, with high TMB associated with enhanced cytolytic activity and prolonged survival [215]. Notably, tumors harboring dMMR or POLE/POLD1 mutations, such as colorectal and endometrial cancers, exhibit exceptionally high TMB and marked sensitivity to ICI [216, 217]. However, TMB's predictive power varies by cancer type. For example, in microsatellite-stable (MSS) colorectal and gastric cancers, tumors with TMB ≥ 10 mut/Mb often exhibit limited response to immune checkpoint inhibitors unless accompanied by specific molecular features, reflecting the dominant immunosuppressive microenvironment and lack of immunogenic neoantigens in MSS tumors [218]. This observation aligns with the broader recognition that TMB's predictive utility is contingent on both mutational load and the functional immunogenicity of the resulting neoantigens, which may be compromised in non-hypermutated or immune-excluded tumors. These discrepancies underscore the need for histology-specific TMB thresholds and validation.

The KEYNOTE-158 trial established TMB's role as a companion diagnostic biomarker. In this study, pembrolizumab achieved an ORR of 29% in TMB-high (≥ 10 mut/Mb) patients across 10 cancer types, including rare malignancies such as small cell neuroendocrine carcinoma and sarcoma [20]. Response rates increased to 37% in tumors with TMB > 13 mut/Mb, suggesting a dose–response relationship between TMB and ICI efficacy [16, 20]. Despite these advances, the 10 mut/Mb threshold remains debated, as some cancers (e.g., melanoma) may benefit from higher cutoffs, while others (e.g., MSI-H tumors) respond robustly even at lower TMB levels [219, 220].

Integrating TMB with PD-L1 expression enhances predictive accuracy, as these biomarkers capture complementary aspects of tumor immunogenicity: TMB quantifies neoantigen load (reflecting genomic instability and potential T-cell recognition), while PD-L1 measures adaptive immune resistance within the tumor microenvironment. In NSCLC, the CheckMate-227 trial demonstrated that patients with TMB-high (≥ 10 mut/Mb) and PD-L1-positive tumors derived the greatest benefit from nivolumab plus ipilimumab, with a median PFS of 7.2 months compared to 3.2 months in TMB-low counterparts [221]. This synergy arises because high TMB increases antigenicity, while PD-L1 expression indicates pre-existing immune engagement. Conversely, in PD-L1-negative tumors, TMB retains predictive value as a stand-alone biomarker, highlighting its independence from PD-L1—particularly valuable in cancers like small cell lung cancer where PD-L1 expression is often absent or heterogeneous. Combined Positive Score (CPS) and Tumor Proportion Score (TPS), which assess PD-L1 expression differently, further refine TMB-based stratification: TPS quantifies only tumor-cell membrane staining (percentage of viable tumor cells), while CPS includes immune cells (number of PD-L1-positive tumor and immune cells per 100 tumor cells), making it more sensitive in cancers with prominent stromal immune infiltration. For instance, in gastric cancer, the phase III KEYNOTE-062 trial showed that TMB-high (≥ 10 mut/Mb) patients with CPS ≥ 1 (a broader cutoff than TPS, validated for gastrointestinal malignancies) exhibited superior survival with pembrolizumab versus chemotherapy [222]. These findings suggest that dual biomarker approaches mitigate the limitations of single-marker strategies—such as spatial heterogeneity in PD-L1 expression or TMB's inability to reflect immune evasion mechanisms—particularly in cancers with heterogeneous PD-L1 expression. However, drawbacks persist: technical variability in TMB quantification and scoring discordance between CPS/TPS assays can lead to inconsistent classifications, while overlapping predictive value sometimes reduces the incremental benefit of combined testing. In conclusion, TMB has emerged as a critical biomarker for ICI response prediction, particularly in TMB-high malignancies. Its integration with PD-L1, CPS, and TPS refines patient selection, though histology-specific validation and technical standardization are imperative. As precision immuno-oncology advances, TMB is poised to serve as a cornerstone in multi-omics frameworks, guiding personalized therapeutic strategies across diverse cancer types.

Challenges and perspectives

Potential advantages and challenges of neoantigens and TMB in immunotherapy

Neoantigens represent a cornerstone of precision immunotherapy due to their capacity to trigger tumor-specific immune responses while sparing healthy tissues [71]. This tumor-exclusive expression pattern fundamentally distinguishes them from tumor-associated antigens that

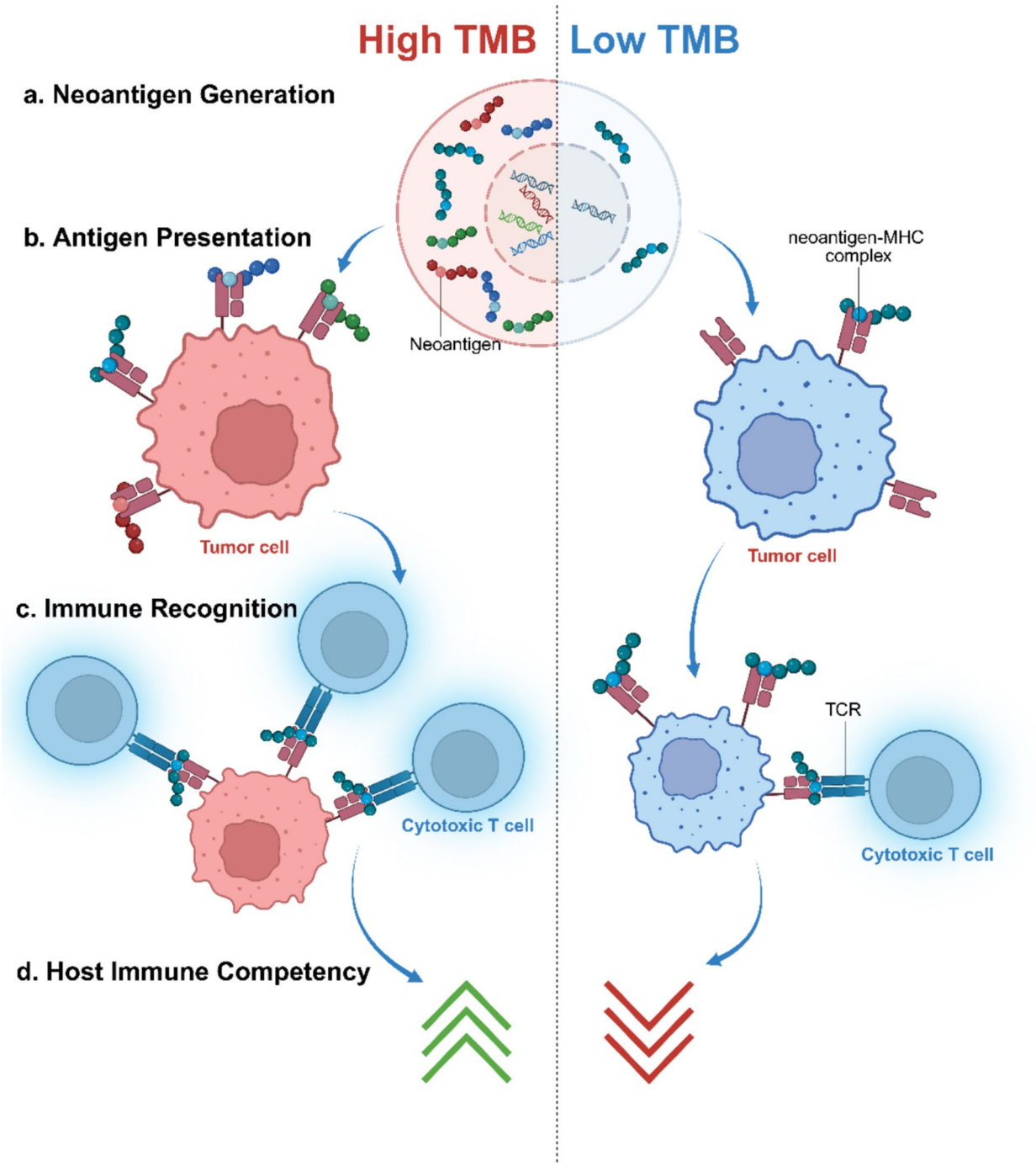

Fig. 4 Mechanisms of TMB in predicting immunotherapy response. The mechanisms of TMB in predicting immunotherapy response can be delineated into four interrelated processes: neoantigen generation, antigen presentation, immune recognition, and host immune competency. **a.** Elevated TMB increases the probability of immunogenic neoantigen generation, as coding-region mutations may produce altered peptides perceived as"non-self"by the immune system. **b.** Elevated TMB increases the probability of neoantigen-MHC binding with sufficient affinity for surface presentation. **c.** In TMB-high tumors, neoantigen diversity also increases the probability of TCR recognition. **d.** Host immune competency, governed by systemic and local factors, also determines the response efficacy of immunotherapy. MHC: major histocompatibility complex, TCR: T cell receptor, TMB: tumor mutational burden. Created with BioRender.com

may exist in normal tissues, thereby minimizing autoimmune complications. The patient-specific nature inherently reduces off-target toxicity and circumvents central or peripheral immune tolerance mechanisms, positioning them as ideal targets for personalized therapies like vaccines and adoptive T-cell therapies [72, 223]. The avoidance of immune tolerance is particularly critical, as conventional tumor antigens often undergo negative selection during thymic education, whereas neoantigens emerge from somatic mutations after immune system maturation. Additionally, TNB strongly correlates with improved responses to ICI, especially in hypermutated cancers such as melanoma and NSCLC, where high TNB is associated with elevated cytolytic activity and durable clinical benefits [67, 79, 116]. This correlation stems from the increased probability of generating immunodominant epitopes when mutational load exceeds a critical threshold, effectively transforming "cold" tumors into "hot" immunogenic microenvironments.

Despite these advantages, neoantigen-based therapies face substantial clinical challenges. Tumor heterogeneity remains a major obstacle, as subclonal neoantigens—expressed only in specific tumor subpopulations—drive immune escape under therapeutic pressure [224]. This spatial and temporal heterogeneity creates evolutionary bottlenecks where therapy-resistant clones lacking targetable neoantigens eventually dominate the tumor ecosystem. Deficiencies in antigen presentation, including B2M loss or MHC downregulation, further obscure neoantigen visibility to T cells, even in tumors with high TMB [28, 175]. Critically, the extreme patient-specificity of somatic mutations generating neoantigens creates a fundamental limitation: each individual's tumor possesses a unique mutanome, with only a little fraction yielding immunogenic epitopes. This intrinsic variability is compounded by the vast polymorphism of HLA alleles across human populations [225]. Specific HLA allotypes exhibit differential binding affinities for peptide

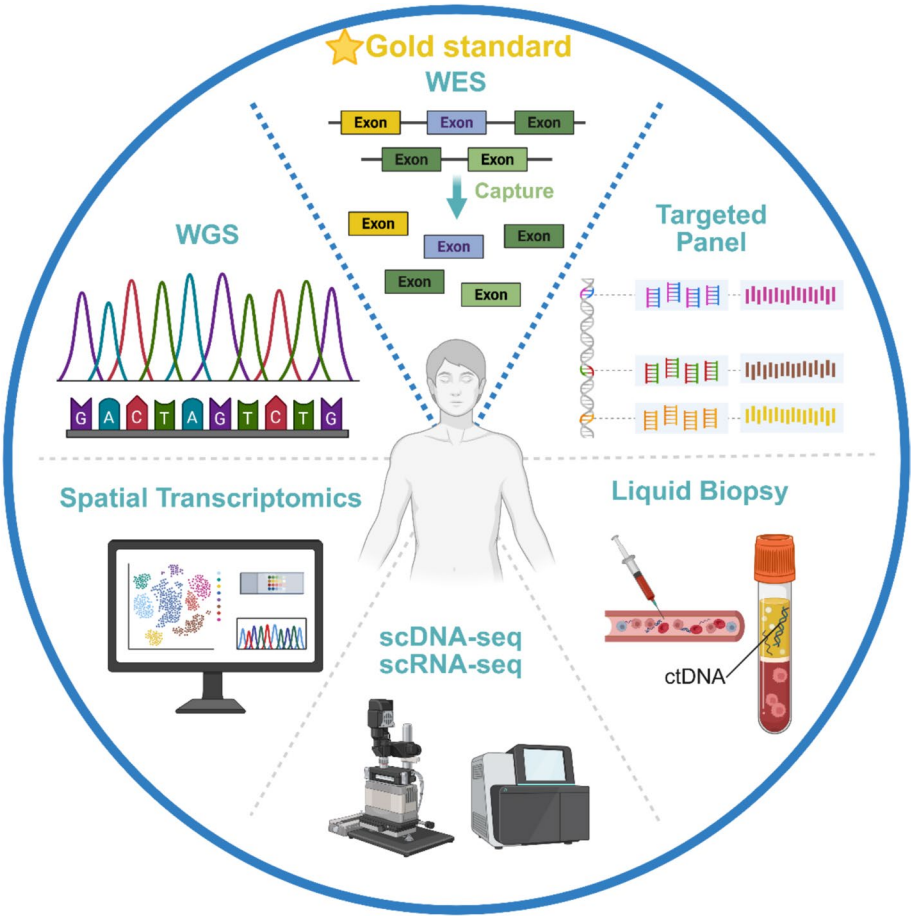

Fig. 5 Detection of TMB in cancer immunotherapy. The primary detection methods of TMB are shown in figure. WES remains the gold standard for comprehensive TMB assessment. Emerging technologies overcome the limitations of WES, such as high costs and lengthy processing times. ctDNA: circulating tumor DNA, scDNA-seq: single-cell DNA sequencing, scRNA-seq: single-cell RNA sequencing, WES: whole-exome sequencing, WGS: whole-genome sequencing. Created with BioRender.com

Table 3 Clinical trials of cancer immunotherapy closely related to tumor mutational burden (selected from ClinicalTrials.gov)

ClinicalTrials.gov ID	Study Title	Phase	Status	First Location	Country	Actual/Estimated Study Start Date	Estimated Enrollment	Conditions	Interventions	Study Model	Primary end point	Secondary end point	Results
NCT05592626	A Study of a Selective T Cell Receptor (TCR) Targeting, Bifunctional Antibody-fusion Molecule STAR0602 in Participants with Advanced Solid Tumors	Phase 1 Phase2	Recruiting	Adventhealth Celebration	United States, Canada, France and Spain	2023/1/4	365	Advanced Solid Tumor	Drug: STAR0602	Sequential Assignment	Number of Participants with Dose-limiting Toxicities (DLTs) in Cycle 1(28 days); AEs and SAEs (Up to 3 years); ORR (Up to 3 years)	DOR; PFS; OS (Up to 3 years); Cmax; Tmax	Unpublished
NCT05086692	A Beta-only IL-2 ImmunoTherapY Study	Phase 1 Phase2	Recruiting	UCSF Helen Diller Family Comprehensive Cancer Center	United States, Australia, Canada and Korea	2021/8/27	115	Advanced Solid Tumor	Drug: MDNA11 Drug: Pembrolizumab (KEYTRUDA®)	Sequential Assignment	MDNA11 Recommended Dose for Expansion for monotherapy (mRDE) and combination (cRDE) (24 months); Incidence of Treatment Related and Emergent Adverse Events (TRAEs and TEAEs) (24 months)	Cmax; Tmax; AUC; ORR; DCR; PFS (Up to 24 months)	Unpublished
NCT05058937	A Study to Examine the Clinical Value of Comprehensive Genomic Profiling Performed by Belgian NGS Laboratories: A Belgian Precision Study of the BSMO in Collaboration with the Cancer Centre	Not Applicable	Recruiting	ASZ Aalst	Belgium	2021/6/1	936	Solid Tumor and;Metastatic Cancer	Diagnostic Test: Comprehensive genomic profiling	Single Group Assignment	Number/prevalence of variants with clinical significance; Number/prevalence of alterations by type (SNVs, CNVs, fusions); Description of patient journey (through study completion, an average of 1 year)	Percentage of patients with successful CGP; PFS ratio; Number of participating NGS laboratories continuing CGP after the study (through study completion, an average of 1 year)	Number/prevalence of variants with clinical significance (81%) versus local NGS testing (21%); Percentage of patients with successful CGP(93%). (124 patients (16%) had a TMB-high score. A high TMB was most frequently observed in lung cancer, melanoma, miscellaneous cancer, CUP, cervical carcinoma, and urothelial carcinomas. For eight patients, an MSI-high status was detected; all eight also exhibited a high TMB) (Ref: [210])

Table 3 (continued)

ClinicalTrials.gov ID	Study Title	Phase	Status	First Location	Country	Actual/Estimated Study Start Date	Estimated Enrollment	Conditions	Interventions	Study Model	Primary end point	Secondary end point	Results
NCT04591431	The Rome Trial from Histology to Target: the Road to Personalize Target Therapy and Immunotherapy	Phase 2	Active, not recruiting	OSPEDALI RIUNITI di ANCONA	Italy	2020/10/7	400	Breast Cancer,Gastrointestinal Cancer,NSCLC and Other Cancer	Drug: Erlotinib Drug: Trastuzumab Drug: Trastuzumab emtansine Drug: Pertuzumab Drug: Lapatinib Drug: Everolimus Drug: Vemurafenib Drug: Cobimetinib Drug: Alectinib Drug: Brigatinib Drug: Palbociclib Drug: Ponatinib Drug: Vismogedib Drug: Itacitinib Drug: Ipatasertib Drug: Entrectinib Drug: Atezolizumab Drug: Nivolumab Drug: Ipilimumab Drug: Pemigatinib Drug: Oncology Drugs Drug: Pralsetinib Drug: Selpercatinib Drug: Talazoparib Drug: Tepotinib Drug: Alpelisib	Parallel Assignment	ORR (42 months)	PFS; TTF; TTNT; OS; QoLs included in the two arms of the study of SoC vs TT (42 months)	Overall population: TT vs SoC (ORR: 17% vs 9.5%; Median PFS:3.7 vs 2.8 mo; Median OS:9.2 vs 7.6 mo) Pts with hTMB/MSS tumors:TT vs SoC(Median PFS:3.6 vs 2.8 mo; 12-month PFS rate:32.7% vs 6.3%) (Ref: [211]) Breast Cancer Group: TT vs SoC (ORR: 20% vs 35%; Median OS:16.8 vs 13.0 mo; PFS:4.7 vs 6.2 mo) Among pts with hTMB (11): TT vs SoC (ORR:40% vs 17%; DCR:60% vs 17%; PFS:16.7 vs 1.9 mo) Subgroup analyses: ESCAT I-II vs ESCAT ≥ III vs SoC (ORR:29% vs 0% vs 35%; Median OS:24.2 vs 7.0 vs 13.0 mo; Median PFS:7.6 vs 3.0 vs 6.2 mo) (Ref: [212])
NCT04484636	PLATON—Platform for Analyzing Targetable Tumor Mutations (pilot-study)	Not Applicable	Completed	HELIOS Klinikum Bad Saarow	Germany	2020/10/28	400	Hepatocellular Cancer,Cholangiocarcinoma,Gallbladder Cancer,Pancreatic Cancer,Oesophageal Cancer and Stomach Cancer	Diagnostic Test: FoundationOne®CDx and FoundationOne®Liquid	Single Group Assignment	Distribution of mutations in patients with HCC, intra- and extrahepatic CCA, GBCA, PDAC and gastric cancer (up to 4 weeks after biospecimen provision)	Heterogeneity of targetable alterations in paraffin embedded specimen vs. cfDNA; Relative frequency of targetable mutations (incl. TMB and MSI status) per disease group; Number of patients receiving therapies in accordance to their genomic profiles (up to 4 weeks after biospecimen provision)	Unpublished

Table 3 (continued)

ClinicalTrials.gov ID	Study Title	Phase	Status	First Location	Country	Actual/Estimated Study Start Date	Estimated Enrollment	Conditions	Interventions	Study Model	Primary end point	Secondary end point	Results
NCT04263688	Multicenter Observational Study of Advanced Non-small Cell Lung Cancer with Malignant Pleural Effusion	Observational	Unknown status	Guangzhou Institute of Respiratory Disease	China	2020/3/1	300	NSCLC	Other: Non-Intervention	Cohort	Objective response rate (ORR) (Up to 2021); Progression-free survival (PFS) (Up to 2021); Overall survival (OS) (Up to 2022)	Not provided	Unpublished
NCT03911557	Durvalumab and Tremelimumab Combination in Somatically Hypermutated Recurrent Solid Tumors	Phase 2	Active, not recruiting	Markey Cancer Center	United States	2019/6/8	14	Solid Tumor	Drug: Durvalumab and Tremelimumab	Single Group Assignment	Time-to-progression ratio (TTP) (2 years)	Progression-free survival (PFS) (2 years)	Unpublished
NCT03373955	A Prospective Study of Constructing Immune Repertoire to Monitor the Therapeutic Effect in NSCLC Patients	Observational	Unknown status	West China Hospital, Sichuan University	China	2017/11/23	60	NSCLC	Drug: pembrolizumab	Case-Only	T cell repertoire (baseline, 2 months, 6 months, 1 year)	peripheral blood circulating tumor DNA (baseline, 2 months, 6 months, 1 year)	Unpublished

AE adverse event, *AUC* area under the plasma concentration–time curve, *CCA* cholangiocarcinoma, *cfDNA* cell free DNA, *CGP* comprehensive genomic profiling, *Cmax* maximum concentration, *CNV* copy number variant, *cRDE* recommended dose for expansion for combination, *CUP* cancer of unknown primary, *DCR* disease control rate, *DLT* dose-limiting toxicity, *DOR* during of response, *ESCAT* the ESMO Scale for Clinical Actionability of molecular Targets, *GBCA* gallbladder cancer, *HCC* hepatic cell carcinoma, *IL-2* interlukin-2, *mRDE* recommended dose for expansion for monotherapy, *MSI* microsatellite instability, *MSS* microsatellite-stable, *NGS* next-generation sequencing, *NSCLC* non-small cell lung cancer, *ORR* objective response rate, *OS* overall survival, *PDAC* pancreatic ductal adenocarcinoma, *PFS* progression-free survival, *Pts* patients, *QoL* quality of life, *SAE* severe adverse event, *SNV* single-nucleotide variant, *SoC* standard of care, *TCR* T cell receptor, *TEAE* treatment emergent adverse event, *Tmax* time to peak drug concentration, *TMB* tumor mutational burden, *TRAE* treatment related adverse event, *TT* targeted therapy, *TTF* time to treatment failure, *TTNT* time to next treatment, *TTP* time-to-progression ratio

sequences, meaning identical mutations may generate immunogenic neoantigens in some patients but remain immunologically inert in others due to HLA mismatch [226]. Consequently, the combinatorial complexity of patient-specific mutations interacting with diverse HLA haplotypes severely restricts the universal applicability of neoantigen-targeting therapies, rendering most neoantigens private to individual patients. To overcome these barriers, innovative strategies are emerging. One approach focuses on "shared neoantigens" derived from recurrent driver mutations commonly found across patients with specific cancer types [156, 227]. These public epitopes offer potential for off-the-shelf therapies targeting broader patient cohorts [228]. Complementarily, leveraging HLA supertypes—groups of HLA alleles with similar peptide-binding preferences—can enhance population coverage [229, 230]. For instance, designing vaccines targeting epitopes presented by HLA supertype alleles could theoretically benefit larger patient subsets despite individual HLA variations [229, 230]. Such strategies aim to transform neoantigen therapeutics from purely personalized paradigms toward population-level solutions.

In terms of the translational and implementation challenges, significant translational hurdles must be overcome to realize the full potential of neoantigen-based strategies. Personalized neoantigen therapies face complex manufacturing logistics, including rapid turnaround times for vaccine production and scalable infrastructure for adoptive cell therapies, which currently limit widespread accessibility. Assay standardization remains critical for both biomarkers, particularly for TMB quantification across sequencing platforms and TNB validation in clinical settings. For TNB to serve as a robust stratification or response marker, it must demonstrate consistent reproducibility across institutions, establish validated predictive thresholds linked to clinical outcomes, and prove cost-effectiveness relative to existing biomarkers. While TNB offers superior specificity over TMB by filtering immunogenic mutations, neither biomarker alone suffices to capture the multifaceted nature of treatment response. Combining TNB/TMB with indicators of immune activation—such as T-cell infiltration, PD-L1 expression, or MSI-H/MSS status—will likely yield more reliable predictive models. Furthermore, TMB's utility as a standalone biomarker is constrained by tumor heterogeneity, temporal variations during therapy, and discordance between tissue and liquid biopsy measurements. These biological and technical gaps underscore the need for dynamic monitoring approaches and integrated frameworks that account for tumor-immune coevolution. Ultimately, the path forward necessitates not only refining the biomarkers themselves but also developing multifactorial frameworks that synthesize diverse data streams. Addressing these gaps through collaborative standardization initiatives and health economics research will be pivotal for equitable clinical implementation. Furthermore, prospective clinical trials explicitly designed to validate these integrated approaches and their impact on patient outcomes are essential to bridge the current translational divide.

The immunosuppressive TME, enriched with Treg, MDSC, and inhibitory cytokines, also suppresses neoantigen-specific T-cell activity [188–191]. Notably, this suppression operates through both direct cell–cell contact mechanisms and paracrine signaling networks that establish regional immune privilege zones within tumors. Moreover, current prediction pipelines frequently overlook post-translational modifications, alternative splicing, or non-canonical antigen presentation, resulting in overestimated immunogenic neoantigens [30, 231, 232]. These compounding individualization factors further exacerbate the computational oversimplification arises from the predominant focus on exome-derived mutations while neglecting the complex proteomic processing required for actual immunogenicity. Collectively, these biological and computational hurdles necessitate integrative strategies to refine neoantigen selection and address resistance.

Enhancing neoantigen prediction accuracy and immunogenicity

Improving neoantigen identification demands advances in multi-omics integration and computational modeling. The current paradigm shift recognizes that neoantigen immunogenicity is not merely a function of mutation presence but requires coordinated expression, processing, presentation, and T-cell recognition. Current approaches predominantly rely on WES and tools like NetMHCpan to predict MHC binding affinity [40, 62]. However, only a minority of predicted neoantigens are naturally processed and presented on MHC molecules, underscoring the gap between in silico predictions and in vivo immunogenicity [233]. This discrepancy highlights the crucial role of proteasomal cleavage patterns and peptide transport efficiency—biological filters largely absent in current algorithms. Emerging solutions combine genomics with immunopeptidomics—MS-based profiling of MHC-bound peptides—to directly identify presented neoantigens [57]. By anchoring predictions to empirically verified MHC ligands, this approach bypasses theoretical assumptions about antigen processing machinery. For instance, integrating MS data with RNA sequencing enhances detection of splice variant- or frameshift-derived neoantigens, which conventional pipelines often miss [234]. Such integration effectively

bridges the genotype–phenotype divide by correlating transcriptional output with actual peptide presentation.

Machine learning models trained on immunopeptidomic datasets now prioritize neoantigens using features beyond MHC binding, such as TCR recognition probability, peptide stability, and dissimilarity to self-antigens [44, 46, 235]. These multidimensional models simulate the immunological "fitness" of neoantigens by quantifying their likelihood to complete the entire immune recognition cascade. These models can also incorporate tumor-specific variables, including HLA diversity, mutation clonality, and immune contexture, to better predict functional immunogenicity [236–238]. For example, clonality-adjusted prediction weights account for the therapeutic relevance of targeting truncal versus subclonal mutations. Experimental validation remains critical: high-throughput TCR screening and in vitro T-cell activation assays confirm neoantigen-induced immune responses [239, 240]. These functional assays serve as essential reality checks by quantifying the magnitude and specificity of T-cell responses against predicted epitopes. Meanwhile, scRNA-seq and spatial transcriptomics map neoantigen-specific T-cell clones within the TME, elucidating their spatial distribution and functional states [241, 242]. This spatial resolution reveals microenvironmental niches where neoantigen-specific T-cells become functionally impaired, informing combination therapy strategies.

Combining neoantigen-based therapies with conventional treatments

The integration of neoantigen-based therapies with conventional treatments—such as chemotherapy, radiotherapy, and ICI—may exploit synergistic mechanisms to improve therapeutic efficacy and overcome resistance, positioning this strategy as a pivotal focus in contemporary oncology research. This combinatorial approach leverages the complementary strengths of each modality: conventional therapies debulk tumors and modulate microenvironments, while neoantigen-targeted therapies provide immunological specificity. Chemotherapy and radiotherapy induce immunogenic cell death, releasing tumor-derived antigens and activating antigen-presenting cells (e.g., dendritic cells), thereby priming adaptive immunity for neoantigen-specific responses [120, 243]. The resulting "antigen storm" enhances cross-presentation of therapy-exposed neoantigens, effectively converting tumor debris into endogenous vaccines. For instance, the abscopal effect of radiotherapy, which triggers systemic antitumor immunity, could synergize with neoantigen vaccines when administered sequentially to enhance localized treatment outcomes [244]. Timing optimization is crucial here, as radiation-induced inflammation may create temporal windows of enhanced immune receptivity. Concurrently, ICI mitigate T-cell exhaustion, sustaining the cytotoxic activity of neoantigen-reactive T lymphocytes. This PD-1/PD-L1 axis blockade essentially "releases the brakes" on neoantigen-specific clones that have infiltrated tumors but become functionally anergic. Preclinical evidence indicates that such combination regimens augment effector T-cell infiltration within the TME, thereby prolonging immune-mediated tumor control [245].

Within biomarker-guided precision oncology, TMB serves as a predictive biomarker to stratify patients for combination therapies. This stratification acknowledges the biological continuum of tumor immunogenicity, where therapeutic strategies must adapt to each patient's neoantigen landscape. TMB-high tumors, characterized by greater neoantigen diversity, demonstrate increased susceptibility to immunotherapies, whereas TMB-low malignancies may benefit from chemotherapy- or radiotherapy-induced antigen exposure to counteract neoantigen paucity [24, 246, 247]. In TMB-low scenarios, conventional therapies essentially function as neoantigen amplifiers by inducing DNA damage and subsequent mutation generation. Furthermore, optimizing the temporal sequencing and dosing of personalized regimens is critical; for instance, post-chemotherapy immune microenvironment remodeling may enhance the therapeutic window for neoantigen vaccine efficacy [248]. his phased approach allows chemotherapy to first eliminate immunosuppressive elements like MDSC, creating a more permissive environment for vaccine-primed T cells.

Despite these advances, challenges persist, including immunosuppressive off-target effects of conventional therapies, tumor clonal evolution-driven antigen escape, and the logistical complexity of personalized therapeutic platforms. The dynamic nature of tumor-immune coevolution demands real-time monitoring approaches to adjust combination regimens as resistance mechanisms emerge. Future research must prioritize elucidating mechanistic synergies between treatment modalities, standardizing biomarker validation, and deploying computational tools to refine combination strategies. Advanced systems biology approaches could decode the nonlinear interactions between chemotherapy-induced stress responses and neoantigen presentation dynamics. Innovations in neoantigen prediction algorithms, multi-omics integration, and scalable manufacturing technologies are poised to advance this paradigm toward more durable and precise cancer therapies, ultimately improving clinical outcomes. The ultimate goal resides in creating adaptive treatment ecosystems

where conventional and immunotherapeutic approaches mutually reinforce each other through precisely orchestrated molecular and cellular interactions.

Conclusions

The exploration of neoantigens and TMB has transformed the landscape of cancer immunotherapy, offering novel opportunities for precision oncology. Neoantigens, tumor-specific antigens arising from somatic mutations, represent ideal targets for personalized therapies due to their absence in normal tissues, thereby minimizing off-target toxicity. Concurrently, TMB serves as a robust biomarker for predicting immunotherapy response by estimating tumor immunogenicity. Together, these parameters have reshaped our understanding of tumor-immune interactions and propelled the development of innovative therapeutic strategies.

Neoantigen-based therapies exhibit significant clinical potential. Early-phase trials demonstrate synergistic effects between neoantigen vaccines and ICI, with vaccines expanding tumor-reactive T-cell clones and ICI reversing T-cell exhaustion. Advances in CAR-T engineering have similarly enhanced specificity and persistence in solid tumors, overcoming challenges such as antigen heterogeneity and immunosuppressive microenvironments. These breakthroughs highlight the necessity of tailoring therapies to individual tumor mutational profiles. TMB has emerged as a critical biomarker for identifying patients likely to benefit from ICI, particularly in hypermutated cancers such as melanoma, NSCLC, and dMMR tumors. The FDA's tumor-agnostic approval of pembrolizumab for TMB-high solid tumors represented a milestone in biomarker-driven oncology. However, TMB's predictive utility remains context-dependent, requiring integration with complementary biomarkers such as PD-L1 expression, immune gene signatures, and HLA status. Liquid biopsy and ultra-sensitive sequencing panels have improved TMB measurement scalability, enabling dynamic monitoring of mutational load and resistance mechanisms. Despite progress, challenges persist in standardizing TMB thresholds, addressing tumor heterogeneity, and differentiating immunogenic from non-functional mutations.

The interplay between neoantigens and TMB underscores the complexity of tumor immunogenicity. While TMB quantifies mutation abundance, only a subset generates immunogenic neoantigens capable of eliciting T-cell responses. This discrepancy emphasizes the need for multi-omics approaches combining genomics, immunopeptidomics, and single-cell technologies to refine neoantigen prediction. Machine learning models trained on MHC-eluted ligand datasets and TCR recognition patterns have improved prediction accuracy, bridging in silico algorithms and in vivo immunogenicity. Furthermore, spatial transcriptomics and single-cell sequencing elucidate tumor-immune dynamics, revealing niches where neoantigen-specific T cells are functionally suppressed. Nonetheless, clinical translation faces hurdles: tumor heterogeneity and clonal evolution drive antigen escape, while defects in antigen presentation render neoantigens undetectable to immune surveillance. Future strategies should prioritize combinatorial approaches, such as combining neoantigen vaccines with chemotherapy or radiotherapy to induce immunogenic cell death and amplify antigen exposure. Additionally, real-time monitoring via ctDNA may enable adaptive therapy adjustments to counteract resistance.

Advancing neoantigen-based immunotherapy will require integrating artificial intelligence, synthetic biology, and multi-omics platforms. Standardizing biomarker validation, optimizing manufacturing scalability, and fostering interdisciplinary collaboration are essential to make these therapies widely accessible. Ultimately, the goal is to establish adaptive treatment ecosystems where conventional and immunotherapeutic modalities synergize to achieve durable remission. As the field evolves, neoantigens and TMB will remain cornerstones of precision immuno-oncology, guiding the development of therapies that harness the immune system's full potential to combat cancer.

Abbreviations

AE	Adverse event
AML	Acute myeloid leukemia
AUC	Area under the plasma concentration–time curve
BAM	Binary Alignment/Map format
BED	Biologically effective dose
B2M	β2-Microglobulin
bTMB	Blood tumor mutational burden
CAR	Chimeric antigen receptor
CBR	Clinical benefit rate
CCA	Cholangiocarcinoma
CDR3	Complementarity determining region 3
cfDNA	Cell free DNA
CGP	Comprehensive genomic profiling
Cmax	Maximum concentration
CNV	Copy number variant
CPS	Combined Positive Score
CR	Complete response
CRC	Colorectal cancer
cRDE	Recommended dose for expansion for combination
ctDNA	Circulating tumor DNA
CTL	Cytotoxic T lymphocyte
CTLA-4	Cytotoxic T-lymphocyte-associated protein 4
CUP	Cancer of unknown primary
DC	Dendritic cell
DCR	Disease control rate
DFS	Disease-free survival
DFS2	Disease-free survival 2
DIPG	Diffuse intrinsic pontine glioma
DLT	Dose-limiting toxicity
DMFS	Distant metastasis-free survival
DMG	Diffuse midline glioma
dMMR	DNA mismatch repair deficiency
DNMT	DNA methyl transferase

DoR	Duration of response
DTH	Delayed type hypersensitivity
EFS	Event free survival
EGFR	Epidermal growth factor receptor
EGFRvIII	Epidermal growth factor receptor variant III
ESCAT	The ESMO Scale for Clinical Actionability of molecular Targets
ESCC	Esophageal squamous cell carcinoma
EORTC	The European Organization for Research and Treatment of Cancer
FASTQ	Fast Quality format
FDA	Food and Drug Administration
FFS	Freedom from surgery
GBCA	Gallbladder cancer
GBM	Glioblastoma
GD2	Disialoganglioside
GM-CSF	Granulocyte–macrophage colony-stimulating factor
GUI	Graphical user interface
HCC	Hepatic cell carcinoma
HDAC	Histone deacetylase
HER2	Human epidermal growth factor receptor 2
HLA	Human leukocyte antigen
HNSCC	Head and neck squamous cell carcinoma
HSV	Herpes simplex virus
ICB	Immune checkpoint blockade
ICI	Immune checkpoint inhibitor
IFN-γ	Interferon-γ
IL	Interlukin, indel: insertions/deletions
irAE	Immune-related adverse event
IRC	Independent review committee
iRECIST	Immune-based Response Evaluation Criteria in Solid Tumors
JAK-STAT	Janus kinase-signal transducer and activator of transcription
KRAS	Kirsten rat sarcoma viral oncogene homolog
LAG-3	Lymphocyte activation gene-3
LCSS	Lung cancer specific survival
LNP	Lipid nanoparticle
LOH	Loss of heterozygosity
mCRC	Metastasis colorectal cancer
MDSC	Myeloid-derived suppressor cell
MHC	Major histocompatibility complex
mPR	Major pathological response
MRD	Minimal residual disease
mRDE	Recommended dose for expansion for monotherapy
MS	Mass spectrometry
MSI-H	High microsatellite instability
MSS	Microsatellite-stable
MTD	Maximum tolerated dose
mut/Mb	Mutations per megabase
NCCN	National Comprehensive Cancer Network
NGS	Next-generation sequencing
NK	Natural killer
NSCLC	Non-small cell lung cancer
oHSV	Oncolytic herpes simplex virus
ORR	Objective response rate
OS	Overall survival
OV	Oncolytic virus
pCR	Pathological complete response
PDAC	Pancreatic ductal adenocarcinoma
PD-1	Programmed cell death-1
PD-L1	Programmed cell death ligand-1
PFS	Progression-free survival
pMMR	Mismatch-repair-proficient
POLE/POLD1	Polymerase epsilon/delta
PR	Partial response
Pts	Patients
pVAC-Seq	Personalized Variant Antigens by Cancer Sequencing
QLQ-C30	Quality of Life Questionnaire-Core 30
QoL	Quality of life
RECIST	The Response Evaluation Criteria in Solid Tumors
RFS	Recurrence-free survival
RIG-I	Retinoic acid-inducible gene I
RP2D	Recommended phase 2 dose
SAE	Severe adverse event
scDNA-seq	Single-cell DNA sequencing
scRNA-seq	Single-cell RNA sequencing
SD	Stable disease
SHERPA	Systematic HLA Epitope Ranking Pan Algorithm
SLP	Synthetic long peptide
SNV	Single-nucleotide variant
SoC	Standard of care
SynNotch	Synthetic Notch
TACE	Transhepatic arterial chemotherapy and embolization
Tan-MUC1	Tumor-associated mucin 1
TCR	T cell receptor
TEAE	Treatment emergent adverse event
TEIM-Res	TCR–Epitope Interaction Modelling at Residue Level
TESAE	Treatment-emergent serious adverse event
TGF-β	Transforming growth factor-β
TIL	Tumor-infiltrating lymphocyte
TIminer	Tumor Immunology miner
TLR	Toll-like receptor
Tmax	Time to peak drug concentration
TMB	Tumor mutational burden
TME	Tumor microenvironment
TNB	Tumor neoantigen burden
TPS	Tumor Proportion Score
TRAE	Treatment-related adverse event
Treg	Regulatory T cell
TSA	Tumor-specific antigen
TSNAD	Tumor-specific neoantigen detector
TT	Targeted therapy
TTF	Time to treatment failure
TTNT	Time to next treatment
TTP	Time to progression
TTR	Time to remission
UMI	Unique molecular identifier
VAF	Variant allele frequency
VCF	Variant call format
WES	Whole-exome sequencing
WGS	Whole-genome sequencing

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Authors' contributions
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Data availability
No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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References:

1. Gaikwad S, Agrawal MY, Kaushik I, Ramachandran S, Srivastava SK. Immune checkpoint proteins: Signaling mechanisms and molecular interactions in cancer immunotherapy. Semin Cancer Biol. 2022;86:137–50. https://doi.org/10.1016/j.semcancer.2022.03.014.
2. Zhu C, Liu C, Wu Q, Sheng T, Zhou R, Ren E, Zhang R, Zhao Z, Shi J, Shen X, et al. Remolding the tumor microenvironment by bacteria augments adoptive T cell therapy in advanced-stage solid tumors. Signal Transduct Target Ther. 2024;9:307. https://doi.org/10.1038/s41392-024-02028-3.
3. Newman MJ. Invention and characterization of a systemically administered, attenuated and killed bacteria-based multiple immune receptor agonist for anti-tumor immunotherapy. Front Immunol. 2024;15:1462221. https://doi.org/10.3389/fimmu.2024.1462221.
4. Chen Z, Hu T, Zhou J, Gu X, Chen S, Qi Q, Wang L. Overview of tumor immunotherapy based on approved drugs. Life Sci. 2024;340: 122419. https://doi.org/10.1016/j.lfs.2024.122419.
5. Bader JE, Voss K, Rathmell JC. Targeting Metabolism to Improve the Tumor Microenvironment for Cancer Immunotherapy. Mol Cell. 2020;78:1019–33. https://doi.org/10.1016/j.molcel.2020.05.034.
6. Bai R, Cui J. Development of Immunotherapy Strategies Targeting Tumor Microenvironment Is Fiercely Ongoing. Front Immunol. 2022;13: 890166. https://doi.org/10.3389/fimmu.2022.890166.
7. Ahn R, Cui Y, White FM. Antigen discovery for the development of cancer immunotherapy. Semin Immunol. 2023;66: 101733. https://doi.org/10.1016/j.smim.2023.101733.
8. Leko V, Rosenberg SA. Identifying and Targeting Human Tumor Antigens for T Cell-Based Immunotherapy of Solid Tumors. Cancer Cell. 2020;38:454–72. https://doi.org/10.1016/j.ccell.2020.07.013.
9. Morad G, Helmink BA, Sharma P, Wargo JA. Hallmarks of response, resistance, and toxicity to immune checkpoint blockade. Cell. 2021;184:5309–37. https://doi.org/10.1016/j.cell.2021.09.020.
10. Goodman RS, Jung S, Balko JM, Johnson DB. Biomarkers of immune checkpoint inhibitor response and toxicity: Challenges and opportunities. Immunol Rev. 2023;318:157–66. https://doi.org/10.1111/imr.13249.
11. Wang SJ, Dougan SK, Dougan M. Immune mechanisms of toxicity from checkpoint inhibitors. Trends Cancer. 2023;9:543–53. https://doi.org/10.1016/j.trecan.2023.04.002.
12. Yang R, Cui J. Advances and applications of RNA vaccines in tumor treatment. Mol Cancer. 2024;23:226. https://doi.org/10.1186/s12943-024-02141-5.
13. Li J, Xiao Z, Wang D, Jia L, Nie S, Zeng X, Hu W. The screening, identification, design and clinical application of tumor-specific neoantigens for TCR-T cells. Mol Cancer. 2023;22:141. https://doi.org/10.1186/s12943-023-01844-5.
14. Lu D, Chen Y, Jiang M, Wang J, Li Y, Ma K, Sun W, Zheng X, Qi J, Jin W, et al. KRAS G12V neoantigen specific T cell receptor for adoptive T cell therapy against tumors. Nat Commun. 2023;14:6389. https://doi.org/10.1038/s41467-023-42010-1.
15. Fan T, Zhang M, Yang J, Zhu Z, Cao W, Dong C. Therapeutic cancer vaccines: advancements, challenges, and prospects. Signal Transduct Target Ther. 2023;8:450. https://doi.org/10.1038/s41392-023-01674-3.
16. Budczies J, Kazdal D, Menzel M, Beck S, Kluck K, Altburger C, Schwab C, Allgauer M, Ahadova A, Kloor M, et al. Tumour mutational burden: clinical utility, challenges and emerging improvements. Nat Rev Clin Oncol. 2024;21:725–42. https://doi.org/10.1038/s41571-024-00932-9.
17. Keenan TE, Burke KP, Van Allen EM. Genomic correlates of response to immune checkpoint blockade. Nat Med. 2019;25:389–402. https://doi.org/10.1038/s41591-019-0382-x.
18. Yarchoan M, Hopkins A, Jaffee EM. Tumor Mutational Burden and Response Rate to PD-1 Inhibition. N Engl J Med. 2017;377:2500–1. https://doi.org/10.1056/NEJMc1713444.
19. Rizvi NA, Hellmann MD, Snyder A, Kvistborg P, Makarov V, Havel JJ, Lee W, Yuan J, Wong P, Ho TS, et al. Cancer immunology. Mutational landscape determines sensitivity to PD-1 blockade in non-small cell lung cancer. Science. 2015;348:124–8. https://doi.org/10.1126/science.aaa1348.
20. Marabelle A, Fakih M, Lopez J, Shah M, Shapira-Frommer R, Nakagawa K, Chung HC, Kindler HL, Lopez-Martin JA, Miller WH Jr, et al. Association of tumour mutational burden with outcomes in patients with advanced solid tumours treated with pembrolizumab: prospective biomarker analysis of the multicohort, open-label, phase 2 KEYNOTE-158 study. Lancet Oncol. 2020;21:1353–65. https://doi.org/10.1016/S1470-2045(20)30445-9.
21. Sholl LM, Hirsch FR, Hwang D, Botling J, Lopez-Rios F, Bubendorf L, Mino-Kenudson M, Roden AC, Beasley MB, Borczuk A, et al. The Promises and Challenges of Tumor Mutation Burden as an Immunotherapy Biomarker: A Perspective from the International Association for the Study of Lung Cancer Pathology Committee. J Thorac Oncol. 2020;15:1409–24. https://doi.org/10.1016/j.jtho.2020.05.019.
22. Huber F, Arnaud M, Stevenson BJ, Michaux J, Benedetti F, Thevenet J, Bobisse S, Chiffelle J, Gehert T, Muller M, et al. A comprehensive proteogenomic pipeline for neoantigen discovery to advance personalized cancer immunotherapy. Nat Biotechnol. 2024. https://doi.org/10.1038/s41587-024-02420-y.
23. Hu C, Zhao L, Liu W, Fan S, Liu J, Liu Y, Liu X, Shu L, Liu X, Liu P, et al. Genomic profiles and their associations with TMB, PD-L1 expression, and immune cell infiltration landscapes in synchronous multiple primary lung cancers. J Immunother Cancer. 2021;9. https://doi.org/10.1136/jitc-2021-003773.
24. Palmeri M, Mehnert J, Silk AW, Jabbour SK, Ganesan S, Popli P, Riedlinger G, Stephenson R, de Meritens AB, Leiser A, et al. Real-world application of tumor mutational burden-high (TMB-high) and microsatellite instability (MSI) confirms their utility as immunotherapy biomarkers. ESMO Open. 2022;7: 100336. https://doi.org/10.1016/j.esmoop.2021.100336.
25. Schumacher TN, Schreiber RD. Neoantigens in cancer immunotherapy. Science. 2015;348:69–74. https://doi.org/10.1126/science.aaa4971.
26. Zamora AE, Crawford JC, Thomas PG. Hitting the Target: How T Cells Detect and Eliminate Tumors. J Immunol. 2018;200:392–9. https://doi.org/10.4049/jimmunol.1701413.
27. Turajlic S, Litchfield K, Xu H, Rosenthal R, McGranahan N, Reading JL, Wong YNS, Rowan A, Kanu N, Al Bakir M, et al. Insertion-and-deletion-derived tumour-specific neoantigens and the immunogenic phenotype: a pan-cancer analysis. Lancet Oncol. 2017;18:1009–21. https://doi.org/10.1016/S1470-2045(17)30516-8.
28. Holder AM, Dedeilia A, Sierra-Davidson K, Cohen S, Liu D, Parikh A, Boland GM. Defining clinically useful biomarkers of immune checkpoint inhibitors in solid tumours. Nat Rev Cancer. 2024;24:498–512. https://doi.org/10.1038/s41568-024-00705-7.
29. Peng M, Mo Y, Wang Y, Wu P, Zhang Y, Xiong F, Guo C, Wu X, Li Y, Li X, et al. Neoantigen vaccine: an emerging tumor immunotherapy. Mol Cancer. 2019;18:128. https://doi.org/10.1186/s12943-019-1055-6.
30. Xie N, Shen G, Gao W, Huang Z, Huang C, Fu L. Neoantigens: promising targets for cancer therapy. Signal Transduct Target Ther. 2023;8:9. https://doi.org/10.1038/s41392-022-01270-x.
31. Vitale I, Sistigu A, Manic G, Rudqvist NP, Trajanoski Z, Galluzzi L. Mutational and Antigenic Landscape in Tumor Progression and Cancer Immunotherapy. Trends Cell Biol. 2019;29:396–416. https://doi.org/10.1016/j.tcb.2019.01.003.
32. Chan TA, Yarchoan M, Jaffee E, Swanton C, Quezada SA, Stenzinger A, Peters S. Development of tumor mutation burden as an immunotherapy biomarker: utility for the oncology clinic. Ann Oncol. 2019;30:44–56. https://doi.org/10.1093/annonc/mdy495.
33. Gubin MM, Artyomov MN, Mardis ER, Schreiber RD. Tumor neoantigens: building a framework for personalized cancer immunotherapy. J Clin Invest. 2015;125:3413–21. https://doi.org/10.1172/JCI80008.

34. Catalano I, Grassi E, Bertotti A, Trusolino L. Immunogenomics of Colorectal Tumors: Facts and Hypotheses on an Evolving Saga. Trends Cancer. 2019;5:779–88. https://doi.org/10.1016/j.trecan.2019.10.006.
35. Rojas LA, Sethna Z, Soares KC, Olcese C, Pang N, Patterson E, Lihm J, Ceglia N, Guasp P, Chu A, et al. Personalized RNA neoantigen vaccines stimulate T cells in pancreatic cancer. Nature. 2023;618:144–50. https://doi.org/10.1038/s41586-023-06063-y.
36. Wood MA, Weeder BR, David JK, Nellore A, Thompson RF. Burden of tumor mutations, neoepitopes, and other variants are weak predictors of cancer immunotherapy response and overall survival. Genome Med. 2020;12:33. https://doi.org/10.1186/s13073-020-00729-2.
37. Kamali MJ, Salehi M, Fath MK. Advancing personalized immunotherapy for melanoma: Integrating immunoinformatics in multi-epitope vaccine development, neoantigen identification via NGS, and immune simulation evaluation. Comput Biol Med. 2025;188: 109885. https://doi.org/10.1016/j.compbiomed.2025.109885.
38. Wagutu G, Gitau J, Mwangi K, Murithi M, Melly E, Harris AR, Sayed S, Ambs S, Makokha F. Whole exome-seq and RNA-seq data reveal unique neoantigen profiles in Kenyan breast cancer patients. Front Oncol. 2024;14:1444327. https://doi.org/10.3389/fonc.2024.1444327.
39. Hundal J, Carreno BM, Petti AA, Linette GP, Griffith OL, Mardis ER, Griffith M. pVAC-Seq: A genome-guided in silico approach to identifying tumor neoantigens. Genome Med. 2016;8:11. https://doi.org/10.1186/s13073-016-0264-5.
40. Reynisson B, Alvarez B, Paul S, Peters B, Nielsen M. NetMHCpan-4.1 and NetMHCIIpan-4.0: improved predictions of MHC antigen presentation by concurrent motif deconvolution and integration of MS MHC eluted ligand data. Nucleic Acids Res. 2020;48:W449–54. https://doi.org/10.1093/nar/gkaa379.
41. Zhou Z, Lyu X, Wu J, Yang X, Wu S, Zhou J, Gu X, Su Z, Chen S. TSNAD: an integrated software for cancer somatic mutation and tumour-specific neoantigen detection. R Soc Open Sci. 2017;4: 170050. https://doi.org/10.1098/rsos.170050.
42. Bais P, Namburi S, Gatti DM, Zhang X, Chuang JH. CloudNeo: a cloud pipeline for identifying patient-specific tumor neoantigens. Bioinformatics. 2017;33:3110–2. https://doi.org/10.1093/bioinformatics/btx375.
43. Tappeiner E, Finotello F, Charoentong P, Mayer C, Rieder D, Trajanoski Z. TIminer: NGS data mining pipeline for cancer immunology and immunotherapy. Bioinformatics. 2017;33:3140–1. https://doi.org/10.1093/bioinformatics/btx377.
44. Gfeller D, Schmidt J, Croce G, Guillaume P, Bobisse S, Genolet R, Queiroz L, Cesbron J, Racle J, Harari A. Improved predictions of antigen presentation and TCR recognition with MixMHCpred2.2 and PRIME2.0 reveal potent SARS-CoV-2 CD8(+) T-cell epitopes. Cell Syst. 2023;14:72–83. https://doi.org/10.1016/j.cels.2022.12.002.
45. Bulik-Sullivan B, Busby J, Palmer CD, Davis MJ, Murphy T, Clark A, Busby M, Duke F, Yang A, Young L, et al. Deep learning using tumor HLA peptide mass spectrometry datasets improves neoantigen identification. Nat Biotechnol. 2018. https://doi.org/10.1038/nbt.4313.
46. Pyke RM, Mellacheruvu D, Dea S, Abbott C, Zhang SV, Phillips NA, Harris J, Bartha G, Desai S, McClory R, et al. Precision Neoantigen Discovery Using Large-Scale Immunopeptidomes and Composite Modeling of MHC Peptide Presentation. Mol Cell Proteomics. 2023;22: 100506. https://doi.org/10.1016/j.mcpro.2023.100506.
47. Wang X, Wu T, Jiang Y, Chen T, Pan D, Jin Z, Xie J, Quan L, Lyu Q. RPEMHC: improved prediction of MHC-peptide binding affinity by a deep learning approach based on residue-residue pair encoding. Bioinformatics. 2024;40. https://doi.org/10.1093/bioinformatics/btad785.
48. Zhao Y, Yu J, Su Y, Shu Y, Ma E, Wang J, Jiang S, Wei C, Li D, Huang Z, et al. A unified deep framework for peptide–major histocompatibility complex–T cell receptor binding prediction. Nature Machine Intelligence. 2025;7:650–60. https://doi.org/10.1038/s42256-025-01002-0.
49. Peng X, Lei Y, Feng P, Jia L, Ma J, Zhao D, Zeng J. Characterizing the interaction conformation between T-cell receptors and epitopes with deep learning. Nature Machine Intelligence. 2023;5:395–407. https://doi.org/10.1038/s42256-023-00634-4.
50. Zhang J, Mardis ER, Maher CA. INTEGRATE-neo: a pipeline for personalized gene fusion neoantigen discovery. Bioinformatics. 2017;33:555–7. https://doi.org/10.1093/bioinformatics/btw674.
51. Vijg J, Dong X. Pathogenic Mechanisms of Somatic Mutation and Genome Mosaicism in Aging. Cell. 2020;182:12–23. https://doi.org/10.1016/j.cell.2020.06.024.
52. Li L, Goedegebuure P, Mardis ER, Ellis MJ, Zhang X, Herndon JM, Fleming TP, Carreno BM, Hansen TH, Gillanders WE. Cancer genome sequencing and its implications for personalized cancer vaccines. Cancers (Basel). 2011;3:4191–211. https://doi.org/10.3390/cancers3044191.
53. Li L, Goedegebuure SP, Gillanders WE. Preclinical and clinical development of neoantigen vaccines. Ann Oncol. 2017;28:xii11-xii17. https://doi.org/10.1093/annonc/mdx681.
54. Richard G, Princiotta MF, Bridon D, Martin WD, Steinberg GD, De Groot AS. Neoantigen-based personalized cancer vaccines: the emergence of precision cancer immunotherapy. Expert Rev Vaccines. 2022;21:173–84. https://doi.org/10.1080/14760584.2022.2012456.
55. Jurtz V, Paul S, Andreatta M, Marcatili P, Peters B, Nielsen M. NetMHCpan-4.0: Improved Peptide-MHC Class I Interaction Predictions Integrating Eluted Ligand and Peptide Binding Affinity Data. J Immunol. 2017;199:3360–3368. https://doi.org/10.4049/jimmunol.1700893.
56. Bassani-Sternberg M, Braunlein E, Klar R, Engleitner T, Sinitcyn P, Audehm S, Straub M, Weber J, Slotta-Huspenina J, Specht K, et al. Direct identification of clinically relevant neoepitopes presented on native human melanoma tissue by mass spectrometry. Nat Commun. 2016;7:13404. https://doi.org/10.1038/ncomms13404.
57. Chong C, Coukos G, Bassani-Sternberg M. Identification of tumor antigens with immunopeptidomics. Nat Biotechnol. 2022;40:175–88. https://doi.org/10.1038/s41587-021-01038-8.
58. Bennett HM, Stephenson W, Rose CM, Darmanis S. Single-cell proteomics enabled by next-generation sequencing or mass spectrometry. Nat Methods. 2023;20:363–74. https://doi.org/10.1038/s41592-023-01791-5.
59. Ade CM, Sporn MJ, Das S, Yu Z, Hanada KI, Qi YA, Maity T, Zhang X, Guha U, Andresson T, et al. Identification of neoepitope reactive T-cell receptors guided by HLA-A*03:01 and HLA-A*11:01 immunopeptidomics. J Immunother Cancer. 2023;11. https://doi.org/10.1136/jitc-2023-007097.
60. Zhang X, Qi Y, Zhang Q, Liu W. Application of mass spectrometry-based MHC immunopeptidome profiling in neoantigen identification for tumor immunotherapy. Biomed Pharmacother. 2019;120: 109542. https://doi.org/10.1016/j.biopha.2019.109542.
61. Sha D, Jin Z, Budczies J, Kluck K, Stenzinger A, Sinicrope FA. Tumor Mutational Burden as a Predictive Biomarker in Solid Tumors. Cancer Discov. 2020;10:1808–25. https://doi.org/10.1158/2159-8290.CD-20-0522.
62. Yang L, Ou YN, Wu BS, Liu WS, Deng YT, He XY, Chen YL, Kang J, Fei CJ, Zhu Y, et al. Large-scale whole-exome sequencing analyses identified protein-coding variants associated with immune-mediated diseases in 350,770 adults. Nat Commun. 2024;15:5924. https://doi.org/10.1038/s41467-024-49782-0.
63. Zhang W, Wang R, Fang H, Ma X, Li D, Liu T, Chen Z, Wang K, Hao S, Yu Z, et al. Influence of low tumor content on tumor mutational burden estimation by whole-exome sequencing and targeted panel sequencing. Clin Transl Med. 2021;11: e415. https://doi.org/10.1002/ctm2.415.
64. Krebs E, Weymann D, Ho C, Bosdet I, Laskin J, Lim HJ, Yip S, Karsan A, Hanna TP, Pollard S, et al. Real-world cost-effectiveness of multi-gene panel sequencing to inform therapeutic decisions for advanced non-small cell lung cancer: a population-based study. Lancet Reg Health Am. 2024;40: 100936. https://doi.org/10.1016/j.lana.2024.100936.
65. Marcus L, Fashoyin-Aje LA, Donoghue M, Yuan M, Rodriguez L, Gallagher PS, Philip R, Ghosh S, Theoret MR, Beaver JA, et al. FDA Approval Summary: Pembrolizumab for the Treatment of Tumor Mutational Burden-High Solid Tumors. Clin Cancer Res. 2021;27:4685–9. https://doi.org/10.1158/1078-0432.CCR-21-0327.
66. Pearlman AH, Hwang MS, Konig MF, Hsiue EH, Douglass J, DiNapoli SR, Mog BJ, Bettegowda C, Pardoll DM, Gabelli SB, et al. Targeting public neoantigens for cancer immunotherapy. Nat Cancer. 2021;2:487–97. https://doi.org/10.1038/s43018-021-00210-y.
67. McGranahan N, Furness AJ, Rosenthal R, Ramskov S, Lyngaa R, Saini SK, Jamal-Hanjani M, Wilson GA, Birkbak NJ, Hiley CT, et al. Clonal neoantigens elicit T cell immunoreactivity and sensitivity to immune checkpoint blockade. Science. 2016;351:1463–9. https://doi.org/10.1126/science.aaf1490.

68. Wang P, Chen Y, Wang C. Beyond Tumor Mutation Burden: Tumor Neoantigen Burden as a Biomarker for Immunotherapy and Other Types of Therapy. Front Oncol. 2021;11: 672677. https://doi.org/10.3389/fonc.2021.672677.
69. Garcia-Garijo A, Fajardo CA, Gros A. Determinants for Neoantigen Identification Front Immunol. 2019;10:1392. https://doi.org/10.3389/fimmu.2019.01392.
70. Wang Y, Shi T, Song X, Liu B, Wei J. Gene fusion neoantigens: Emerging targets for cancer immunotherapy. Cancer Lett. 2021;506:45–54. https://doi.org/10.1016/j.canlet.2021.02.023.
71. Rosenthal R, Cadieux EL, Salgado R, Bakir MA, Moore DA, Hiley CT, Lund T, Tanic M, Reading JL, Joshi K, et al. Neoantigen-directed immune escape in lung cancer evolution. Nature. 2019;567:479–85. https://doi.org/10.1038/s41586-019-1032-7.
72. Inderberg EM, Walchli S, Myhre MR, Trachsel S, Almasbak H, Kvalheim G, Gaudernack G. T cell therapy targeting a public neoantigen in microsatellite instable colon cancer reduces in vivo tumor growth. Oncoimmunology. 2017;6: e1302631. https://doi.org/10.1080/2162402X.2017.1302631.
73. Gubin MM, Zhang X, Schuster H, Caron E, Ward JP, Noguchi T, Ivanova Y, Hundal J, Arthur CD, Krebber WJ, et al. Checkpoint blockade cancer immunotherapy targets tumour-specific mutant antigens. Nature. 2014;515:577–81. https://doi.org/10.1038/nature13988.
74. Durgeau A, Virk Y, Corgnac S, Mami-Chouaib F. Recent Advances in Targeting CD8 T-Cell Immunity for More Effective Cancer Immunotherapy. Front Immunol. 2018;9:14. https://doi.org/10.3389/fimmu.2018.00014.
75. Lu YJ, Barreira-Silva P, Boyce S, Powers J, Cavallo K, Behar SM. CD4 T cell help prevents CD8 T cell exhaustion and promotes control of Mycobacterium tuberculosis infection. Cell Rep. 2021;36: 109696. https://doi.org/10.1016/j.celrep.2021.109696.
76. Farhood B, Najafi M, Mortezaee K. CD8(+) cytotoxic T lymphocytes in cancer immunotherapy: A review. J Cell Physiol. 2019;234:8509–21. https://doi.org/10.1002/jcp.27782.
77. Zhang S, Huang C, Li Y, Li Z, Zhu Y, Yang L, Hu H, Sun Q, Liu M, Cao S. Anti-cancer immune effect of human colorectal cancer neoantigen peptide based on MHC class I molecular affinity screening. Front Immunol. 2024;15:1473145. https://doi.org/10.3389/fimmu.2024.1473145.
78. Finnigan JP, Newman JH, Patskovsky Y, Patskovska L, Ishizuka AS, Lynn GM, Seder RA, Krogsgaard M, Bhardwaj N. Structural basis for self-discrimination by neoantigen-specific TCRs. Nat Commun. 2024;15:2140. https://doi.org/10.1038/s41467-024-46367-9.
79. Lauss M, Donia M, Harbst K, Andersen R, Mitra S, Rosengren F, Salim M, Vallon-Christersson J, Torngren T, Kvist A, et al. Mutational and putative neoantigen load predict clinical benefit of adoptive T cell therapy in melanoma. Nat Commun. 2017;8:1738. https://doi.org/10.1038/s41467-017-01460-0.
80. Holm JS, Funt SA, Borch A, Munk KK, Bjerregaard AM, Reading JL, Maher C, Regazzi A, Wong P, Al-Ahmadie H, et al. Neoantigen-specific CD8 T cell responses in the peripheral blood following PD-L1 blockade might predict therapy outcome in metastatic urothelial carcinoma. Nat Commun. 1935;2022:13. https://doi.org/10.1038/s41467-022-29342-0.
81. Veatch JR, Singhi N, Srivastava S, Szeto JL, Jesernig B, Stull SM, Fitzgibbon M, Sarvothama M. Yechan-Gunja, S.; James, S.E.; et al. A therapeutic cancer vaccine delivers antigens and adjuvants to lymphoid tissues using genetically modified T cells. J Clin Invest. 2021;131. https://doi.org/10.1172/JCI144195.
82. Jin P, Shen J, Zhao M, Yu J, Jin W, Jiang G, Li Z, He M, Liu X, Wu S, et al. Driver mutation landscape of acute myeloid leukemia provides insights for neoantigen-based immunotherapy. Cancer Lett. 2024;611: 217427. https://doi.org/10.1016/j.canlet.2024.217427.
83. Yang A, Zhou M, Gao Y, Zhang Y. Mechanisms of CD8(+) T cell exhaustion and its clinical significance in prognosis of anti-tumor therapies: A review. Int Immunopharmacol. 2025;159: 114843. https://doi.org/10.1016/j.intimp.2025.114843.
84. Zhong T, Sun S, Zhao M, Zhang B, Xiong H. The mechanisms and clinical significance of CD8(+) T cell exhaustion in anti-tumor immunity. Cancer Biol Med. 2025;22. https://doi.org/10.20892/j.issn.2095-3941.2024.0628.
85. Franco F, Jaccard A, Romero P, Yu YR, Ho PC. Metabolic and epigenetic regulation of T-cell exhaustion. Nat Metab. 2020;2:1001–12. https://doi.org/10.1038/s42255-020-00280-9.
86. Vadakekolathu J, Rutella S. Escape from T-cell-targeting immunotherapies in acute myeloid leukemia. Blood. 2024;143:2689–700. https://doi.org/10.1182/blood.2023019961.
87. Nair R, Somasundaram V, Kuriakose A, Krishn SR, Raben D, Salazar R, Nair P. Deciphering T-cell exhaustion in the tumor microenvironment: paving the way for innovative solid tumor therapies. Front Immunol. 2025;16:1548234. https://doi.org/10.3389/fimmu.2025.1548234.
88. Wei SC, Duffy CR, Allison JP. Fundamental Mechanisms of Immune Checkpoint Blockade Therapy. Cancer Discov. 2018;8:1069–86. https://doi.org/10.1158/2159-8290.CD-18-0367.
89. Li G, Wu T, Li H, Wei C, Sun Y, Gao P, Huang X, Liu Z, Li J, Wang Y, et al. Construction of a tumor mutational burden-derived LncRNA prognostic computational framework associated with therapy sensitivity in skin cutaneous melanoma. J Transl Med. 2024;22:966. https://doi.org/10.1186/s12967-024-05732-4.
90. Kang S, Mansurov A, Kurtanich T, Chun HR, Slezak AJ, Volpatti LR, Chang K, Wang T, Alpar A,; Refvik KC, et al. Engineered IL-7 synergizes with IL-12 immunotherapy to prevent T cell exhaustion and promote memory without exacerbating toxicity. Sci Adv. 2023;9:eadh9879. https://doi.org/10.1126/sciadv.adh9879.
91. Christodoulou I, Ho WJ, Marple A, Ravich JW, Tam A, Rahnama R, Fearnow A, Rietberg C, Yanik S, Solomou EE, et al. Engineering CAR-NK cells to secrete IL-15 sustains their anti-AML functionality but is associated with systemic toxicities. J Immunother Cancer. 2021;9. https://doi.org/10.1136/jitc-2021-003894.
92. Merino A, Brunstein CC, Shanley R, Rashid F, Wangen R, Bachanova V, Juckett M, Maakaron J, Felices M, Weisdorf D, et al. N-803, an IL-15 Superagonist Complex as Maintenance Therapy After Allogeneic Donor Stem Cell Transplant for Acute Myeloid Leukemia or Myelodysplastic Syndrome; A Phase 2 Trial. Transplant Cell Ther. 2024;30:1206 e1201–1206 e1212. https://doi.org/10.1016/j.jtct.2024.09.023.
93. Xu T, Fang Y, Gu Y, Xu D, Hu T, Yu T, Xu YY, Shen HY, Ma P, Shu Y. HDAC inhibitor SAHA enhances antitumor immunity via the HDAC1/JAK1/FGL1 axis in lung adenocarcinoma. J Immunother Cancer. 2024;12. https://doi.org/10.1136/jitc-2024-010077.
94. Salmon JM, Todorovski I, Stanley KL, Bruedigam C, Kearney CJ, Martelotto LG, Rossello F, Semple T, Arnau GM, Zethoven M, et al. Epigenetic Activation of Plasmacytoid DCs Drives IFNAR-Dependent Therapeutic Differentiation of AML. Cancer Discov. 2022;12:1560–79. https://doi.org/10.1158/2159-8290.CD-20-1145.
95. Goyal A, Bauer J, Hey J, Papageorgiou DN, Stepanova E, Daskalakis M, Scheid J, Dubbelaar M, Klimovich B, Schwarz D, et al. DNMT and HDAC inhibition induces immunogenic neoantigens from human endogenous retroviral element-derived transcripts. Nat Commun. 2023;14:6731. https://doi.org/10.1038/s41467-023-42417-w.
96. Mai Z, Kongjia L, Wang X, Xie X, Pang L, Yang H, Wen J, Fu J. Impaired TGF-beta signaling via AHNAK family mutations elicits an esophageal cancer subtype with sensitivities to genotoxic therapy and immunotherapy. Cancer Immunol Immunother. 2024;73:225. https://doi.org/10.1007/s00262-024-03798-z.
97. Han Y, Dong C, Hu M, Wang X, Wang G. Unlocking the adenosine receptor mechanism of the tumour immune microenvironment. Front Immunol. 2024;15:1434118. https://doi.org/10.3389/fimmu.2024.1434118.
98. McLaughlin M, Patin EC, Pedersen M, Wilkins A, Dillon MT, Melcher AA, Harrington KJ. Inflammatory microenvironment remodelling by tumour cells after radiotherapy. Nat Rev Cancer. 2020;20:203–17. https://doi.org/10.1038/s41568-020-0246-1.
99. Ngwa W, Irabor OC, Schoenfeld JD, Hesser J, Demaria S, Formenti SC. Using immunotherapy to boost the abscopal effect. Nat Rev Cancer. 2018;18:313–22. https://doi.org/10.1038/nrc.2018.6.
100. Wang G, Kang X, Chen KS, Jehng T, Jones L, Chen J, Huang XF, Chen SY. An engineered oncolytic virus expressing PD-L1 inhibitors activates tumor neoantigen-specific T cell responses. Nat Commun. 2020;11:1395. https://doi.org/10.1038/s41467-020-15229-5.
101. Ye K, Li F, Wang R, Cen T, Liu S, Zhao Z, Li R, Xu L, Zhang G, Xu Z, et al. An armed oncolytic virus enhances the efficacy of tumor-infiltrating lymphocyte therapy by converting tumors to artificial antigen-presenting cells in situ. Mol Ther. 2022;30:3658–76. https://doi.org/10.1016/j.ymthe.2022.06.010.

102. Gogenur M, Balsevicius L, Bulut M, Colak N, Justesen TF, Fiehn AK, Jensen MB, Host-Rasmussen K, Cappelen B, Gaggar S, et al. Neoadjuvant intratumoral influenza vaccine treatment in patients with proficient mismatch repair colorectal cancer leads to increased tumor infiltration of CD8+ T cells and upregulation of PD-L1: a phase 1/2 clinical trial. J Immunother Cancer. 2023;11. https://doi.org/10.1136/jitc-2023-006774.
103. Feola S, Hamdan F, Russo S, Chiaro J, Fusciello M, Feodoroff M, Antignani G, D'Alessio F, Molsa R, Stigzelius V, et al. Novel peptide-based oncolytic vaccine for enhancement of adaptive antitumor immune response via co-engagement of innate Fcgamma and Fcalpha receptors. J Immunother Cancer. 2024;12. https://doi.org/10.1136/jitc-2023-008342.
104. Hu H, Tang L, Zhao Y, Cheng J, Huang M, You Y, Zou P, Lei Q, Zhu X, Guo AY. Single-cell analysis of the survival mechanisms of fratricidal CAR-T targeting of T cell malignancies. Mol Ther Nucleic Acids. 2024;35:102225. https://doi.org/10.1016/j.omtn.2024.102225.
105. Jiang T, Jin H, Ji X, Zheng X, Xu CX, Zhang PJ. Drivers of centrosome abnormalities: Senescence progression and tumor immune escape. Semin Cancer Biol. 2025;110:56–64. https://doi.org/10.1016/j.semcancer.2025.01.008.
106. Ranjan K, Rajendran BK, Deen IU, Costantini A, de Rodas ML, Desai SS, Scallo F, Gianino N, Ferrone S, Schalper KA. IL-4 mediated TAP2 downregulation is a dominant and reversible mechanism of immune evasion and immunotherapy resistance in non-small cell lung cancer. Mol Cancer. 2025;24:80. https://doi.org/10.1186/s12943-025-02276-z.
107. Mosca L, de Angelis A, Ronchi A, De Chiara A, Fazioli F, Ruosi C, Altucci L, Conte M, de Nigris F. Sarcoma Common MHC-I Haplotype Restricts Tumor-Specific CD8+ T Cell Response. Cancers (Basel). 2022;14. https://doi.org/10.3390/cancers14143414.
108. Yao L, Zhou L, Xuan Y, Zhang P, Wang X, Wang T, Meng T, Xue Y, Ma X, Shah AS, et al. The proteasome activator REGgamma counteracts immunoproteasome expression and autoimmunity. J Autoimmun. 2019;103: 102282. https://doi.org/10.1016/j.jaut.2019.05.010.
109. Overall SA, Toor JS, Hao S, Yarmarkovich M, Sara MOR, Morozov GI, Nguyen S, Japp AS, Gonzalez N, Moschidi D, et al. High throughput pMHC-I tetramer library production using chaperone-mediated peptide exchange. Nat Commun. 1909;2020:11. https://doi.org/10.1038/s41467-020-15710-1.
110. Mishra B, Ivashkiv LB. Interferons and epigenetic mechanisms in training, priming and tolerance of monocytes and hematopoietic progenitors. Immunol Rev. 2024;323:257–75. https://doi.org/10.1111/imr.13330.
111. Russell MA, Richardson SJ, Morgan NG. The role of the interferon/JAK-STAT axis in driving islet HLA-I hyperexpression in type 1 diabetes. Front Endocrinol (Lausanne). 2023;14:1270325. https://doi.org/10.3389/fendo.2023.1270325.
112. Song TY, Long M, Zhao HX, Zou MW, Fan HJ, Liu Y, Geng CL, Song MF, Liu YF, Chen JY, et al. Tumor evolution selectively inactivates the core microRNA machinery for immune evasion. Nat Commun. 2021;12:7003. https://doi.org/10.1038/s41467-021-27331-3.
113. Wang S, Sun Z, Wang C, Zhang A, Zhang C, Hou S, Lin N, Li Q. The JAK1/STAT3 pathway mediates the effects of SERPINH1 on glioma EMT. Int Immunopharmacol. 2025;157: 114731. https://doi.org/10.1016/j.intimp.2025.114731.
114. Yang W, Lee KW, Srivastava RM, Kuo F, Krishna C, Chowell D, Makarov V, Hoen D, Dalin MG, Wexler L, et al. Immunogenic neoantigens derived from gene fusions stimulate T cell responses. Nat Med. 2019;25:767–75. https://doi.org/10.1038/s41591-019-0434-2.
115. Gao B, Wang Y, Zhang X, Jiang H, Han F, Li C, Lu S. Identification and validation of inflammatory subtypes in intrahepatic cholangiocellular carcinoma. J Transl Med. 2024;22:730. https://doi.org/10.1186/s12967-024-05529-5.
116. Van Allen EM, Miao D, Schilling B, Shukla SA, Blank C, Zimmer L, Sucker A, Hillen U, Foppen MHG, Goldinger SM, et al. Genomic correlates of response to CTLA-4 blockade in metastatic melanoma. Science. 2015;350:207–11. https://doi.org/10.1126/science.aad0095.
117. Le DT, Durham JN, Smith KN, Wang H, Bartlett BR, Aulakh LK, Lu S, Kemberling H, Wilt C, Luber BS, et al. Mismatch repair deficiency predicts response of solid tumors to PD-1 blockade. Science. 2017;357:409–13. https://doi.org/10.1126/science.aan6733.
118. Ott PA, Hu Z, Keskin DB, Shukla SA, Sun J, Bozym DJ, Zhang W, Luoma A, Giobbie-Hurder A, Peter L, et al. An immunogenic personal neoantigen vaccine for patients with melanoma. Nature. 2017;547:217–21. https://doi.org/10.1038/nature22991.
119. Ott PA, Hu-Lieskovan S, Chmielowski B, Govindan R, Naing A, Bhardwaj N, Margolin K, Awad MM, Hellmann MD, Lin JJ, et al. A Phase Ib Trial of Personalized Neoantigen Therapy Plus Anti-PD-1 in Patients with Advanced Melanoma, Non-small Cell Lung Cancer, or Bladder Cancer. Cell. 2020;183:347–362 e324. https://doi.org/10.1016/j.cell.2020.08.053.
120. Lhuillier C, Rudqvist NP, Yamazaki T, Zhang T, Charpentier M, Galluzzi L, Dephoure N, Clement CC, Santambrogio L, Zhou XK, et al. Radiotherapy-exposed CD8+ and CD4+ neoantigens enhance tumor control. J Clin Invest. 2021;131. https://doi.org/10.1172/JCI138740.
121. Thibaudin M, Fumet JD, Chibaudel B, Bennouna J, Borg C, Martin-Babau J, Cohen R, Fonck M, Taieb J, Limagne E, et al. First-line durvalumab and tremelimumab with chemotherapy in RAS-mutated metastatic colorectal cancer: a phase 1b/2 trial. Nat Med. 2023;29:2087–98. https://doi.org/10.1038/s41591-023-02497-z.
122. Ji D, Yi H, Zhang D, Zhan T, Li Z, Li M, Jia J, Qiao M, Xia J, Zhai Z, et al. Somatic Mutations and Immune Alternation in Rectal Cancer Following Neoadjuvant Chemoradiotherapy. Cancer Immunol Res. 2018;6:1401–16. https://doi.org/10.1158/2326-6066.CIR-17-0630.
123. Wan JCM, Sasieni P, Rosenfeld N. Promises and pitfalls of multi-cancer early detection using liquid biopsy tests. Nat Rev Clin Oncol. 2025. https://doi.org/10.1038/s41571-025-01033-x.
124. Jia Q, Chiu L, Wu S, Bai J, Peng L, Zheng L, Zang R, Li X, Yuan B, Gao Y, et al. Tracking Neoantigens by Personalized Circulating Tumor DNA Sequencing during Checkpoint Blockade Immunotherapy in Non-Small Cell Lung Cancer. Adv Sci (Weinh). 2020;7:1903410. https://doi.org/10.1002/advs.201903410.
125. Zhang Z, Lu M, Qin Y, Gao W, Tao L, Su W, Zhong J. Neoantigen: A New Breakthrough in Tumor Immunotherapy. Front Immunol. 2021;12:672356. https://doi.org/10.3389/fimmu.2021.672356.
126. Chen F, Zou Z, Du J, Su S, Shao J, Meng F, Yang J, Xu Q, Ding N, Yang Y, et al. Neoantigen identification strategies enable personalized immunotherapy in refractory solid tumors. J Clin Invest. 2019;129:2056–70. https://doi.org/10.1172/JCI99538.
127. Li L, Zhang X, Wang X, Kim SW, Herndon JM, Becker-Hapak MK, Carreno BM, Myers NB, Sturmoski MA, McLellan MD, et al. Optimized polyepitope neoantigen DNA vaccines elicit neoantigen-specific immune responses in preclinical models and in clinical translation. Genome Med. 2021;13:56. https://doi.org/10.1186/s13073-021-00872-4.
128. Chiapperino L, Graber N, Panese F. A precision immuno-oncology turn? Hybridizing cancer genomics and immunotherapy through neoantigens-based adoptive cell therapies. Soc Stud Sci. 2024:3063127241303720. https://doi.org/10.1177/03063127241303720.
129. Alban TJ, Riaz N, Parthasarathy P, Makarov V, Kendall S, Yoo SK, Shah R, Weinhold N, Srivastava R, Ma X, et al. Neoantigen immunogenicity landscapes and evolution of tumor ecosystems during immunotherapy with nivolumab. Nat Med. 2024;30:3209–22. https://doi.org/10.1038/s41591-024-03240-y.
130. Qin Y, Huo M, Liu X, Li SC. Biomarkers and computational models for predicting efficacy to tumor ICI immunotherapy. Front Immunol. 2024;15:1368749. https://doi.org/10.3389/fimmu.2024.1368749.
131. Ma R, Li Z, Chiocca EA, Caligiuri MA, Yu J. The emerging field of oncolytic virus-based cancer immunotherapy. Trends Cancer. 2023;9:122–39. https://doi.org/10.1016/j.trecan.2022.10.003.
132. Liu Z, Lv J, Dang Q, Liu L, Weng S, Wang L, Zhou Z, Kong Y, Li H, Han Y, et al. Engineering neoantigen vaccines to improve cancer personalized immunotherapy. Int J Biol Sci. 2022;18:5607–23. https://doi.org/10.7150/ijbs.76281.
133. Reparaz D, Aparicio B, Llopiz D, Hervas-Stubbs S, Sarobe P. Therapeutic Vaccines against Hepatocellular Carcinoma in the Immune Checkpoint Inhibitor Era: Time for Neoantigens? Int J Mol Sci. 2022;23. https://doi.org/10.3390/ijms23042022.
134. Weber JS, Carlino MS, Khattak A, Meniawy T, Ansstas G, Taylor MH, Kim KB, McKean M, Long GV, Sullivan RJ, et al. Individualised neoantigen therapy mRNA-4157 (V940) plus pembrolizumab versus pembrolizumab monotherapy in resected melanoma (KEYNOTE-942): a randomised, phase 2b study. Lancet. 2024;403:632–44. https://doi.org/10.1016/S0140-6736(23)02268-7.

135. Gainor JF, Patel MR, Weber JS, Gutierrez M, Bauman JE, Clarke JM, Julian R, Scott AJ, Geiger JL, Kirtane K, et al. T-cell Responses to Individualized Neoantigen Therapy mRNA-4157 (V940) Alone or in Combination with Pembrolizumab in the Phase 1 KEYNOTE-603 Study. Cancer Discov. 2024;14:2209–23. https://doi.org/10.1158/2159-8290.CD-24-0158.
136. Gustafsson C, Govindarajan S, Minshull J. Codon bias and heterologous protein expression. Trends Biotechnol. 2004;22:346–53. https://doi.org/10.1016/j.tibtech.2004.04.006.
137. Thess A, Grund S, Mui BL, Hope MJ, Baumhof P, Fotin-Mleczek M, Schlake T. Sequence-engineered mRNA Without Chemical Nucleoside Modifications Enables an Effective Protein Therapy in Large Animals. Mol Ther. 2015;23:1456–64. https://doi.org/10.1038/mt.2015.103.
138. Guasp P, Reiche C, Sethna Z, Balachandran VP. RNA vaccines for cancer: Principles to practice. Cancer Cell. 2024;42:1163–84. https://doi.org/10.1016/j.ccell.2024.05.005.
139. Alspach E, Lussier DM, Miceli AP, Kizhvatov I, DuPage M, Luoma AM, Meng W, Lichti CF, Esaulova E, Vomund AN, et al. MHC-II neoantigens shape tumour immunity and response to immunotherapy. Nature. 2019;574:696–701. https://doi.org/10.1038/s41586-019-1671-8.
140. Yang X, Fan J, Wu Y, Ma Z, Huang J, Zhang Y, Zhou Z, Mo F, Liu X, Yuan H, et al. Synthetic multiepitope neoantigen DNA vaccine for personalized cancer immunotherapy. Nanomedicine. 2021;37: 102443. https://doi.org/10.1016/j.nano.2021.102443.
141. Hung SI, Chu MT, Hou MM, Lee YS, Yang CK, Chu SY, Liu FY, Hsu HC, Pao SC, Teng YC, et al. Personalized neoantigen-based T cell therapy triggers cytotoxic lymphocytes expressing polyclonal TCR against metastatic ovarian cancer. Biomed Pharmacother. 2023;169: 115928. https://doi.org/10.1016/j.biopha.2023.115928.
142. Zhang X, Xu Z, Dai X, Zhang X, Wang X. Research progress of neoantigen-based dendritic cell vaccines in pancreatic cancer. Front Immunol. 2023;14:1104860. https://doi.org/10.3389/fimmu.2023.1104860.
143. Kembuan GJ, Kim JY, Maus MV, Jan M. Targeting solid tumor antigens with chimeric receptors: cancer biology meets synthetic immunology. Trends Cancer. 2024;10:312–31. https://doi.org/10.1016/j.trecan.2024.01.003.
144. Hou AJ, Chen LC, Chen YY. Navigating CAR-T cells through the solid-tumour microenvironment. Nat Rev Drug Discov. 2021;20:531–50. https://doi.org/10.1038/s41573-021-00189-2.
145. Flugel CL, Majzner RG, Krenciute G, Dotti G, Riddell SR, Wagner DL, Abou-El-Enein M. Overcoming on-target, off-tumour toxicity of CAR T cell therapy for solid tumours. Nat Rev Clin Oncol. 2023;20:49–62. https://doi.org/10.1038/s41571-022-00704-3.
146. Cho JH, Okuma A, Sofjan K, Lee S, Collins JJ, Wong WW. Engineering advanced logic and distributed computing in human CAR immune cells. Nat Commun. 2021;12:792. https://doi.org/10.1038/s41467-021-21078-7.
147. An Z, Aksoy O, Zheng T, Fan QW, Weiss WA. Epidermal growth factor receptor and EGFRvIII in glioblastoma: signaling pathways and targeted therapies. Oncogene. 2018;37:1561–75. https://doi.org/10.1038/s41388-017-0045-7.
148. Eskilsson E, Rosland GV, Talasila KM, Knappskog S, Keunen O, Sottoriva A, Foerster S, Solecki G, Taxt T, Jirik R, et al. EGFRvIII mutations can emerge as late and heterogenous events in glioblastoma development and promote angiogenesis through Src activation. Neuro Oncol. 2016;18:1644–55. https://doi.org/10.1093/neuonc/now113.
149. O'Rourke DM, Nasrallah MP, Desai A, Melenhorst JJ, Mansfield K, Morrissette JJD, Martinez-Lage M, Brem S, Maloney E, Shen A, et al. A single dose of peripherally infused EGFRvIII-directed CAR T cells mediates antigen loss and induces adaptive resistance in patients with recurrent glioblastoma. Sci Transl Med. 2017;9. https://doi.org/10.1126/scitranslmed.aaa0984.
150. Choe JH, Watchmaker PB, Simic MS, Gilbert RD, Li AW, Krasnow NA, Downey KM, Yu W, Carrera DA, Celli A, et al. SynNotch-CAR T cells overcome challenges of specificity, heterogeneity, and persistence in treating glioblastoma. Sci Transl Med. 2021;13. https://doi.org/10.1126/scitranslmed.abe7378.
151. Hyrenius-Wittsten A, Su Y, Park M, Garcia JM, Alavi J, Perry N, Montgomery G, Liu B, Roybal KT. SynNotch CAR circuits enhance solid tumor recognition and promote persistent antitumor activity in mouse models. Sci Transl Med. 2021;13. https://doi.org/10.1126/scitranslmed.abd8836.
152. Agrawal B, Krantz MJ, Reddish MA, Longenecker BM. Cancer-associated MUC1 mucin inhibits human T-cell proliferation, which is reversible by IL-2. Nat Med. 1998;4:43–9. https://doi.org/10.1038/nm0198-043.
153. Schulz G, Cheresh DA, Varki NM, Yu A, Staffileno LK, Reisfeld RA. Detection of ganglioside GD2 in tumor tissues and sera of neuroblastoma patients. Cancer Res. 1984;44:5914–20.
154. Del Bufalo F, De Angelis B, Caruana I, Del Baldo G, De Ioris MA, Serra A, Mastronuzzi A, Cefalo MG, Pagliara D, Amicucci M, et al. GD2-CART01 for Relapsed or Refractory High-Risk Neuroblastoma. N Engl J Med. 2023;388:1284–95. https://doi.org/10.1056/NEJMoa2210859.
155. Majzner RG, Heitzeneder S, Mackall CL. Harnessing the Immunotherapy Revolution for the Treatment of Childhood Cancers. Cancer Cell. 2017;31:476–85. https://doi.org/10.1016/j.ccell.2017.03.002.
156. Rappaport AR, Kyi C, Lane M, Hart MG, Johnson ML, Henick BS, Liao CY, Mahipal A, Shergill A, Spira AI, et al. A shared neoantigen vaccine combined with immune checkpoint blockade for advanced metastatic solid tumors: phase 1 trial interim results. Nat Med. 2024;30:1013–22. https://doi.org/10.1038/s41591-024-02851-9.
157. Yarchoan M, Gane EJ, Marron TU, Perales-Linares R, Yan J, Cooch N, Shu DH, Fertig EJ, Kagohara LT, Bartha G, et al. Personalized neoantigen vaccine and pembrolizumab in advanced hepatocellular carcinoma: a phase 1/2 trial. Nat Med. 2024;30:1044–53. https://doi.org/10.1038/s41591-024-02894-y.
158. Kelly RJ, Landon BV, Zaidi AH, Singh D, Canzoniero JV, Balan A, Hales RK, Voong KR, Battafarano RJ, Jobe BA, et al. Neoadjuvant nivolumab or nivolumab plus LAG-3 inhibitor relatlimab in resectable esophageal/gastroesophageal junction cancer: a phase Ib trial and ctDNA analyses. Nat Med. 2024;30:1023–34. https://doi.org/10.1038/s41591-024-02877-z.
159. Lopez J, Powles T, Braiteh F, Siu LL, LoRusso P, Friedman CF, Balmanoukian AS, Gordon M, Yachnin J, Rottey S, et al. Autogene cevumeran with or without atezolizumab in advanced solid tumors: a phase 1 trial. Nat Med. 2025;31:152–64. https://doi.org/10.1038/s41591-024-03334-7.
160. Poh A. First Oncolytic Viral Therapy for Melanoma. Cancer Discov. 2016;6:6. https://doi.org/10.1158/2159-8290.CD-NB2015-158.
161. Todo T, Ito H, Ino Y, Ohtsu H, Ota Y, Shibahara J, Tanaka M. Intratumoral oncolytic herpes virus G47Δ for residual or recurrent glioblastoma: a phase 2 trial. Nat Med. 2022;28:1630–9. https://doi.org/10.1038/s41591-022-01897-x.
162. Ribas A, Dummer R, Puzanov I, VanderWalde A, Andtbacka RHI, Michielin O, Olszanski AJ, Malvehy J, Cebon J, Fernandez E, et al. Oncolytic Virotherapy Promotes Intratumoral T Cell Infiltration and Improves Anti-PD-1 Immunotherapy. Cell. 2017;170:1109–1119 e1110. https://doi.org/10.1016/j.cell.2017.08.027.
163. Park SH, Breitbach CJ, Lee J, Park JO, Lim HY, Kang WK, Moon A, Mun JH, Sommermann EM, Maruri Avidal L, et al. Phase 1b Trial of Biweekly Intravenous Pexa-Vec (JX-594), an Oncolytic and Immunotherapeutic Vaccinia Virus in Colorectal Cancer. Mol Ther. 2015;23:1532–40. https://doi.org/10.1038/mt.2015.109.
164. Breitbach CJ, Burke J, Jonker D, Stephenson J, Haas AR, Chow LQ, Nieva J, Hwang TH, Moon A, Patt R, et al. Intravenous delivery of a multimechanistic cancer-targeted oncolytic poxvirus in humans. Nature. 2011;477:99–102. https://doi.org/10.1038/nature10358.
165. Gallego Perez-Larraya J, Garcia-Moure M, Labiano S, Patino-Garcia A, Dobbs J, Gonzalez-Huarriz M, Zalacain M, Marrodan L, Martinez-Velez N, Puigdelloses M, et al. Oncolytic DNX-2401 Virus for Pediatric Diffuse Intrinsic Pontine Glioma. N Engl J Med. 2022;386:2471–81. https://doi.org/10.1056/NEJMoa2202028.
166. Hellums EK, Markert JM, Parker JN, He B, Perbal B, Roizman B, Whitley RJ, Langford CP, Bharara S, Gillespie GY. Increased efficacy of an interleukin-12-secreting herpes simplex virus in a syngeneic intracranial murine glioma model. Neuro Oncol. 2005;7:213–24. https://doi.org/10.1215/S1152851705000074.
167. Kowalsky SJ, Liu Z, Feist M, Berkey SE, Ma C, Ravindranathan R, Dai E, Roy EJ, Guo ZS, Bartlett DL. Superagonist IL-15-Armed Oncolytic Virus Elicits Potent Antitumor Immunity and Therapy That Are Enhanced with PD-1 Blockade. Mol Ther. 2018;26:2476–86. https://doi.org/10.1016/j.ymthe.2018.07.013.
168. Melcher A, Harrington K, Vile R. Oncolytic virotherapy as immunotherapy. Science. 2021;374:1325–6. https://doi.org/10.1126/science.abk3436.

169. Jansen AM, van Wezel T, van den Akker BE, Ventayol Garcia M, Ruano D, Tops CM, Wagner A, Letteboer TG, Gomez-Garcia EB, Devilee P, et al. Combined mismatch repair and POLE/POLD1 defects explain unresolved suspected Lynch syndrome cancers. Eur J Hum Genet. 2016;24:1089–92. https://doi.org/10.1038/ejhg.2015.252.
170. Guan J, Lu C, Jin Q, Lu H, Chen X, Tian L, Zhang Y, Ortega J, Zhang J, Siteni S, et al. MLH1 Deficiency-Triggered DNA Hyperexcision by Exonuclease 1 Activates the cGAS-STING Pathway. Cancer Cell. 2021;39:109–121 e105. https://doi.org/10.1016/j.ccell.2020.11.004.
171. Devlin JR, Alonso JA, Ayres CM, Keller GLJ, Bobisse S, Vander Kooi CW, Coukos G, Gfeller D, Harari A, Baker BM. Structural dissimilarity from self drives neoepitope escape from immune tolerance. Nat Chem Biol. 2020;16:1269–1276. https://doi.org/10.1038/s41589-020-0610-1.
172. Claeys A, Van den Eynden J. MHC class II genotypes are independent predictors of anti-PD1 immunotherapy response in melanoma. Commun Med (Lond). 2024;4:184. https://doi.org/10.1038/s43856-024-00612-w.
173. Rooney MS, Shukla SA, Wu CJ, Getz G, Hacohen N. Molecular and genetic properties of tumors associated with local immune cytolytic activity. Cell. 2015;160:48–61. https://doi.org/10.1016/j.cell.2014.12.033.
174. Liu T, Tan J, Wu M, Fan W, Wei J, Zhu B, Guo J, Wang S, Zhou P, Zhang H, et al. High-affinity neoantigens correlate with better prognosis and trigger potent antihepatocellular carcinoma (HCC) activity by activating CD39(+)CD8(+) T cells. Gut. 2021;70:1965–77. https://doi.org/10.1136/gutjnl-2020-322196.
175. Sade-Feldman M, Jiao YJ, Chen JH, Rooney MS, Barzily-Rokni M, Eliane JP, Bjorgaard SL, Hammond MR, Vitzthum H, Blackmon SM, et al. Resistance to checkpoint blockade therapy through inactivation of antigen presentation. Nat Commun. 2017;8:1136. https://doi.org/10.1038/s41467-017-01062-w.
176. Chen PL, Roh W, Reuben A, Cooper ZA, Spencer CN, Prieto PA, Miller JP, Bassett RL, Gopalakrishnan V, Wani K, et al. Analysis of Immune Signatures in Longitudinal Tumor Samples Yields Insight into Biomarkers of Response and Mechanisms of Resistance to Immune Checkpoint Blockade. Cancer Discov. 2016;6:827–37. https://doi.org/10.1158/2159-8290.CD-15-1545.
177. Hu X, Deng X, Xie J, Tang H, Zou Y. Heterogeneous PD-L1 expression in metastases impacts immunotherapy response. EBioMedicine. 2023;97:104816. https://doi.org/10.1016/j.ebiom.2023.104816.
178. Luksza M, Sethna ZM, Rojas LA, Lihm J, Bravi B, Elhanati Y, Soares K, Amisaki M, Dobrin A, Hoyos D, et al. Neoantigen quality predicts immunoediting in survivors of pancreatic cancer. Nature. 2022;606:389–95. https://doi.org/10.1038/s41586-022-04735-9.
179. Liang H, Huang J, Ao X, Guo W, Chen Y, Lu D, Lv Z, Tan X, He W, Jiang M, et al. TMB and TCR Are Correlated Indicators Predictive of the Efficacy of Neoadjuvant Chemotherapy in Breast Cancer. Front Oncol. 2021;11:740427. https://doi.org/10.3389/fonc.2021.740427.
180. Burbach BJ, O'Flanagan SD, Shao Q, Young KM, Slaughter JR, Rollins MR, Street TJL, Granger VE, Beura LK, Azarin SM, et al. Irreversible electroporation augments checkpoint immunotherapy in prostate cancer and promotes tumor antigen-specific tissue-resident memory CD8+ T cells. Nat Commun. 2021;12:3862. https://doi.org/10.1038/s41467-021-24132-6.
181. Ichimiya S, Fujimura A, Masuda M, Masuda S, Yasumatsu R, Umebayashi M, Tanaka H, Koya N, Nakagawa S, Yew PY, et al. Contribution of pre-existing neoantigen-specific T cells to a durable complete response after tumor-pulsed dendritic cell vaccine plus nivolumab therapy in a patient with metastatic salivary duct carcinoma. Immunol Invest. 2022;51:1498–514. https://doi.org/10.1080/08820139.2021.1973491.
182. Lopez de Rodas M, Villalba-Esparza M, Sanmamed MF, Chen L, Rimm DL, Schalper KA. Biological and clinical significance of tumour-infiltrating lymphocytes in the era of immunotherapy: a multidimensional approach. Nat Rev Clin Oncol. 2025. https://doi.org/10.1038/s41571-024-00984-x.
183. Westcott PMK, Sacks NJ, Schenkel JM, Ely ZA, Smith O, Hauck H, Jaeger AM, Zhang D, Backlund CM, Beytagh MC, et al. Low neoantigen expression and poor T-cell priming underlie early immune escape in colorectal cancer. Nat Cancer. 2021;2:1071–85. https://doi.org/10.1038/s43018-021-00247-z.
184. Gros A, Parkhurst MR, Tran E, Pasetto A, Robbins PF, Ilyas S, Prickett TD, Gartner JJ, Crystal JS, Roberts IM, et al. Prospective identification of neoantigen-specific lymphocytes in the peripheral blood of melanoma patients. Nat Med. 2016;22:433–8. https://doi.org/10.1038/nm.4051.
185. Pauken KE, Godec J, Odorizzi PM, Brown KE, Yates KB, Ngiow SF, Burke KP, Maleri S, Grande SM, Francisco LM, et al. The PD-1 Pathway Regulates Development and Function of Memory CD8(+) T Cells following Respiratory Viral Infection. Cell Rep. 2020;31: 107827. https://doi.org/10.1016/j.celrep.2020.107827.
186. Hu Z, Leet DE, Allesoe RL, Oliveira G, Li S, Luoma AM, Liu J, Forman J, Huang T, Iorgulescu JB, et al. Personal neoantigen vaccines induce persistent memory T cell responses and epitope spreading in patients with melanoma. Nat Med. 2021;27:515–25. https://doi.org/10.1038/s41591-020-01206-4.
187. Ciardiello F, Ciardiello D, Martini G, Napolitano S, Tabernero J, Cervantes A. Clinical management of metastatic colorectal cancer in the era of precision medicine. CA Cancer J Clin. 2022;72:372–401. https://doi.org/10.3322/caac.21728.
188. Wang X, Lamberti G, Di Federico A, Alessi J, Ferrara R, Sholl ML, Awad MM, Vokes N, Ricciuti B. Tumor mutational burden for the prediction of PD-(L)1 blockade efficacy in cancer: challenges and opportunities. Ann Oncol. 2024;35:508–22. https://doi.org/10.1016/j.annonc.2024.03.007.
189. Xiong L, He X, Wang L, Dai P, Zhao J, Zhou X, Tang H. Hypoxia-associated prognostic markers and competing endogenous RNA coexpression networks in lung adenocarcinoma. Sci Rep. 2022;12:21340. https://doi.org/10.1038/s41598-022-25745-7.
190. Zhu ZY, Tang N, Wang MF, Zhou JC, Wang JL, Ren HZ, Shi XL. Comprehensive Pan-Cancer Genomic Analysis Reveals PHF19 as a Carcinogenic Indicator Related to Immune Infiltration and Prognosis of Hepatocellular Carcinoma. Front Immunol. 2021;12: 781087. https://doi.org/10.3389/fimmu.2021.781087.
191. Zhang X, Shi M, Chen T, Zhang B. Characterization of the Immune Cell Infiltration Landscape in Head and Neck Squamous Cell Carcinoma to Aid Immunotherapy. Mol Ther Nucleic Acids. 2020;22:298–309. https://doi.org/10.1016/j.omtn.2020.08.030.
192. Barber DL, Wherry EJ, Masopust D, Zhu B, Allison JP, Sharpe AH, Freeman GJ, Ahmed R. Restoring function in exhausted CD8 T cells during chronic viral infection. Nature. 2006;439:682–7. https://doi.org/10.1038/nature04444.
193. Xu L, Hu Y, Liu W. Pyroptosis-Mediated Molecular Subtypes are Characterized by Distinct Tumor Microenvironment Infiltration Characteristics in Breast Cancer. J Inflamm Res. 2022;15:345–62. https://doi.org/10.2147/JIR.S349186.
194. Montesion M, Murugesan K, Jin DX, Sharaf R, Sanchez N, Guria A, Minker M, Li G, Fisher V, Sokol ES, et al. Somatic HLA Class I Loss Is a Widespread Mechanism of Immune Evasion Which Refines the Use of Tumor Mutational Burden as a Biomarker of Checkpoint Inhibitor Response. Cancer Discov. 2021;11:282–92. https://doi.org/10.1158/2159-8290.CD-20-0672.
195. McGranahan N, Rosenthal R, Hiley CT, Rowan AJ, Watkins TBK, Wilson GA, Birkbak NJ, Veeriah S, Van Loo P, Herrero J, et al. Allele-Specific HLA Loss and Immune Escape in Lung Cancer Evolution. Cell. 2017;171:1259–1271 e1211. https://doi.org/10.1016/j.cell.2017.10.001.
196. Pavelescu LA, Enache RM, Rosu OA, Profir M, Cretoiu SM, Gaspar BS. Predictive Biomarkers and Resistance Mechanisms of Checkpoint Inhibitors in Malignant Solid Tumors. Int J Mol Sci. 2024;25. https://doi.org/10.3390/ijms25179659.
197. Chida K, Kawazoe A, Kawazu M, Suzuki T, Nakamura Y, Nakatsura T, Kuwata T, Ueno T, Kuboki Y, Kotani D, et al. A Low Tumor Mutational Burden and PTEN Mutations Are Predictors of a Negative Response to PD-1 Blockade in MSI-H/dMMR Gastrointestinal Tumors. Clin Cancer Res. 2021;27:3714–24. https://doi.org/10.1158/1078-0432.CCR-21-0401.
198. Merker JD, Oxnard GR, Compton C, Diehn M, Hurley P, Lazar AJ, Lindeman N, Lockwood CM, Rai AJ, Schilsky RL, et al. Circulating Tumor DNA Analysis in Patients With Cancer: American Society of Clinical Oncology and College of American Pathologists Joint Review. J Clin Oncol. 2018;36:1631–41. https://doi.org/10.1200/JCO.2017.76.8671.
199. Fridland S, Choi J, Nam M, Schellenberg SJ, Kim E, Lee G, Yoon N, Chae YK. Assessing tumor heterogeneity: integrating tissue and circulating

tumor DNA (ctDNA) analysis in the era of immuno-oncology - blood TMB is not the same as tissue TMB. J Immunother Cancer. 2021;9. https://doi.org/10.1136/jitc-2021-002551.
200. Topham JT, O'Callaghan CJ, Feilotter H, Kennecke HF, Lee YS, Li W, Banks KC, Quinn K, Renouf DJ, Jonker DJ, et al. Circulating Tumor DNA Identifies Diverse Landscape of Acquired Resistance to Anti-Epidermal Growth Factor Receptor Therapy in Metastatic Colorectal Cancer. J Clin Oncol. 2023;41:485–96. https://doi.org/10.1200/JCO.22.00364.
201. Zang J, Zhang R, Jin D, Xie F, Shahatiaili A, Wu G, Zhang Y, Zhao Z, Du P, Jia S, et al. Integrated longitudinal circulating tumor DNA profiling predicts immunotherapy response of metastatic urothelial carcinoma in the POLARIS-03 trial. J Pathol. 2023;261:198–209. https://doi.org/10.1002/path.6166.
202. Raiber-Moreau EA, Portella G, Butler MG, Clement O, Konigshofer Y, Hadfield J. Development and validation of blood tumor mutational burden reference standards. Genes Chromosomes Cancer. 2023;62:121–30. https://doi.org/10.1002/gcc.23100.
203. Jing W, Li L, Zhang X, Wu S, Zhao J, Hou Q, Wu H, Ma W, Li S, Liu H, et al. Genetic Profiling of Breast Cancer with and Without Preexisting Metabolic Disease. Transl Oncol. 2020;13:245–53. https://doi.org/10.1016/j.tranon.2019.09.008.
204. Chaudhary R, Quagliata L, Martin JP, Alborelli I, Cyanam D, Mittal V, Tom W, Au-Young J, Sadis S, Hyland F. A scalable solution for tumor mutational burden from formalin-fixed, paraffin-embedded samples using the Oncomine Tumor Mutation Load Assay. Transl Lung Cancer Res. 2018;7:616–30. https://doi.org/10.21037/tlcr.2018.08.01.
205. Han Y, Wang D, Peng L, Huang T, He X, Wang J, Ou C. Single-cell sequencing: a promising approach for uncovering the mechanisms of tumor metastasis. J Hematol Oncol. 2022;15:59. https://doi.org/10.1186/s13045-022-01280-w.
206. Leader AM, Grout JA, Maier BB, Nabet BY, Park MD, Tabachnikova A, Chang C, Walker L, Lansky A, Le Berichel J, et al. Single-cell analysis of human non-small cell lung cancer lesions refines tumor classification and patient stratification. Cancer Cell. 2021;39:1594–1609 e1512. https://doi.org/10.1016/j.ccell.2021.10.009.
207. Duan Q, Tang C, Ma Z, Chen C, Shang X, Yue J, Jiang H, Gao Y, Xu B. Genomic Heterogeneity and Clonal Evolution in Gastroesophageal Junction Cancer Revealed by Single Cell DNA Sequencing. Front Oncol. 2021;11: 672020. https://doi.org/10.3389/fonc.2021.672020.
208. Loh JW, Lee JY, Lim AH, Guan P, Lim BY, Kannan B, Lee ECY, Gu NX, Ko TK, Ng CC, et al. Spatial transcriptomics reveal topological immune landscapes of Asian head and neck angiosarcoma. Commun Biol. 2023;6:461. https://doi.org/10.1038/s42003-023-04856-5.
209. Vega DM, Yee LM, McShane LM, Williams PM, Chen L, Vilimas T, Fabrizio D, Funari V, Newberg J, Bruce LK, et al. Aligning tumor mutational burden (TMB) quantification across diagnostic platforms: phase II of the Friends of Cancer Research TMB Harmonization Project. Ann Oncol. 2021;32:1626–36. https://doi.org/10.1016/j.annonc.2021.09.016.
210. Volders PJ, Aftimos P, Dedeurwaerdere F, Martens G, Canon JL, Beniuga G, Froyen G, Van Huysse J, De Pauw R, Prenen H, et al. A nationwide comprehensive genomic profiling and molecular tumor board platform for patients with advanced cancer. NPJ Precis Oncol. 2025;9:66. https://doi.org/10.1038/s41698-025-00858-0.
211. Botticelli A, Scagnoli S, Conte P, Cremolini C, Ascierto PA, Cappuzzo F, Aglietta M, Mazzuca F, Capoluongo E, Blandino G, et al. LBA7 The Rome trial from histology to target: The road to personalize targeted therapy and immunotherapy. Annals of Oncology. 2024;35. https://doi.org/10.1016/j.annonc.2024.08.2245.
212. Botticelli A, Scagnoli S, Guarneri V, Del Mastro L, Puglisi F, Pisegna S, Crimini E, Verkhovskaia S, Biffoni M, Conte PF, et al. 15P The ROME trial: Personalized target therapy and immunotherapy in the breast cancer subgroup. ESMO Open. 2025;10. https://doi.org/10.1016/j.esmoop.2025.104569.
213. Carbone DP, Reck M, Paz-Ares L, Creelan B, Horn L, Steins M, Felip E, van den Heuvel MM, Ciuleanu TE, Badin F, et al. First-Line Nivolumab in Stage IV or Recurrent Non-Small-Cell Lung Cancer. N Engl J Med. 2017;376:2415–26. https://doi.org/10.1056/NEJMoa1613493.
214. Hellmann MD, Paz-Ares L, Bernabe Caro R, Zurawski B, Kim SW, Carcereny Costa E, Park K, Alexandru A, Lupinacci L, de la Mora Jimenez E, et al. Nivolumab plus Ipilimumab in Advanced Non-Small-Cell Lung Cancer. N Engl J Med. 2019;381:2020–31. https://doi.org/10.1056/NEJMoa1910231.
215. Bhave P, Ahmed T, Lo SN, Shoushtari A, Zaremba A, Versluis JM, Mangana J, Weichenthal M, Si L, Lesimple T, et al. Efficacy of anti-PD-1 and ipilimumab alone or in combination in acral melanoma. J Immunother Cancer. 2022;10. https://doi.org/10.1136/jitc-2022-004668.
216. Lizardo DY, Kuang C, Hao S, Yu J, Huang Y, Zhang L. Immunotherapy efficacy on mismatch repair-deficient colorectal cancer: From bench to bedside. Biochim Biophys Acta Rev Cancer. 2020;1874: 188447. https://doi.org/10.1016/j.bbcan.2020.188447.
217. Budczies J, Seidel A, Christopoulos P, Endris V, Kloor M, Gyorffy B, Seliger B, Schirmacher P, Stenzinger A, Denkert C. Integrated analysis of the immunological and genetic status in and across cancer types: impact of mutational signatures beyond tumor mutational burden. Oncoimmunology. 2018;7: e1526613. https://doi.org/10.1080/2162402X.2018.1526613.
218. Wang J, Xiu J, Farrell A, Baca Y, Arai H, Battaglin F, Kawanishi N, Soni S, Zhang W, Millstein J, et al. Mutational analysis of microsatellite-stable gastrointestinal cancer with high tumour mutational burden: a retrospective cohort study. Lancet Oncol. 2023;24:151–61. https://doi.org/10.1016/S1470-2045(22)00783-5.
219. McGrail DJ, Pilie PG, Rashid NU, Voorwerk L, Slagter M, Kok M, Jonasch E, Khasraw M, Heimberger AB, Lim B, et al. High tumor mutation burden fails to predict immune checkpoint blockade response across all cancer types. Ann Oncol. 2021;32:661–72. https://doi.org/10.1016/j.annonc.2021.02.006.
220. Lin ZF, Qin LX, Chen JH. Biomarkers for response to immunotherapy in hepatobiliary malignancies. Hepatobiliary Pancreat Dis Int. 2022;21:413–9. https://doi.org/10.1016/j.hbpd.2022.08.002.
221. Hellmann MD, Ciuleanu TE, Pluzanski A, Lee JS, Otterson GA, Audigier-Valette C, Minenza E, Linardou H, Burgers S, Salman P, et al. Nivolumab plus Ipilimumab in Lung Cancer with a High Tumor Mutational Burden. N Engl J Med. 2018;378:2093–104. https://doi.org/10.1056/NEJMoa1801946.
222. Shitara K, Van Cutsem E, Bang YJ, Fuchs C, Wyrwicz L, Lee KW, Kudaba I, Garrido M, Chung HC, Lee J, et al. Efficacy and Safety of Pembrolizumab or Pembrolizumab Plus Chemotherapy vs Chemotherapy Alone for Patients With First-line, Advanced Gastric Cancer: The KEYNOTE-062 Phase 3 Randomized Clinical Trial. JAMA Oncol. 2020;6:1571–80. https://doi.org/10.1001/jamaoncol.2020.3370.
223. Lybaert L, Lefever S, Fant B, Smits E, De Geest B, Breckpot K, Dirix L, Feldman SA, van Criekinge W, Thielemans K, et al. Challenges in neoantigen-directed therapeutics. Cancer Cell. 2023;41:15–40. https://doi.org/10.1016/j.ccell.2022.10.013.
224. Roerden M, Castro AB, Cui Y, Harake N, Kim B, Dye J, Maiorino L, White FM, Irvine DJ, Litchfield K, et al. Neoantigen architectures define immunogenicity and drive immune evasion of tumors with heterogenous neoantigen expression. J Immunother Cancer. 2024;12. https://doi.org/10.1136/jitc-2024-010249.
225. Jaeckel E, Friedman SL, Hudecek M, Protzer U. Chimeric Antigen Receptor (CAR) T-cell Therapy: Engineering Immune Cells To Treat Liver Diseases. J Hepatol. 2025. https://doi.org/10.1016/j.jhep.2025.06.007.
226. Chen Y, Mason GH, Scourfield DO, Greenshields-Watson A, Haigh TA, Sewell AK, Long HM, Gallimore AM, Rizkallah P, MacLachlan BJ, et al. Structural definition of HLA class II-presented SARS-CoV-2 epitopes reveals a mechanism to escape pre-existing CD4(+) T cell immunity. Cell Rep. 2023;42: 112827. https://doi.org/10.1016/j.celrep.2023.112827.
227. Goloudina A, Le Chevalier F, Authie P, Charneau P, Majlessi L. Shared neoantigens for cancer immunotherapy. Mol Ther Oncol. 2025;33: 200978. https://doi.org/10.1016/j.omton.2025.200978.
228. Mauriello A, Cavalluzzo B, Ragone C, Tagliamonte M, Buonaguro L. Shared neoantigens' atlas for off-the-shelf cancer vaccine development. J Transl Med. 2025;23:558. https://doi.org/10.1186/s12967-025-06478-3.
229. Bedran G, Gasser HC, Weke K, Wang T, Bedran D, Laird A, Battail C, Zanzotto FM, Pesquita C, Axelson H, et al. The Immunopeptidome from a Genomic Perspective: Establishing the Noncanonical Landscape of MHC Class I-Associated Peptides. Cancer Immunol Res. 2023;11:747–62. https://doi.org/10.1158/2326-6066.CIR-22-0621.
230. Wang S, Guo L, Liu D, Liu W, Wu Y. HLAsupE: an integrated database of HLA supertype-specific epitopes to aid in the development of vaccines

with broad coverage of the human population. BMC Immunol. 2016;17:17. https://doi.org/10.1186/s12865-016-0156-x.
231. Rospo G, Chila R, Matafora V, Basso V, Lamba S, Bartolini A, Bachi A, Di Nicolantonio F, Mondino A, Germano G, et al. Non-canonical antigens are the largest fraction of peptides presented by MHC class I in mismatch repair deficient murine colorectal cancer. Genome Med. 2024;16:15. https://doi.org/10.1186/s13073-023-01275-3.
232. Venkataramany AS, Schieffer KM, Lee K, Cottrell CE, Wang PY, Mardis ER, Cripe TP, Chandler DS. Alternative RNA splicing defects in pediatric cancers: new insights in tumorigenesis and potential therapeutic vulnerabilities. Ann Oncol. 2022;33:578–92. https://doi.org/10.1016/j.annonc.2022.03.011.
233. Gebert J, Gelincik O, Oezcan-Wahlbrink M, Marshall JD, Hernandez-Sanchez A, Urban K, Long M, Cortes E, Tosti E, Katzenmaier EM, et al. Recurrent Frameshift Neoantigen Vaccine Elicits Protective Immunity With Reduced Tumor Burden and Improved Overall Survival in a Lynch Syndrome Mouse Model. Gastroenterology. 2021;161:1288–1302 e1213. https://doi.org/10.1053/j.gastro.2021.06.073.
234. Tretter C, de Andrade Kratzig N, Pecoraro M, Lange S, Seifert P, von Frankenberg C, Untch J, Zuleger G, Wilhelm M, Zolg DP, et al. Proteogenomic analysis reveals RNA as a source for tumor-agnostic neoantigen identification. Nat Commun. 2023;14:4632. https://doi.org/10.1038/s41467-023-39570-7.
235. Wang F, Sangfuang N, McCoubrey LE, Yadav V, Elbadawi M, Orlu M, Gaisford S, Basit AW. Advancing oral delivery of biologics: Machine learning predicts peptide stability in the gastrointestinal tract. Int J Pharm. 2023;634: 122643. https://doi.org/10.1016/j.ijpharm.2023.122643.
236. Strazar M, Park J, Abelin JG, Taylor HB, Pedersen TK, Plichta DR, Brown EM, Eraslan B, Hung YM, Ortiz K, et al. HLA-II immunopeptidome profiling and deep learning reveal features of antigenicity to inform antigen discovery. Immunity. 2023;56:1681–1698 e1613. https://doi.org/10.1016/j.immuni.2023.05.009.
237. Arends CM, Jaiswal S. Gene-Specific Machine Learning Models to Classify Driver Mutations in Clonal Hematopoiesis. Cancer Discov. 2024;14:1581–3. https://doi.org/10.1158/2159-8290.CD-24-0751.
238. He LN, Li H, Du W, Fu S, Luo L, Chen T, Zhang X, Chen C, Jiang Y, Wang Y, et al. Machine learning-based risk model incorporating tumor immune and stromal contexture predicts cancer prognosis and immunotherapy efficacy. iScience. 2023;26:107058. https://doi.org/10.1016/j.isci.2023.107058.
239. Moravec Z, Zhao Y, Voogd R, Cook DR, Kinrot S, Capra B, Yang H, Raud B, Ou J, Xuan J, et al. Discovery of tumor-reactive T cell receptors by massively parallel library synthesis and screening. Nat Biotechnol. 2025;43:214–22. https://doi.org/10.1038/s41587-024-02210-6.
240. Hsiue EH, Wright KM, Douglass J, Hwang MS, Mog BJ, Pearlman AH, Paul S, DiNapoli SR, Konig MF, Wang Q, et al. Targeting a neoantigen derived from a common TP53 mutation. Science. 2021;371. https://doi.org/10.1126/science.abc8697.
241. Tan CL, Lindner K, Boschert T, Meng Z, Rodriguez Ehrenfried A, De Roia A, Haltenhof G, Faenza A, Imperatore F, Bunse L, et al. Prediction of tumor-reactive T cell receptors from scRNA-seq data for personalized T cell therapy. Nat Biotechnol. 2025;43:134–42. https://doi.org/10.1038/s41587-024-02161-y.
242. Kraemer AI, Chong C, Huber F, Pak H, Stevenson BJ, Muller M, Michaux J, Altimiras ER, Rusakiewicz S, Simo-Riudalbas L, et al. The immunopeptidome landscape associated with T cell infiltration, inflammation and immune editing in lung cancer. Nat Cancer. 2023;4:608–28. https://doi.org/10.1038/s43018-023-00548-5.
243. Li R, Hao Y, Roche K, Chen G, Pan W, Wang AZ, Min Y. Chemotherapy-induced nanovaccines implement immunogenicity equivalence for improving cancer chemoimmunotherapy. Biomaterials. 2023;301: 122290. https://doi.org/10.1016/j.biomaterials.2023.122290.
244. Au KM, Swinnea JS, Wang AZ. Intratumoral Injectable Redox-Responsive Immune Niche Improves the Abscopal Effect in Radiotherapy. Adv Mater. 2025;37: e2411330. https://doi.org/10.1002/adma.202411330.
245. Anandappa AJ, Wu CJ, Ott PA. Directing Traffic: How to Effectively Drive T Cells into Tumors. Cancer Discov. 2020;10:185–97. https://doi.org/10.1158/2159-8290.CD-19-0790.
246. Jin Y, Lin C, Shi X, He Q, Yan J, Yu X, Chen M. Impact of clinical and molecular features on efficacy and outcome of patients with non-small cell lung cancer receiving second-line osimertinib. BMC Cancer. 2022;22:586. https://doi.org/10.1186/s12885-022-09683-1.
247. Ota N, Yoshimoto Y, Darwis NDM, Sato H, Ando K, Oike T, Ohno T. High tumor mutational burden predicts worse prognosis for cervical cancer treated with radiotherapy. Jpn J Radiol. 2022;40:534–41. https://doi.org/10.1007/s11604-021-01230-5.
248. He M, Xu W, Dan Y, Pan Y, Li Y, Chen M, Dong CM. Mannosylated Fluoropolypeptide Nanovaccines Remodeling Tumor Immunosuppressive Microenvironment to Achieve Highly Potent Cancer Immunotherapy. Adv Healthc Mater. 2024;13: e2401354. https://doi.org/10.1002/adhm.202401354.

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